PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 4

Cost and Management Accounting

Short Notes

For every lesson, in syllabus order: the key idea, the formats and formulas, the answer frame, the common mistakes with their fixes, and the recap.

66 lessons in 15 chapters · phone edition · made 2026-10-20 from the PassBook lessons

The labs (step-through simulators) are not printed: open the lesson in the PassBook app for the interactive version.

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CHAPTER 1

Introduction to Cost and Management Accounting

1.1 Objectives, scope and users of cost and management accounting

Key idea

Cost and management accounting exists to ascertain the cost of each cost object and to use it for pricing, cost control, cost reduction and decisions. It serves users inside the entity (who manage it) and outside it (who regulate, audit, own or lend to it).

ANSWER FRAME

"Describe the main objectives of a cost and management accounting system"

1. **Ascertainment of cost**: accumulated, assigned and ascertained for each cost object (unit, job, operation, process, department, service).
2. **Determination of selling price and profitability**: a basis for price fixation and rate negotiation, even where prices are set by external factors.
3. **Cost control**: discipline in expenditure, in four steps: pre-determined standard; measurement of actual performance; comparison; analysis of variance and action.

4. **Cost reduction**: the definition ("real and permanent reduction in the unit cost ... without impairing their suitability for the use intended or diminution in the quality"), value added and non-value added activities, continuous research.
5. **Assisting management in decision making**: relevant information for planning, implementing, measuring, controlling and evaluating.

In a case ("in a manufacturing organisation"), add one line of the case's facts under each heading.

"Give examples of internal and external users and explain how they are concerned" (5 marks)

Part of the answer	What earns the mark
Opening line	The information is used by different stakeholders, broadly categorised into internal and external users
Each internal user	Its name as ICAI gives it (policy makers, managers, operational level staff, employees) and its uses
Each external user	Its name (regulatory authorities, auditors, shareholders, creditors and lenders) and what it is concerned with
Number asked	Exactly as many of each as the question asks, the strongest ones first

Three answers to "Explain two external users"

Loses marks

"Auditors and banks."

Names only, and "banks" is not ICAI's term.

Passes

"Auditors use cost data for audit. Creditors and lenders use it to give loans."

Right users, thin uses.

Full marks

"Auditors need costing information and reports reviewed by management, for a financial or cost audit. Creditors and lenders need regular reports on the net debt position and stock balances."

ICAI's names, each with its concern.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: cost reduction that only kept costs on budget**
   Fix: cost reduction is a *real and permanent* reduction in the *unit* cost without impairing suitability or quality; it challenges the standards themselves. Test it against the three-fold assumption: a saving in unit cost, a permanent saving, quality unaffected if not improved.

2. **The scope that lost two functions**
   Fix: seven functions, and the Study Material calls **cost reports** the ultimate function of cost accounting (SM 1.8). Use the seven verbs.

3. **Auditors inside, employees outside**
   Fix: ICAI's internal users are policy makers, managers, operational level staff and employees; the external users are regulatory authorities, auditors, shareholders, and creditors and lenders (SM 1.12).

RECAP

- Five objectives, in ICAI's order: ascertainment of cost, determination of selling price and profitability, cost control, cost reduction, and assisting management in decision making.

- Scope, seven functions: costing, cost accounting, cost analysis, cost comparisons, cost control, cost reports (the ultimate function) and statutory compliances.
- Internal users: policy makers, managers, operational level staff, employees. External users: regulatory authorities, auditors, shareholders, creditors and lenders.
- Cost control: pre-determined standard → measure actual (same basis) → compare and report → analyse variance and act (amend standards if needed).
- Cost reduction: real and permanent reduction in unit cost without impairing suitability or quality; three-fold assumption: saving in unit cost, permanent, quality unaffected if not improved.
- Costing: "the technique and process of ascertaining costs". Cost, as a noun: expenditure (actual or notional) incurred on or attributable to a specified article, product or activity.
- Cost leadership needs a robust cost and management accounting system. (No navigation app, no shortcut.)

1.2 Cost accounting department and installing a costing system

Key idea

Study the business before installing the system, build the system so that people can and do use it, and remember that it must be worth its cost. A costing system that management does not trust produces reports that nobody reads.

ANSWER FRAME

"Discuss the essential features of a good cost accounting system" (5 marks)

Six headings, each with one line of ICAI's explanation: informative and simple (tailor-made, no unnecessary details); accurate and authentic; uniformity and consistency (benchmarking, comparability); integrated and inclusive (financial accounting, taxation, statistics, operational research); flexible and adaptive (technological, reporting, regulatory changes); trust on the system (active role of management).

"Factors to be considered before installing a costing system" (5 marks)

Factor	The line that earns the mark
(a) Objective	Fixing prices or establishing cost control

Factor	The line that earns the mark
(b) Nature of business or industry	Methods follow the industry's needs (refinery: process-wise accounts)
(c) Organisational hierarchy	Top: corporate strategy; strategic: pricing, diversification; operational: standard quantities, idle time
(d) Knowing the product	By-products; perishables: marginal costing
(e) Knowing the production process	Scientific apportionment needs knowledge of the process
(f) Information synchronisation	Other departments' needs (stock statement to the bank, tax returns)
(g) Method of maintenance of cost records	Inter-locking or reconciliation through control accounts
(h) Statutory compliances and audit	Statutory records, cost accounting standards
(i) Information attributes	Complete, accurate, timely, relevant

In a case ("as a consultant"), apply each factor to the facts in one more line.

Three answers to "practical difficulties and how to overcome them"

Loses marks

"Staff do not like change, so train them."

One vague difficulty, one remedy.

Passes

The four difficulties named, with no remedies, or remedies with no difficulties.

Half the question answered.

Full marks

The four difficulties, each with its reason, then the four remedies: sell the idea, behavioural approach, training, regular meetings.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: "Install it first, study the business later"**
   Fix: factors (nine) come before installation: objective, nature of business, hierarchy, product, process, synchronisation, cost records, statutory compliances and audit, information attributes. Essential features (six) are qualities of the finished system.

2. **Marginal costing for everything**
   Fix: ICAI's point is specific: products with by-products need by-product accounting; for perishable or short shelf-life products, marginal costing is appropriate, to know the contribution and minimum selling price.

3. **Limitations that are not ICAI's**
   Fix: the Study Material's three are expensive, requirement of reconciliation, and duplication of work (SM 1.24).

RECAP

- Nine factors to study before installing: objective, nature of business or industry, organisational hierarchy, knowing the product, knowing the production process, information synchronisation,

method of maintenance of cost records, statutory compliances and audit, and information attributes.

- Six essential features of a good system: informative and simple, accurate and authentic, uniformity and consistency, integrated and inclusive, flexible and adaptive, and trust on the system.
- Three limitations of cost accounting: it is expensive, it requires reconciliation with the financial accounts, and it duplicates work. So the first question is whether the system will be profitable to have.
- Five functions: collect cost by element; assign to cost objects; set budgets and standards, compare, report; information for decisions (MIS, the main function); evaluate responsibility centres.
- Perishables: marginal costing (contribution, minimum price). By-products: a system that accounts for them.
- Practical difficulties (ICAI's RTP and MTP answers): no top management support, staff resistance, user departments' non-co-operation, shortage of trained staff; remedies: sell the idea, behavioural approach, training, regular meetings.
- A system nobody trusts is a 46-page report nobody reads.

1.3 Cost, financial and management accounting, and financial management

Key idea

Financial accounting reports the financial performance of the entity, as a whole or by segment, to outsiders, from historical data, usually yearly, in a set format. Cost accounting ascertains the cost of each cost object for cost control and decisions, from historical and predetermined costs, as and when required. Management accounting is wider and forward-looking; all three serve financial management's decisions.

ANSWER FRAME

"Distinguish between cost accounting and management accounting" (4–5 marks)

A three-column table, Basis | Cost Accounting | Management Accounting, with the six bases in ICAI's order: nature, objective, area, recording of data, development, rules and regulation. "Any five" means the first five done well; a sixth row costs little and covers a weak one.

"Distinguish between financial accounting and cost accounting"

The same table with seven bases: objective, nature, recording of data, users of information, analysis of cost

and profit, time period, presentation of information. Write the key words examiners look for: "financial performance of an entity" against "cost control and decision making"; "historical" against "historical and pre-determined"; "segment-wise or as a whole" against "each cost object"; "usually for a year" against "as and when required"; "set format" against "no set formats".

"Explain the relationship of cost and management accounting with other disciplines"

1. Cost accounting provides inputs to financial accounting: cost of goods sold and inventories.
2. Management accounting takes inputs from cost accounts, financial accounts, statistical and operation management tools.
3. Cost and management accounting is an application of financial management for decision making purposes.
4. A labelled sketch: three overlapping circles (financial, management and cost accounting) on financial management.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: "Management accounting is cost accounting with a bigger salary"**
   Fix: management accounting records qualitative as well as quantitative information, is wider in area (financial accounting, budgeting, taxation, planning), projects the future and follows no specific rules. Six bases, and none of them is "users" or "time period".

2. **Predetermined costs in the financial accounts**
   Fix: financial accounting records historical data; cost accounting uses both historical and pre-determined costs (standards and budgets are part of it).

3. **The yearly cost report in a statutory format**
   Fix: cost reports and statements are prepared as and when required, in general with no set format, generally for internal management (sometimes for regulatory authorities also).

RECAP

- Cost accounting provides inputs to financial accounting (costs of goods sold and inventories); management accounting takes inputs from cost accounts, financial accounts, statistical and operation management tools.

- Cost vs management accounting, six bases: nature (quantitative / both), objective (cost of product or service / planning and co-ordination), area (cost ascertainment / wider), recording (past and present / future), development (industrial revolution / modern business), rules (follows / none).
- Financial vs cost accounting, seven bases: objective, nature, recording of data (historical / historical and pre-determined), users (outsiders / internal management), analysis (segment or whole / each cost object), time period (usually a year / as and when required), presentation (set format / no set format).
- Cost and management accounting is an application of financial management for decision making purposes.
- A segment profit can hide a loss-making cost object: only the cost records show it. (The cafés' ₹26,40,000 hid Bhopal's loss of ₹8,40,000.)

1.4 Cost units, cost centres, cost objects and responsibility centres

Key idea

The cost object is what you cost, the cost unit is what you cost it per, and the cost driver is what makes the cost rise. Responsibility centres say who answers for what: costs (cost centre), revenue (revenue centre), both (profit centre), or profit plus capital investment (investment centre).

ANSWER FRAME

"Define responsibility centre and discuss its types" (5 marks)

1. The definition: management delegates responsibility and authority to departments or persons, who are held responsible for performance in terms of expenditure, revenue, profitability and return on investment, measured against set standards.
2. **Cost centres**: accountable for costs under their control, measured against standards or budgets; (a) standard cost centre (measurable output, identifiable input-output ratio, responsible for adverse variances); (b) discretionary cost centre (output not measurable in financial terms; inputs compared with the allocated budget; R&D, advertisement).
3. **Revenue centres**: accountable for revenue (sales department); may incur selling expenditure such as

commission.

4. **Profit centres**: revenue and expenditure; judged on profitability (decentralised branches).
5. **Investment centres**: profitability plus authority over capital investment; judged on ROI besides profit (Maharatna, Navratna, Miniratna PSUs).

"Identify the responsibility centre, with reasons" (case)

Step	What to write
1	The type, in ICAI's name (standard cost centre, not "a production centre")
2	The fact in the case that decides it ("can buy trucks within its own capital budget")
3	The Study Material's test it meets ("authority to make capital investment decisions; measured on ROI")
4	Where a near-miss is likely, why not the next type ("not an investment centre: capital spending needs head-office approval")

"State the cost unit"

A two-column table, industry and cost unit, in the order the question lists them, using the Study Material's units; for a service, a composite unit where the service depends on two measures (tonne-km, passenger-km, patient day).

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the cost unit that was a building**
   Fix: the cost unit is the unit of product, service or time in which cost is expressed: for a hotel, room or meal (the CIMA list in the Study Material; ICAI's January 2025 answer also accepts guest days or room days). Ask "per what?" and the answer is the cost unit.

2. **Every branch an investment centre**
   Fix: a branch whose manager answers for revenue and costs is a profit centre ("decentralised branches", SM 1.23). Only the authority to make capital investment decisions, with performance measured on ROI, makes it an investment centre.

3. **The R&D lab on a standard cost**
   Fix: where output cannot be measured in financial terms, no input-output ratio exists: it is a discretionary cost centre, judged against its allocated budget.

4. **Cost driver or cost unit?**
   Fix: the cost driver is the factor that causes the cost (number of purchase orders for procurement cost); the cost unit is the measure the cost is expressed in (tonne-km for transport).

RECAP

- Cost object: anything for which a separate measurement of cost is required (product, service, project, customer, brand category, activity, department, programme); the nucleus of cost classification.
- Cost unit: a unit of product, service or time (or a combination) in which costs are ascertained or expressed; usually a physical unit (number, weight, area, volume, length, time, value); an order, contract or batch can be one.
- Responsibility centres answer for expenditure, revenue, profitability or return on investment: cost centres (costs under their control), revenue centres (revenue), profit centres (revenue and expenditure, judged on profitability) and investment centres (profit plus capital investment decisions, judged on ROI besides profit).
- Standard cost centre: measurable output, identifiable input-output ratio. Discretionary cost centre: output not measurable in financial terms, inputs compared with the allocated budget (R&D, advertisement).
- Cost driver: a factor influencing the level of cost (number of set-ups, purchase orders, inspection hours, tests).
- Cost units to know: power and electricity kWh; steel ton; transport passenger-km (tonne-km for goods); brick-making 1,000 bricks; brewing barrel; hospitals

patient day; education course, enrolled or successful student; credit control accounts maintained.

- Profit centre example: decentralised branches. Investment centre example: Maharatna, Navratna and Miniratna PSUs. (Papa buys the banquet hall; Mama-ji only stays within budget.)

1.5 Elements and classification of cost

Key idea

Classification is grouping costs by a common characteristic. ICAI uses six bases, and every cost sits in all six at once: the one you use depends on the question being asked and on the cost object (a box, a café, a department) you are costing.

FORMATS AND FORMULAS

From a list of costs to a cost statement

ICAI questions hand you a jumbled list and ask you to classify it, then total it. File each item first by element (direct or indirect), then by function. The statement below follows the order of ICAI's specimen cost sheet (SM 6.8) for a month with no opening or closing stocks. *Reproduce this layout from memory.*

Particulars	Amount (₹)
Direct material	xxx
Direct employee (labour) cost	xxx
Direct expenses	xxx
Prime cost	**xxx**

Particulars	Amount (₹)
Add: Works or factory overheads	xxx
Works or factory cost	**xxx**
Add: Administration overheads related to production	xxx
Cost of production (= cost of goods sold)	**xxx**
Add: General administration overheads	xxx
Add: Selling overheads	xxx
Add: Distribution overheads	xxx
Cost of sales	**xxx**

With stocks, quality control, research and development, recoveries and primary packing, the full format is in the cost sheet for a manufacturer. The Chapter 1 chart (SM 1.27) adds all administration overheads before cost of goods sold; cost of sales comes out the same, but in a cost sheet answer keep general administration after cost of goods sold, as SM 6.7–6.8 does.

∑ **Formula**

The element identities

$$\text{Prime cost} = \text{Direct material} + \text{Direct employee cost} + \text{Direct expenses}$$

$$\text{Overheads} = \text{Indirect material} + \text{Indirect employee cost} + \text{Indirect expenses}$$

Total cost = prime cost + overheads, whichever way the overheads are grouped (SM 1.25–1.26).

∑ **Formula**

The functional build-up (SM 1.27)

$$\text{Works cost} = \text{Prime cost} + \text{Factory overheads}$$

$$\text{Cost of sales} = \text{Works cost} + \text{Administration} + \text{S\&D overheads}$$

S&D = selling and distribution.

Definition

Direct materials

Materials which are present in the finished product (cost object) or can be economically identified in the product. A direct material used in small quantities, where identifying the quantity is not economically feasible, is treated as indirect; materials used for purposes ancillary to the business are indirect materials. (SM 1.25)

Definition

Direct labour

Labour which can be economically identified or attributed wholly to a cost object: employees engaged on the actual production of the product or in carrying out the operations converting raw materials into finished product. (SM 1.26)

Definition

Overheads

The aggregate of indirect material costs, indirect labour costs and indirect expenses, subdivided into production or works, administration, selling and distribution overheads. (SM 1.26)

ANSWER FRAME

"Discuss any five types of costs used in managerial decision making"

Write each as the term, ICAI's meaning, and one example. The five most useful, in ICAI's order: **differential cost** (change in total cost due to a change in activity level, technology, process or method), **opportunity cost** (value of the sacrifice or benefit of the opportunity forgone), **out-of-pocket cost** (portion of total cost involving cash outflow; avoidable if the proposal is not accepted), **shut down cost** (costs that continue while a plant is temporarily shut down, such as rent, rates and depreciation) and **sunk cost** (historical cost that plays no role in current decisions, such as the written-down value of a machine being replaced) (SM 1.34–1.35).

"Differentiate controllable and uncontrollable costs"

Point	Meaning
Controllable cost	Can be influenced by the manager heading the responsibility centre
Uncontrollable cost	Cannot be influenced by the action of a specified member

Point

Usual examples

Controllable cost

Direct material, direct labour, direct expenses, some overheads

Uncontrollable cost

Share of a service department's cost apportioned to another department

Point

Responsibility

Controllable cost

The centre's manager is held accountable for it, as a cost centre is for costs under its control (SM 1.22)

Uncontrollable cost

The specified member cannot influence it, so it is not used to judge that member

Point

Key point

Controllable cost

No cost is uncontrollable for everyone; the classification is in relation to a particular individual (SM 1.33)

Uncontrollable cost

"Discuss cost classification based on variability and controllability"

1. Variability: fixed, variable and semi-variable costs, each with ICAI's meaning and an example (SM 1.27–1.28).
2. Controllability: controllable and uncontrollable costs, with ICAI's meaning and the tool-room example (SM 1.33).

3. A closing line: the distinction is sometimes left to judgment; no cost is uncontrollable, only in relation to a particular individual.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the royalty that ended up in overheads**
   Fix: expenses that can be traced to the cost object in an economically feasible manner are direct expenses and belong in prime cost (SM 6.4).

2. **The written-down value that sneaked into the decision**
   Fix: it is a sunk cost, already incurred, and plays no role in the decision; only future costs that differ between the choices count, here the ₹1,50,000 scrap value.

3. **"Uncontrollable" stamped on a cost for everyone**
   Fix: controllability is always in relation to a particular individual; the decision to buy the machine was controllable by someone (SM 1.33).

RECAP

- Classification = grouping costs by common characteristics; six bases in ICAI's order: element,

function, behaviour, controllability, normality, managerial decision making (SM 1.24).

- Elements: material, labour, other expenses (each direct or indirect) and overheads; overheads = indirect material + indirect labour + indirect expenses.
- Direct means economically traceable to the cost object; tiny quantities and ancillary materials are indirect even if physically present.
- Prime cost = direct material + direct labour + direct expenses; + factory overheads = works cost; with no stocks, works cost = cost of goods sold; + general administration, selling and distribution overheads = cost of sales (SM 6.8).
- Functions: production, administration, selling, distribution, research and development.
- Controllability is always in relation to a particular individual; apportioned costs are uncontrollable for the receiving manager.
- Abnormal cost is charged to Costing Profit and Loss Account.
- Decisions: sunk costs are ignored; opportunity cost = benefit forgone; out-of-pocket = cash outflow, avoidable; shut-down costs continue while closed. (Same rupee, different list.)

1.6 Cost behaviour: fixed, variable, semi-variable and step costs

Key idea

Behaviour describes **totals** against activity within a range: fixed totals stay put, variable totals move in proportion, semi-variable totals do both, step totals stay flat and then jump. **Per unit**, the picture flips: fixed cost per unit falls as output rises, variable cost per unit stays the same.

FORMATS AND FORMULAS

Reproduce the high-low layout and these formulas from memory.

High-low working	Activity	Total cost (₹)
At the highest volume	xxx	xxx
At the lowest volume	xxx	xxx
Difference	**xxx**	**xxx**

Variable cost rate = difference in cost ÷ difference in activity; fixed cost = total cost at either level − variable rate × that level's activity. Check that both levels give the same fixed cost.

∑ **Formula**

Total cost within the relevant range

$$\text{Total cost} = \text{Fixed cost} + \text{Variable cost per unit} \times \text{Output}$$

Valid only "within certain output and turnover limits"; add any step or additional fixed cost separately.

∑ **Formula**

High-low method

$$\text{Variable cost rate} = \frac{\text{Cost at highest} - \text{cost at lowest}}{\text{Highest volume} - \text{lowest volume}}$$

Cost at highest and cost at lowest = total cost at the highest and at the lowest volume. The Study Material states it with sales value as the volume, so the rate is a percentage of sales (SM 1.30); with units or hours as the volume, the rate is per unit or per hour.

∑ **Formula**

Comparison by period or level of activity

$$\text{Variable cost per unit} = \frac{\text{Change in amount of expense}}{\text{Change in quantity of output}}$$

Fixed cost = expense at either level − variable cost per unit × output at that level (SM 1.31–1.32).

∑ **Formula**

Least squares method

$$y = mx + c$$

y = total cost, x = volume of output, m = variable element of cost per unit, c = total fixed cost (SM 1.32).

∑ **Formula**

Analytical method

$$\text{Variable part} = \text{Degree of variability} \times \text{Semi-variable expense}$$

$$\text{New total} = \text{Fixed part} + \text{Variable part} \times \frac{\text{New output}}{\text{Old output}}$$

Formula

Step cost

$$\text{Number of steps} = \frac{\text{Activity}}{\text{Activity per step}}, \text{ rounded up}$$

$$\text{Step cost} = \text{Number of steps} \times \text{Cost of one step}$$

Definition

Fixed costs

Costs which are incurred for a period, and which, within certain output and turnover limits, tend to be unaffected by fluctuations in the levels of activity (output or turnover). They do not tend to increase or decrease with the changes in output. (SM 1.27)

Definition

Variable costs

Costs which tend to vary with the volume of activity. Any increase in the activity results in an increase in the variable cost and vice-versa. (SM 1.28)

Definition

Semi-variable costs

Costs which contain both fixed and variable components and are thus partly affected by fluctuations in the level of activity. (SM 1.28)

ANSWER FRAME

"Discuss cost classification based on variability"

1. Fixed costs: ICAI's definition ("incurred for a period ... within certain output and turnover limits ... unaffected by fluctuations in the levels of activity") and examples: rent, insurance of factory building.
2. Variable costs: definition ("tend to vary with the volume of activity") and examples: direct material, direct labour.
3. Semi-variable costs: definition ("contain both fixed and variable components") and examples: telephone bills, gas and electricity.
4. A sketch of each, cost on the Y-axis and output on the X-axis.

"Explain the methods of segregating semi-variable costs"

Name all five in ICAI's order (graphical, high-low, analytical, comparison by period or level of activity, least squares), then two or three lines on each with its formula, and close with ICAI's verdict that least squares

is the best method (SM 1.29–1.32). A one-line numerical illustration for high-low earns the marks for "explain".

The numerical parts

Loses marks

"Power at 90% = ₹2,10,000 × 90 ÷ 60 = ₹3,15,000."

Treats a semi-variable cost as variable: the fixed ₹60,000 has been multiplied too.

Passes

"Variable rate ₹2.50 a kg, so power at 90% = ₹2,85,000."

Right figure, but the rate and the fixed part appear from nowhere.

Full marks

A working note for each item: change in cost ÷ change in output = rate; fixed = total − rate × output; check at the other level; then the estimate. Step costs shown as number of steps × cost per step, with the rounding stated.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the rent that doubled with the chai**

   Fix: split first. Only variable costs and the variable parts of semi-variable costs move with output; fixed costs stay, and step costs move only when a band is crossed.

2. **The supervisor who became 2.25 rupees a kilogram**

   Fix: when the question says "one for every ..." or "for every 20% increase ... or part thereof", count the steps and round up ("or proportionately" is pro rata instead).

3. **High-low with the wrong two months**

   Fix: high-low uses the highest and lowest volumes; comparison by period uses the two periods the question names.

RECAP

- Fixed: incurred for a period, unaffected by activity within limits (rent, insurance). Variable: varies with activity (direct material, direct labour). Semi-variable: both parts (telephone, gas, electricity) (SM 1.27–1.28).
- Step cost: flat within a band of activity, jumps at the next band; count steps and round up.

- Per unit: fixed cost per unit falls as output rises; variable cost per unit stays the same.
- Five methods in ICAI's order: graphical, high-low, analytical, comparison by period or level of activity, least squares (the best).
- High-low: variable rate = change in cost ÷ change in volume between the highest and lowest volumes; fixed = total − rate × volume; check at the other level.
- Comparison by period: change in expense ÷ change in output for the two periods given.
- Least squares: y = mx + c (m variable cost per unit, c total fixed cost).
- Split before you flex: never scale a whole semi-variable cost in proportion to output. (The rent does not double with the chai.)

1.7 Cost control and cost reduction

Key idea

Cost control keeps costs to the standard (it accepts the standard). **Cost reduction** challenges the standard and lowers it, and counts only if the saving is in unit cost, permanent, and leaves utility and quality unaffected.

ANSWER FRAME

"Distinguish between cost control and cost reduction" (5 marks)

A two-column table with ICAI's five points in ICAI's order: aim (maintain standards / challenge standards), conditions (lowest cost under existing conditions / no condition permanent), emphasis (past and present / present and future), function (preventive / corrective, operates even with efficient cost control), end (ends when targets achieved / no visible end, continuous). If time allows, add one line each: the definition of cost reduction and the four steps of cost control.

"DISCUSS some more differences" (when the question gives one)

Leave out the point already given (for example the emphasis) and write the other four. Repeating the given point earns nothing.

"Explain cost reduction" or "What are the assumptions of cost reduction?"

1. ICAI's definition, word for word.
2. The approach: no cost is termed as lowest.
3. The two actions: value analysis (value added and non-value added activities; Value Chain Analysis by Michael Porter) and continuous research and study.
4. The three-fold assumptions: saving in unit cost; permanent nature; utility and quality unaffected, if not improved.

Loses marks

"Cost control controls cost; cost reduction reduces cost."

Restates the names; no point of difference.

Passes

Three or four correct points in a list, in any order, some in the student's own words.

Correct, but misses points and ICAI's phrases.

Full marks

All five points in a two-column table, in ICAI's order, in ICAI's words ("existing conditions", "preventive", "corrective", "no visible end").

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the preventive and the corrective swapped**
   Fix: ICAI's table says cost control is a **preventive** function and cost reduction a **corrective** function that operates even when an efficient cost control system exists (SM 1.6), and the suggested answers of May 2024 and May 2026 print exactly these words. Write them as ICAI gives them, whatever your intuition says.

2. **Calling every saving a cost reduction**
   Fix: test all three assumptions: a saving in unit cost, of a permanent nature, with utility and quality unaffected. Fail one, and it is not cost reduction.

3. **"Under best conditions"**
   Fix: ICAI's words are "under **existing** conditions"; it is cost reduction that refuses to accept existing conditions as permanent.

RECAP

- Cost control: maintaining discipline in expenditure, so that expenditures are in consonance with predetermined standards and variations are noted and reported continuously.

- Four steps of cost control, in order: set the standard, measure actual on the same basis, compare and report the variance, analyse and act (amend the standard if necessary).
- Cost reduction: real and permanent reduction in unit cost without impairing suitability for the use intended or diminution in quality.
- Three-fold assumptions: saving in unit cost; permanent; utility and quality unaffected, if not improved.
- Two actions: value analysis (eliminate non-value added activities; Porter's Value Chain Analysis) and continuous research and study.
- Five differences: maintain vs challenge standards; existing conditions vs no condition permanent; past and present vs present and future; preventive vs corrective; ends at target vs no visible end.
- Reduction lowers the standard; control then guards the new standard. (The ₹6.50 cup became the ₹6.10 cup.)

1.8 Methods and techniques of costing, and digital costing

Key idea

The **method** follows the nature of the output (job, batch, contract, process, single output, service, or a mix). A **technique** (uniform, marginal, standard, historical, absorption) is used with any method. **Digital costing** changes how cost data is collected, stored, analysed and reported, and links the entity with its suppliers, customers and market.

ANSWER FRAME

"State the method of costing and the suggested unit of cost"

A three-column table, one row per industry. ICAI's pairings from the Study Material and suggested answers:

Industry	Method	Cost unit
Transport	Operating (service) costing	Passenger-km, tonne-km
Power, electricity supply	Operating (service) costing	Kilowatt-hour (kWh)
Hospital	Operating (service) costing	Patient day

Industry	Method	Cost unit
Hotel	Operating (service) costing	Room, meal
Bridge, road, building	Contract costing	Each contract
Printing, interior decoration, advertising	Job costing	Each job
Bricks	Single or output costing	1,000 bricks
Coal	Single or output costing	Tonne
Cement	Single or output costing	Ton, per bag
Chemicals, oil refinery	Process costing	Litre, kilogram, ton; barrel
Biscuits, pharmaceuticals, bakery	Batch costing	Batch
Bicycles, car assembly	Multiple costing	Number

(Methods: SM 1.36–1.38 and the suggested answers in Exam angle; cost units: SM 1.20–1.21, and cost units.)

"Describe any five benefits of the Digital Costing System" (5 marks)

Open with a two-line definition (collects, classifies and accounts data like the conventional system; differs in collection, storage, analysis and reporting; links production, procurement and inventory with suppliers, customers and the market). Then five of the eight benefits, each with ICAI's key phrase: **cost with certainty** on a cost object; **time spent** for incentive plans; **material requirement planning** and JIT; **non-value-added activities** eliminated; **real-time** standards and variances; **Marked to Market** standards; **demand prediction** for budgets; **cost behaviour** for cost-benefit analysis.

"Explain the impact of IT in cost accounting" (5 marks)

ICAI's seven points in order: integration through ERP and simpler reconciliation; paperless documents; procurement over intranet and extranet, enabling JIT; codified, accurate and timely cost information, customised job-wise or process-wise; uniformity of reports, budgets and standards; real-time variance reports; elimination of non-value-added activities. Five well-explained points earn the marks.

Loses marks

"Bicycle: unit costing."

The industry name was read, not the output: parts and assembly need different methods.

Passes

"Bicycle: multiple costing."

Correct, but a bare name; an "explain your reasoning" question gives no marks for the reason.

Full marks

"Bicycle: multiple costing. The parts are costed by job or batch costing and the assembly by single or output costing; the combination is multiple costing. Cost unit: number of bicycles."

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: marginal costing listed as a method**
   Fix: methods are single or output, batch, job, contract, process, operating and multiple costing (SM 1.36–1.37); uniform, marginal, standard, historical and absorption costing are techniques (SM 1.38–1.39).

2. **Operating costing and operation costing treated as one**
   Fix: operating costing is for concerns rendering services (transport, water supply, retail trade); operation costing is process costing applied to each mechanical operation (SM 1.37). ICAI's January 2021 (old scheme) answer printed "Operation/ Service Costing" for an airline; write the Study Material's term, operating (service) costing.

3. **Digital costing answered as "computerised accounts"**
   Fix: use ICAI's points: links with suppliers, customers and the market; AI and machine learning on Big Data; real-time standards and variances; Marked to Market standards; demand prediction; elimination of non-value-added activities (SM 1.18–1.19).

RECAP

- Methods (SM 1.36–1.37): single or output, batch, job, contract, process (operation), operating, multiple. The method follows the nature of the output.
- Batch cost per unit = cost of the batch ÷ units in the batch; operation costing = process costing for mechanical operations.
- Techniques (SM 1.38–1.39): uniform, marginal, standard costing and variance analysis, historical

(post, continuous), absorption. Standard costing can be used with any method.

- Uniform costing's two advantages: inter-firm comparison; an industry cost for tax relief or protection.
- IT impact (SM 1.17–1.18): ERP integration, paperless, extranet procurement and JIT, codified timely cost, uniformity, real-time variances, NVA elimination.
- Digital costing (SM 1.18): same functions, different collection, storage, analysis and reporting; links production, procurement and inventory with suppliers, customers and the market.
- Its benefits (SM 1.19): cost with certainty, time data for incentives, MRP and JIT, NVA elimination, real-time standards, M2M standards, demand prediction, cost behaviour analysis. (Containers, lenses, and now sensors.)

CHAPTER 2

Cost Sheet

2.1 Cost sheet for a manufacturer

Key idea

A cost sheet classifies cost by function and builds it up in a fixed order: prime cost → works cost → cost of production → cost of goods sold → cost of sales. Each stock is adjusted at the tier where it sits: raw material in materials consumed, work-in-process in works cost, finished goods in cost of goods sold.

FORMATS AND FORMULAS

ICAI's specimen format for a manufacturing entity (SM 6.8–6.9), in ICAI's words and line order. *Reproduce it from memory: examiners mark the subtotals.* ICAI prints two amount columns, **Total Cost (₹)** and **Cost per unit (₹)**; one is shown here so the format reads on a phone.

Particulars	Total cost (₹)
1. Direct materials consumed: Opening Stock of Raw Material; Add: Additions/ Purchases; Less: Closing stock of Raw Material	xxx
2. Direct employee (labour) cost	xxx
3. Direct expenses	xxx

Particulars	Total cost (₹)
4. Prime Cost (1+2+3)	**xxx**
5. Add: Works/ Factory Overheads	xxx
6. Gross Works Cost (4+5)	**xxx**
7. Add: Opening Work in Process	xxx
8. Less: Closing Work in Process	(xxx)
9. Works/ Factory Cost (6+7−8)	**xxx**
10. Add: Quality Control Cost	xxx
11. Add: Research and Development Cost	xxx
12. Add: Administrative Overheads (relating to production activity)	xxx
13. Less: Credit for Recoveries/Scrap/By-Products/misc. income	(xxx)
14. Add: Packing cost (primary)	xxx
15. Cost of Production (9+10+11+12−13+14)	**xxx**
16. Add: Opening stock of finished goods	xxx
17. Less: Closing stock of finished goods	(xxx)
18. Cost of Goods Sold (15+16−17)	**xxx**
19. Add: Administrative Overheads (General)	xxx

Particulars	Total cost (₹)
20. Add: Marketing Overheads: Selling Overheads; Distribution Overheads	xxx
21. Cost of Sales (18+19+20)	**xxx**

Secondary packing sits with selling and distribution (SM 6.7); interest, if any, on its own line before cost of sales (SM 6.10). Then sales, with profit as the balancing figure (or profit first, then sales, when a margin is given).

Interactive version: PassBook app, lesson Cost sheet for a manufacturer

∑ **Formula**

Prime cost and works cost

$$\text{Prime cost} = \text{Direct materials} + \text{Direct employee cost} + \text{Direct expenses}$$

$$\text{Works cost} = \text{Prime cost} + \text{Factory overheads} + \text{Opening WIP} - \text{Closing WIP}$$

∑ **Formula**

Cost of production, cost of goods sold, cost of sales

$$\begin{aligned}\text{Cost of production} &= \text{Works cost} \\ &+\text{QC} + \text{R\&D} \quad +\text{Production admin} \\ &-\text{Recoveries} \quad +\text{Primary packing}\end{aligned}$$

$$\begin{aligned}\text{COGS} &= \text{Cost of production} \\ &+\text{Opening FG} - \text{Closing FG}\end{aligned}$$

$$\begin{aligned}\text{Cost of sales} &= \text{COGS} + \text{General admin} \\ &+\text{Selling} + \text{Secondary packing} \quad +\text{Distribution}\end{aligned}$$

QC = quality control; R&D = research and development for the process; production admin = administrative overheads related to production; recoveries = realised or realisable value of scrap, waste and by-products; FG = finished goods.

∑ **Formula**

Profit as a percentage of sales or of cost

$$\begin{aligned}\text{Profit} &= x\% \text{ of sales} \\ &= \frac{x}{100 - x} \text{ of cost of sales}\end{aligned}$$

20% on sales is 25% on cost; 25% on sales is 33⅓% on cost. Sales = cost of sales ÷ (1 – x%).

ICAI's examples under each head

The lists to classify from, in ICAI's words and order (SM 6.3–6.8). September 2026 Q6(a) asked exactly this for ten items.

Head	ICAI's examples
Direct material (additions to cost)	(a) freight inwards; (b) insurance and other expenditure directly attributable to procurement; (c) trade discounts or rebates (to be deducted); (d) duties and taxes (if input tax credit is not available or availed)
Direct employee (labour) cost	(a) wages and salary; (b) allowances and incentives; (c) payment for overtimes; (d) bonus / ex-gratia; (e) employer's contribution to welfare funds such as provident fund; (f) other benefits (medical, leave with pay, free or subsidised food, leave travel concession, provisions for retirement benefits)
Direct expenses	(a) utilities such as power and fuel, steam; (b) royalty paid or payable for production or provision of service; (c) hire charges for specific equipment; (d) fee for technical assistance and know-how; (e) amortised cost of moulds, patterns, patents; (f) product or service specific design or drawing; (g) product or service specific software; (h) other expenses directly related with production or provision of service

Head	ICAI's examples
Factory overheads	(a) consumable stores and spares; (b) depreciation of plant and machinery, factory building; (c) lease rent of production assets; (d) repair and maintenance of plant and machinery, factory building; (e) indirect employees cost related with production activities; (f) drawing and designing department cost; (g) insurance of plant and machinery, factory building, stock of raw material and WIP; (h) amortised cost of jigs, fixtures, tooling; (i) service department cost such as tool room, engineering and maintenance, pollution control
Selling overheads	(a) salary and wages of the sales department and of employees directly related with selling; (b) rent, depreciation, maintenance and other cost of the sales department; (c) advertisement, maintenance of website for online sales, market research
Distribution overheads	(a) salary and wages of employees engaged in distribution; (b) transportation and insurance related with distribution; (c) depreciation, hire charges, maintenance and other operating costs of distribution vehicles

Definition

Prime cost

Prime cost represents the total of direct materials costs, direct employee (labour) costs and direct expenses. (SM 6.3)

Definition

Primary and secondary packing cost

Primary packing cost: cost incurred on packing materials which are essential to hold and preserve the product for further processing or its use by consumer. Secondary packing cost: cost incurred on packing materials which is used to store, transport, and promote the product. (SM 6.20)

Definition

Cost of sales

It is the total cost of a product incurred to make the product available to the customer or consumer. It is the aggregate of cost of goods sold, administrative costs, marketing costs and other separate line items of cost which could not form part of cost of production. (SM 6.20)

ANSWER FRAME

The chapter's own theory questions (SM 6.22) and the MTP favourites. Write the points in this order.

Describe how costs are classified on the basis of function. Costs are divided according to the function for which they have been incurred: (i) production / manufacturing cost; (ii) administration cost; (iii) selling cost; (iv) distribution cost; (v) research and development costs. In a cost sheet they are grouped into prime cost, cost of production, cost of goods sold and cost of sales.

Explain the treatment of administration overheads. (1) Only administrative overheads related to production are included in cost of production. (2) General administration overheads are not included in production cost; they are added to cost of goods sold to arrive at cost of sales. (3) General administration includes (SM 6.7): (a) depreciation and maintenance of building, furniture etc. of corporate or general management; (b) salary of administrative employees, accountants, directors, secretaries etc.; (c) rent, rates and taxes, insurance, lighting, office expenses etc.; (d) indirect materials: printing and stationery, office supplies etc.; (e) legal charges, audit fees, corporate office expenses like directors' sitting fees, remuneration and commission, meeting expenses etc.

State the advantages of a cost sheet. The five points of SM 6.10, in that order.

Explain the treatment of the following items. One line per item, in the words of the table in How it works.

Loses marks

One column of figures, closing stock at the cost of goods sold figure, GST added "to be safe", income tax in administration.

Passes

The right tiers in the right order, but stocks valued without showing units produced, and excluded items silently dropped.

Full marks

ICAI's line order, every subtotal named, working notes for units and closing stock, each excluded item listed with its reason.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the GST that sneaked into material cost**
   Fix: GST is material cost only if input tax credit is not available or not availed.

2. **The managing director's salary in cost of production**
   Fix: only administration related to production goes into cost of production; the rest comes after cost of goods sold.

3. **The two kinds of packing swapped**
   Fix: packing that holds and preserves the product is primary (cost of production); packing to store, transport or promote it is secondary (selling and distribution).

4. **Closing stock valued at the wrong rate**
   Fix: this period's cost of production per unit (FIFO), ₹58.

RECAP

- Functional classification: production, administration, selling, distribution, R&D (SM 6.2).
- Prime cost = direct materials consumed + direct employee cost + direct expenses.

- Works cost = prime cost + factory overheads + opening WIP − closing WIP.
- Cost of production = works cost + QC + R&D (process) + production administration − scrap and by-product credit + primary packing.
- Cost of goods sold = cost of production + opening − closing finished goods (closing at this period's cost of production per unit, FIFO).
- Cost of sales = cost of goods sold + general administration + selling + secondary packing + distribution; interest as a separate item.
- Out of cost: abnormal costs, penalties and fines, creditable GST, income tax; subsidies reduce the cost they relate to.
- x% profit on sales = x ÷ (100 − x) on cost. (The tiffin is built from the bottom up.)

Interactive version: PassBook app, lesson Cost sheet for a manufacturer

2.2 Estimated cost sheets, tenders and quotations

Key idea

Price to be quoted = the order's estimated direct costs + overheads at pre-determined recovery rates (from past cost sheets, on ICAI's bases) + costs specific to the order + profit. A margin "on selling price" must be converted before it is added to cost: 20% on price is 25% on cost.

FORMATS AND FORMULAS

The estimated cost sheet follows the cost sheet's own order (cost sheet for a manufacturer), with each overhead line showing its rate and base. *Reproduce it from memory.*

Statement of estimated cost and price to be quoted	Amount (₹)
Direct materials (estimated quantity × expected price)	xxx
Direct wages (estimated hours × expected rate)	xxx
Direct expenses specific to the order	xxx
Prime cost	**xxx**
Add: Factory overheads (F% of direct wages)	xxx
Works / factory cost	**xxx**

Statement of estimated cost and price to be quoted	Amount (₹)
Add: Administration overheads (A% of works cost)	xxx
Cost of production	**xxx**
Add: Selling and distribution overheads (S% of cost of production)	xxx
Add: Costs specific to the order (delivery, special packing)	xxx
Total cost	**xxx**
Add: Profit (x% on price = x ÷ (100 − x) on cost)	xxx
Price to be quoted	**xxx**

∑ **Formula**

Overhead recovery rates from a past cost sheet

$$F = \frac{\text{Factory overheads}}{\text{Direct wages}} \times 100$$

$$A = \frac{\text{Admin overheads}}{\text{Works cost}} \times 100$$

$$S = \frac{\text{S\&D overheads}}{\text{Cost of production}} \times 100$$

S&D = selling and distribution. The base in each denominator is the one the question names; the same base is used for the order.

∑ **Formula**

Pre-determined overhead rate

$$\text{Pre-determined rate} = \frac{\text{Estimated overheads}}{\text{Budgeted units}}$$

Estimated (normal) overheads for the period ÷ budgeted number of units during the period (SM 4.41); with hours or wages as the denominator for an hour rate or a percentage rate.

Formula

Price from a margin, and cost from a price

$$\text{Price} = \frac{\text{Total cost}}{1 - x\%}$$

$$\text{Total cost} = \frac{\text{Selling price}}{100\% + p\%}$$

x = profit as a percentage of selling price; p = profit as a percentage of total cost (SM 4.51).

Definition

Estimated cost

Kohler defines estimated cost as "the expected cost of manufacture, or acquisition, often in terms of a unit of product computed on the basis of information available in advance of actual production or purchase". Estimated costs are prospective costs since they refer to prediction of costs. (SM 1.34)

Definition

Pre-determined cost

A cost which is computed in advance before production or operations start, on the basis of specification of all the factors affecting cost, is known as a pre-determined cost. (SM 1.34)

ANSWER FRAME

For a quotation question, ICAI's suggested answers earn marks in this order:

1. **Working note: recovery rates**, each with its formula and base ("Normal factory overheads ÷ direct labour hours = ₹150 an hour"; "Administration ÷ works cost × 100 = 10%").
2. **Working note: revised figures** (new material price, wage rate, revised overheads).
3. **Statement of estimated cost**, in cost sheet order, each overhead line showing "% of base".
4. **Profit line with the conversion stated** ("20% on price = 25% on cost").
5. **The final line in the question's words**: "Price to be quoted" or "Sales value", and a one-line conclusion ("Hence the price to be quoted is ₹42,23,340").

For theory, the point to write is ICAI's: pre-determined overhead rates are used because "production overheads are to be determined in advance for fixing selling price, quote tender price and to formulate budgets" (SM 4.41), and a cost sheet "facilitates the preparation of cost estimates required for submitting tenders" (SM 6.10).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the 20% that was added to cost**
   Fix: 20% on price = 25% on cost; price = ₹33,78,672 ÷ 80% = ₹42,23,340.

2. **Last year's material price in this year's quotation**
   Fix: revise direct costs first (quantity with its normal loss, then the new price), then apply the rates.

3. **The right rate on the wrong base**
   Fix: a rate is only valid on the base it was computed on; write the base on every overhead line.

4. **Fixed cost per unit carried to a new volume**
   Fix: at a new level, take fixed costs in total and divide by the new volume.

RECAP

- An estimated cost sheet is a cost sheet for the future; its bottom line is the price to be quoted (SM 6.10, advantage iii).
- Recovery rates from last period: factory overheads ÷ direct wages; administration ÷ works cost; selling and distribution ÷ cost of production (or the bases the question names).

- Revise direct costs to expected prices first, then apply each rate to the order's own base.
- Add order-specific costs (delivery, special dies or packing) at their own amounts.
- Price = total cost ÷ (1 − margin on price); x% on price = x ÷ (100 − x) on cost; total cost = price ÷ (100% + p% on cost).
- Two jobs with known prices give two equations in F and A; eliminate, then price the third job (SM 4.49–4.52). A common unknown mark-up cancels in the ratio of the two prices.
- At a new volume, fixed costs stay in total and change per unit; split semi-variable costs first.
- The halwai prices the wedding before the wedding.

Interactive version: PassBook app, lesson Estimated cost sheets, tenders and quotations

2.3 Cost sheet for the service sector

Key idea

A service cost statement has no stock adjustments. It is prepared on functional classification or on variability (fixed or standing charges, variable or running charges, semi-variable or maintenance charges), and the total is divided by the service's cost unit (per meal, passenger-km, patient-day, project) to get cost per unit, then the price.

FORMATS AND FORMULAS

Reproduce the operating cost sheet from memory. Each head (A standing charges, B running charges, C maintenance charges) is followed by its own total and a per-unit figure. A commission or tax on takings is shown either inside running charges (SM Illustration 3, 12.16) or on its own line after the A + B + C total (January 2025 Q4(a): "total cost before commission", then commission); either way its amount is solved from takings.

Operating cost sheet for ... (period)	Total (₹)	Per unit (₹)
A. Standing (fixed)		
Rent, insurance, taxes	xxx	
Salaries (time basis)	xxx	
Depreciation (time basis)	xxx	

Operating cost sheet for ... (period)	Total (₹)	Per unit (₹)
Interest, administration	xxx	
Total standing charges	**xxx**	**xxx**
B. Running (variable)		
Fuel, food, consumables	xxx	
Wages per trip or unit	xxx	
Depreciation (usage basis)	xxx	
Total running charges	**xxx**	**xxx**
C. Maintenance		
Repairs, tyres, spares	xxx	**xxx**
Total cost (A + B + C)	**xxx**	**xxx**
Add: Charges on takings	xxx	xxx
Add: Profit	xxx	xxx
Takings / price	**xxx**	**xxx**

On functional classification the heads are those of the manufacturer's cost sheet (direct material, direct employee cost, direct expenses, overheads), without the work-in-process and finished goods lines.

Σ **Formula**

Cost per unit of service

$$\text{Cost per unit} = \frac{\text{Total cost of the period}}{\text{Total service units of the period}}$$

Units: per meal, per passenger-km, per patient-day, per room-day, per project, per application (SM 12.3–12.4).

Σ **Formula**

Takings when charges are a percentage of takings

$$\text{Takings} = \frac{\text{Total cost}}{1 - c\% - p\%}$$

c = commission, licence fee or tax as a percentage of takings; p = profit as a percentage of takings. The equation behind it: T = cost + cT + pT (SM 12.16).

Definition

Statement of costs for service sectors (SM 12.10)

For preparing a statement of cost or a cost sheet for service sector, costs are usually collected and accumulated for a specified period viz. a month, quarter or a year. The cost statement may be prepared either on the basis of functional classification or on the basis of variability: (1) fixed costs or standing charges; (2) variable costs or operating expenses; (3) semi-variable costs or maintenance expenses.

ANSWER FRAME

Differentiate between service costing and product costing. ICAI's four points, in ICAI's order (SM 12.3), as a table. ICAI's own answer (RTP September 2024 Q19(b)) states each point for services only; the product column is the contrast that makes it a "differentiate" answer.

Basis	Service costing	Product costing
Tangibility	Services are intangible and cannot be stored: no inventory	Products can be stored: inventories of WIP and finished goods
Cost units	Composite cost units (passenger-km, patient-day)	Simple units (per unit, per batch, per job)
Material vs employee cost	Employee cost is the major element	Material cost is usually the major element
Traceability	Indirect costs such as administration are significant; tracing may not be economically feasible	Larger share of costs traced directly

What do you understand by operating costs? Describe their features. ICAI's answer (RTP May 2025 Q20(b)): (1) operating costs are the costs incurred by undertakings which do not manufacture any product but provide a service, such as transport concerns, electricity

undertakings, hospitals and theatres; service costing is also known as operating costing (SM 12.2). (2) They are classified under three categories, with examples: operating and running charges (variable: petrol, diesel, lubricants), maintenance charges (semi-variable: tyres, repairs, spares) and fixed or standing charges (garage rent, insurance, road licence, depreciation, interest on capital, salary of the operating manager) (SM 12.10, 12.12). (3) The cost unit is composite, such as passenger-km or kWh (SM 12.3). Add internal and external applications (SM 12.2) if the marks allow.

Numerical answer, in order: working notes (units of service, period conversions, depreciation basis); the statement with A, B, C heads, totals and per-unit column; the takings equation; the final rate in the question's words ("fare per passenger-km", "fee per trainee").

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the licence fee charged on cost**
   Fix: percentages of takings are solved for: ₹66 ÷ 0.80 = ₹82.50.

2. **Closing stock in a service statement**
   Fix: a service cannot be stored (SM 12.3); the statement runs straight from costs of the period to cost per unit.

3. **Depreciation always fixed**
   Fix: time basis = fixed; activity or usage basis = variable (SM 12.10).

4. **The drop-outs forgotten**
   Fix: cost what actually happens (54 completers, 6 half-used kits, 54 exam fees) and choose the divisor the question asks for: per enrolled trainee for the fee, per certified trainee for the cost of a certificate.

RECAP

- No inventory in services: no WIP or finished goods lines (SM 12.3).
- Two layouts: functional (as for products) or variability: standing (fixed), running (variable), maintenance (semi-variable) charges (SM 12.10).

- No information on semi-variable costs: show fixed and variable heads only.
- Depreciation: time basis fixed, usage basis variable. Interest: separate item of cost of sales, fixed unless otherwise given.
- Cost per unit = total cost of the period ÷ service units of the period (meal, passenger-km, patient-day, project).
- Commission, tax and profit on takings: takings = cost ÷ (1 − c − p).
- Training batches: fixed cost per batch, then per trainee; for a minimum fill, divide fixed cost by (fee × (1 − margin) − variable cost per trainee). A cost shared by several batches (one studio) falls per trainee as batches are added.
- A lunch not eaten at one o'clock is not inventory.

CHAPTER 3

Material Cost

3.1 Procurement procedure, GeM, ERP and stock verification

Key idea

No material moves without the right document from the right department: bill of materials, material requisition, purchase requisition, purchase order, goods received note, then the bill passed against them. Someone independent checks the records against the shelves, without prior notice.

FORMATS AND FORMULAS

Asked as "distinguish" questions. *Reproduce the points, in this order, from memory.*

Bill of materials vs material requisition note (SM 2.8)

Bill of materials	Material requisition note
Prepared by the engineering or planning department.	Prepared by the production or other consuming department.
A complete schedule of component parts and raw materials required for a particular job or work order.	A document asking the store keeper to issue materials to the consuming department.
Often serves as a material requisition: it can replace material requisitions.	Cannot replace a bill of materials.
Can be used for the purpose of quotations.	Useful in arriving at historical cost only.
Helps in keeping a quantitative control on materials drawn through material requisitions.	Shows the material actually drawn from stores.

Bin card vs stores ledger (SM 2.21)

Bin card	Stores ledger
Maintained by the store keeper in the store.	Maintained in the cost accounting department.
Contains only quantitative details of material received, issued and returned to stores.	Contains information both in quantity and value.
Entries are made when the transaction takes place.	Always posted after the transaction.
Each transaction is individually posted.	Transactions may be summarised and then posted.
Inter-department transfers do not appear.	Material transfers from one job to another are recorded for costing purposes.

Document

Purchase requisition

Usually prepared by

Store keeper (regular); departmental head (special)

Copies

—

Document

Purchase order

Usually prepared by

Purchase manager or concerned officer

Copies

Store or indenting dept, receiving dept, cost accounting dept; one on file with the requisitions

Document

Goods received note

Usually prepared by

Receiving department

Copies

Quadruplicate: purchase, store or indenting, receiving, accounting

Document

Material transfer note

Usually prepared by

Not named by ICAI; made at the time of transfer

Copies

Duplicate: cost accounting dept, department making the transfer

Document

Shop credit note

Usually prepared by

Department returning the surplus

Copies

Triplicate: store room, cost accounting dept, department returning it

Definition

Stores ledger

A ledger containing a separate account for each item of material and component stocked in store giving details of the receipts, issues and balance both in terms of quantity and value. A subsidiary ledger to the main cost ledger, posted from goods received notes, material requisition notes etc. (SM 2.20, 2.73)

Formula

Consumption of materials in the financial accounts

$$\text{Consumption} = \text{Opening stock} + \text{Purchases} - \text{Closing stock}$$

Stock is taken at cost or net realisable value, whichever is less (SM 2.72).

ANSWER FRAME

"Distinguish" and "uses" questions

For "distinguish", use the two tables in *Formats and formulas*: five points each, starting with who prepares or

maintains it. For "uses of the bill of materials" (SM 2.7), open with the definition, then:

Department	Use
Marketing (purchase)	Materials are procured on the basis of the specifications mentioned in it.
Production	Production is planned according to the nature and volume of materials required; material requisition lists are prepared accordingly.
Stores	A reference document while issuing materials to the requisitioning department.
Cost / accounting	Used to estimate cost and profit; purchase, issue and usage are compared and verified against it.

"Write a note on continuous stock verification"

1. Definition: physical verification at appropriate intervals, without prior notice, by staff independent of the store and production; the element of surprise is essential.
2. The shortcoming of annual stock-taking: shutdown, or a count of limited character.
3. The five advantages, in ICAI's order (Example 2).

"State with reasons whether true or false"

Verdict first, then the provision or definition that decides it, then the correct position (Example 3). Five statements,

one mark each, as in November 2023 (old scheme) Q6(b).

Three answers to "Material requisition note is prepared by the store keeper. True or false?"

Loses marks

"False."

The reason carries the mark.

Passes

"False. It is prepared by the production department."

Correct, but says nothing of what the document is.

Full marks

"False. The material requisition note is a voucher of authority used to get materials issued from store; it is prepared by the production or other consuming department. The store keeper prepares the purchase requisition for regular materials."

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the store keeper who wrote his own permission slip**
   Fix: the consuming department prepares the material requisition; the store keeper issues against it.

2. **The return that got the wrong paper**
   Fix: to the supplier, a material returned note, with a books adjustment only if the goods were already in the receiving report; to the store, a shop credit note (stores debit note) in triplicate, made out by the returning department. ICAI's summary calls the note for returns to the store a "material return note", so always name the direction.

3. **The surprise check that was announced**
   Fix: continuous verification is by independent staff, at intervals, without notice.

RECAP

- Material control: systematic control over procurement, storage and usage to keep an even flow of material.
- A supplier's invoice is passed after checking quantity against the goods received note and price against the purchase order.

- Bin card: quantity only, kept by the store keeper. Stores ledger: quantity and value, kept by the cost department.
- Documents in order: bill of materials (planning) → material requisition (production) → purchase requisition (store keeper, or departmental head for special items) → RFP / NIT → comparative statement → purchase order → goods received note or material returned note → bill passed.
- GeM: e-market for government buyers; e-bidding, reverse e-auction, demand aggregation; mandatory for government users; L1 bidder selected.
- Copies: GRN quadruplicate; material transfer note duplicate (no store copy); shop credit note triplicate.
- ERP: one entry for all departments; e-copies instead of multiple copies.
- Continuous stock verification: independent staff, intervals, no prior notice; annual counts mean shutdown or a limited count. (Cardamom does not adjust.)

3.2 Valuing material receipts

Key idea

Material cost = purchase price net of trade discounts, rebates, subsidies and recoverable taxes + the costs of bringing it to its present location and condition. Cash discount, demurrage, fines and penalties stay out. A normal shortage raises the cost of each good unit; an abnormal one goes to the Costing Profit and Loss Account.

FORMATS AND FORMULAS

ICAI's table of treatments (SM 2.13–2.15)

Learn the groups and the order: "state the treatment of" questions use exactly these words.

Item	Treatment
Discounts and subsidy	
(i) Trade discount	Deducted from the purchase price if not shown as deduction in the invoice.
(ii) Quantity discount	Like trade discount: deducted if not shown as deduction.
(iii) Cash discount	Not deducted; treated as interest and finance item. Ignored.

Item	Treatment
(iv) Subsidy / grant / incentives	Deducted from the cost of purchase.
Duties and taxes	
(v) Road tax / toll tax	Included in the cost of purchase if paid by the buyer.
(vi) GST	Excluded if credit is available; unless mentioned specifically, not part of cost.
(vii) Customs duty	Added to the purchase cost.
Penalty and charges	
(viii) Demurrage	Penalty imposed by the transporter for delay in loading or unloading: abnormal cost, not included.
(ix) Detention charges / fine	Imposed by a statutory authority for non-compliance: abnormal cost, not included.
(x) Penalty	Of any type, not included.
Other expenditures	
(xi) Insurance charges	Paid for protecting goods in transit: added.

Item	Treatment
(xii) Commission or brokerage	Added.
(xiii) Freight inwards	Added: directly attributable to procurement.
(xiv) Cost of containers	Non-returnable: added. Returnable and refunded: not considered. Refund less than amount paid: only the shortfall added.
(xv) Shortage	Normal: absorbed by good units. Abnormal: not absorbed, debited to Costing P&L.

The statement ICAI's answers use

Reproduce this layout: units in one column, rupees in the other.

Particulars	Units	Amount (₹)
Listed price of materials	xxx	xxx
Less: Trade discount		(xxx)
Add: GST (only if credit is not available)		xxx
Add: Toll tax, freight and insurance, commission		xxx
Add: Cost of returnable containers (deposit less refund)		xxx
Add: Other expenses		xxx
Total cost of material	**xxx**	**xxx**
Less: Normal shortage	(xxx)	—
Total cost of good units	**xxx**	**xxx**
Cost per unit		**xxx**

Notes below the statement explain each item left out (cash discount, demurrage, GST with credit).

∑ **Formula**

Cost per unit of material received

$$\text{Rate} = \frac{\text{Total cost of material}}{\text{Good units}}$$

Good units = units purchased − normal shortage. An abnormal shortage is valued at this rate and charged to the Costing Profit and Loss Account.

∑ **Formula**

"Other expenses at x% of total cost"

$$\text{OE} = C \times \frac{x}{100 - x}$$

OE = other expenses; C = the cost before them. The total cost includes the other expenses themselves, so 2% of total cost is 2/98 of the cost before them (ICAI's convention in MTP May 2024 and RTP September 2024).

ANSWER FRAME

"State the treatment of the following items

associated with the purchase of materials"

One line per item, ICAI's words, one mark each:

Item	Full-marks line
Cash discount	Not deducted from the purchase price; treated as an interest and finance item and ignored.
GST / IGST / CGST	Excluded from the cost of purchase if credit for the same is available; unless mentioned specifically, not part of cost.
Demurrage	A penalty imposed by the transporter for delay in loading or unloading; an abnormal cost, not included.
Basic customs duty	Paid on import of goods; added to the purchase cost.
Returnable containers	Not considered if refunded; if the refund is less than the amount paid, only the shortfall is added.
Shortage	Normal: absorbed by the good units. Abnormal: not absorbed; debited to the Costing Profit and Loss Account.

The numerical: three answers

Loses marks

"Cost per tin = ₹8,26,000 ÷ 25,000 = ₹33.04."

The supplier's bill (price after trade discount plus GST) divided by tins bought: GST in, the costs of the path out, the shortage ignored.

Passes

The right items, but the normal shortage taken out of cost as well as units: (₹7,44,000 − 1,000 × ₹29.76) ÷ 24,000 = ₹29.76.

Close, but the shortage is in the wrong column.

Full marks

ICAI's statement (units and rupees), commission on the listed price, the shortage reducing units only (₹31.00), and a numbered note for each item left out.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the cash discount that went shopping**
   Fix: ignore it in material cost, whether or not it was received.

2. **The demurrage that sneaked into stock**
   Fix: a penalty for delay or non-compliance is an abnormal cost; it never enters the cost of material.

3. **The shortage that was taken out twice**
   Fix: keep the total cost and divide by the good units; the shortage's cost is spread over them.

RECAP

- Material cost: purchase price net of trade discounts, rebates and recoverable taxes, plus costs of bringing it to its present location and condition.
- Added: freight inwards, transit insurance, customs duty, toll tax paid by buyer, commission, non-returnable containers.
- Kept out: cash discount (finance item), demurrage, detention charges and fines, penalties.
- Deducted: trade and quantity discounts (if not already on the invoice), subsidies, grants and incentives.

- GST: out if credit is available (and when the question is silent); in for a buyer who cannot take credit.
- Returnable containers: only the shortfall of the refund is added.
- Rate = total cost ÷ (units − normal shortage); abnormal shortage at that rate to Costing P&L.
- "x% of total cost" = cost before it × x/(100 − x). (The forklift pays for the wedding.)

Interactive version: PassBook app, lesson Valuing material receipts

3.3 Pricing material issues: FIFO, LIFO and average methods

Key idea

A pricing method decides which purchase price each issue carries; the total of issues and closing stock is always the same. In rising prices, FIFO gives the lowest cost of issues and the highest closing stock, LIFO the reverse, and the weighted average falls in between.

FORMATS AND FORMULAS

The stores ledger

ICAI's layout: Date | Receipts (GRN or return note no., Qty, Rate, Amount) | Issues (requisition no., Qty, Rate, Amount) | Balance (Qty, Rate, Amount). *Rule these columns from memory; in this lesson each group is written as "qty @ rate = amount" so it reads on a phone.*

Date

Receipts

GRN no., qty @ rate = amount

Issues

MR no., qty @ rate = amount

Balance

qty @ rate = amount

∑ **Formula**

Weighted average price

$$\text{Rate} = \frac{\text{Total cost in stock}}{\text{Total units in stock}}$$

Recompute after every receipt (SM 2.61, 2.75).

∑ **Formula**

Simple average price

$$\text{Rate} = \frac{\text{Sum of the lot prices}}{\text{Number of lots}}$$

ICAI: total of unit price of each purchase ÷ total number of purchases. Lots already fully issued are left out of the average (SM 2.59, 2.75).

FIFO, LIFO and weighted average when prices are rising

Point	FIFO	LIFO	Weighted average
Cost of material issued	Lowest (old prices)	Highest (latest prices)	In between
Closing stock	Highest, near current price	Lowest, at oldest prices	In between
Profit shown	Highest	Lowest	In between

In falling prices all three rows swap between FIFO and LIFO.

ANSWER FRAME

"Which method would you prefer, and why?" (as in TYK 9, SM 2.96)

1. Compare the cost charged to each job and the closing stock under each method (a small table).
2. Rising prices: FIFO charges least and leaves the highest closing stock (higher profit); LIFO the reverse (lower profit).
3. The weighted average evens out fluctuations: issues and stock show greater uniformity.
4. Conclude with the method you prefer and why: ICAI's own answer prefers the weighted average for the reason in point 3. Add that LIFO is not permitted under AS 2 and Ind AS 2.

"When and why would you recommend LIFO?" (Illustration 12, SM 2.55)

When prices are rising: issues are close to current market price, so the cost of production matches current sales revenue; profit is not unduly high; in falling prices the products appear competitive; and in inflation it shows the correct profit and avoids paying undue taxes to some extent. Add that it is not permitted under AS 2 and Ind AS 2.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the average that never moved**
   Fix: a new rate after every receipt; issues carry the rate standing on their date.

2. **The return priced at today's rate**
   Fix: returns to stores at the original issue price; returns to the vendor at the price of that lot, as ICAI's solutions do.

3. **The transfer that was booked twice**
   Fix: no entry in the stores ledger; adjust the two jobs in the cost accounts.

RECAP

- A pricing method decides which purchase price each issue carries; issues + closing stock = opening stock + receipts under every method.
- FIFO: oldest prices to issues, closing stock at latest prices; suits falling prices.
- LIFO: latest prices to issues, closing stock at oldest prices; suits rising prices; not permitted under AS 2 and Ind AS 2.
- Weighted average = total cost in stock ÷ total quantity, recomputed after every receipt; smooths fluctuations.
- Simple average = total of lot prices ÷ number of lots, ignoring quantities and fully issued lots.

- Rising prices: issues FIFO < weighted average < LIFO; closing stock FIFO > weighted average > LIFO.
- Returns to stores at the original issue price; returns to vendor in the issue column; transfers between jobs: no stores ledger entry.
- Shortages: abnormal in the issue column at the current rate, to Costing P&L; normal absorbed (quantity cut, value kept). (Sugar is sugar; price tags are not.)

Interactive version: PassBook app, lesson Pricing material issues: FIFO, LIFO and average methods

3.4 Stock levels: re-order, minimum, maximum, average and danger levels

Key idea

Order when stock falls to what the maximum usage could need over the maximum lead time (re-order level). The other levels follow from it: minimum = re-order level less normal usage during normal lead time; maximum = re-order level plus re-order quantity less the least usage during the shortest lead time.

FORMATS AND FORMULAS

Write each formula with its words first, then the figures, as ICAI's answers do. Below, usage = consumption and lead time = re-order period, as ICAI uses both pairs of words.

Formula

Re-order level (ROL)

$$\text{ROL} = \text{Max. usage} \times \text{Max. lead time}$$

Alternatively ROL = minimum level + (average consumption × average re-order period) (SM 2.22–2.23).

∑ **Formula**

Minimum stock level

$$\text{Minimum} = \text{ROL} - (\text{Avg. usage} \times \text{Avg. lead time})$$

∑ **Formula**

Maximum stock level

$$\text{Maximum} = \text{ROL} + \text{ROQ} - (\text{Min. usage} \times \text{Min. lead time})$$

Here the re-order quantity may be the EOQ (SM 2.27).

∑ **Formula**

Average stock level

$$\text{Average} = \text{Minimum} + \tfrac{1}{2}\,\text{ROQ} \qquad \text{or} \qquad \frac{\text{Maximum} + \text{Minimum}}{2}$$

Σ **Formula**

Danger level

$$\text{Danger} = \text{Avg. usage} \quad \times \text{Emergency lead time}$$

"Some time minimum consumption is also used" (SM 2.27).

Σ **Formula**

Economic order quantity (when ROQ is not given)

$$\text{EOQ} = \sqrt{\frac{2AO}{C}}$$

A = annual requirement, O = cost per order, C = carrying cost per unit per annum (SM 2.24). Details: EOQ and quantity discounts.

Definition

Re-order stock level

The level which lies between minimum and maximum levels in such a way that before the material ordered is received into the stores, there is sufficient quantity in hand to cover both normal and abnormal consumption situations; the level at which fresh order should be placed for replenishment of stock. (SM 2.22)

Definition

Minimum, maximum and danger levels (SM 2.27)

Minimum: the lowest level of material stock which must be maintained in hand at all times, so that there is no stoppage of production due to non-availability of inventory. **Maximum**: the highest level of quantity for any material which can be held in stock at any time; any quantity beyond it causes extra expenditure on funds, storage and obsolescence. **Danger**: the level at which normal issues of the raw material inventory are stopped and emergency issues only are made. **Buffer stock**: stock kept for contingency, used in case of sudden order.

ANSWER FRAME

"Distinguish between re-order level and re-order

quantity" (SM 2.22–2.23)

Re-order level	Re-order quantity
The stock level at which a fresh order is placed.	The quantity for which the purchase requisition is made.
Answers "when to order".	Answers "how much to order".
Maximum consumption × maximum re-order period.	Set considering the minimum level, re-order level, delivery time and cost; ideally the EOQ, where ordering and carrying costs together are minimum.
Lies between the minimum and maximum levels.	Added to the re-order level in the maximum-level formula.

"Explain the methods of inventory control by setting quantitative levels"

For each level: what it answers (the table in *How it works*), ICAI's definition, the formula. Five levels for five marks; end with a one-line link: all are set so that production is never interrupted (minimum, danger) and funds are not locked up (maximum).

Short notes: danger level; maximum and minimum levels

Definition in ICAI's words, formula, and the action: at the danger level normal issues stop and only emergency

issues are made; near the maximum level the store keeper stops further purchasing; at the re-order level the store keeper initiates the purchase requisition (SM 2.18).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the re-order level built on a good day**
   Fix: ROL = maximum consumption × maximum re-order period (or minimum level + average × average).

2. **The maximum level with the wrong subtraction**
   Fix: the maximum level subtracts the **minimum** consumption over the **minimum** lead time: the least that can be used before the order arrives.

3. **The danger level on the normal lead time**
   Fix: use the lead time for **emergency** purchases.

RECAP

- Re-order level = maximum consumption × maximum re-order period (or minimum level + average consumption × average period).
- Minimum level = re-order level − average consumption × average re-order period.
- Maximum level = re-order level + re-order quantity − minimum consumption × minimum re-order period.

- Average level = minimum + ½ ROQ, or (maximum + minimum) ÷ 2. Danger level = average consumption × emergency lead time.
- Re-order quantity not given: use the EOQ, √(2AO/C).
- Missing minimum consumption: 2 × average − maximum. Same time units throughout.
- Perishable material: maximum level is the lower of the formula and shelf-life consumption.
- Safety stock: optimum where expected stock-out cost + carrying cost is least; ROL = safety stock + lead time consumption. (Kesar's lakh packs: no.)

Interactive version: PassBook app, lesson Stock levels: re-order, minimum, maximum, average and danger levels

3.5 Economic Order Quantity and quantity discounts

Key idea

EOQ is the order size at which annual ordering cost plus annual carrying cost is lowest; in the basic model the two are equal there. A quantity discount is worth taking only if the total cost, purchase price included, is lower at the discount lot than at the EOQ.

FORMATS AND FORMULAS

Reproduce the formula, the four assumptions and the comparison statement from memory.

Formula

Economic Order Quantity

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}}$$

A = annual requirement (the demand for raw material or input for a year); O = cost per order (the cost of placing an order for purchase); C = carrying cost per unit per annum (the cost of carrying average inventory on an annual basis) (SM 2.24).

∑ **Formula**

Carrying cost per unit per annum

$$C = r \times P + F$$

r = sum of the carrying cost rates per annum; P = price per unit; F = any fixed carrying cost per unit per annum. Storage 2%, interest 12% and obsolescence 6% on a ₹20 unit give C = 20% × ₹20 = ₹4 (SM 2.26). A storage cost of ₹2.5 a kg a month is ₹30 a year.

∑ **Formula**

Orders, interval and annual costs at order size Q

$$\text{Number of orders} = \frac{A}{Q}$$

$$\text{Order interval} = \frac{\text{Days in the year}}{\text{Number of orders}}$$

$$\text{Ordering cost} = \frac{A}{Q} \times O$$

$$\text{Carrying cost} = \frac{Q}{2} \times C$$

$$\text{Total cost} = A \times \text{price} \quad + \frac{A}{Q} \times O + \frac{Q}{2} \times C$$

Q ÷ 2 is the average inventory.

Definition

Economic Order Quantity (EOQ)

The size of an order for which total of ordering and carrying cost are minimum. (SM 2.23)

Definition

Ordering cost

Costs associated with the purchase or order of materials, such as cost to invite quotations, documentation works like preparation of purchase orders, employee cost directly attributable to the procurement of material, transportation and inspection cost. (SM 2.23)

Definition

Carrying cost

Costs for holding or carrying inventories in store, such as the cost of fund invested in inventories, cost of storage, insurance cost and obsolescence. (SM 2.23)

Assumptions underlying EOQ, in ICAI's order (SM 2.24):

1. Ordering cost per order and carrying cost per unit per annum are known and they are fixed.
2. Anticipated usage of material in units is known.
3. Cost per unit of the material is constant and is known as well.

4. The quantity of material ordered is received immediately, i.e. the lead time is zero.

The comparison statement for a quantity discount

ICAI prints it as two statements, "(a) When EOQ is ordered" and "(b) When quantity discount is accepted", headed **Evaluation of Profitability of Different Options of Order Quantity**; side by side it reads:

Particulars	(a) At EOQ (₹)	(b) At discount lot (₹)
Purchase cost (A × price)	xxx	xxx
Ordering cost (A ÷ Q × O)	xxx	xxx
Carrying cost (Q ÷ 2 × C)	xxx	xxx
Total cost	**xxx**	**xxx**

End with "Advise – ...". With a schedule of price breaks, ICAI uses one row per order size (SM 2.90); on paper write the columns: annual requirement, order size, number of orders, cost of inventory, ordering cost, carrying cost, total cost.

Note

Rounding the number of orders

If A ÷ Q is not a whole number, the Study Material and recent ICAI answers round the number of orders up ("number of orders cannot be in decimals", SM 2.90; "22.60 Orders or 23 Orders", RTP September 2026 Q5; "rounded off to next whole number", MTP September 2026 Series I). When a question is silent, round up and say so in one line.

ANSWER FRAME

"Explain EOQ" or "Write a short note on EOQ"

1. Re-order quantity: the quantity for which the purchase requisition is made; it should be where the total of carrying cost and ordering cost is at minimum.
2. Definition: the size of an order for which the total of ordering and carrying cost is minimum.
3. Ordering cost, with ICAI's examples; carrying cost, with ICAI's examples.
4. Formula with A, O and C named.
5. The four assumptions, in order.
6. A sketch of the two cost curves and their total, lowest at the EOQ.

The numerical parts: three answers to the same price-ladder question

Loses marks

"Order 2,400 quintals: it has the lowest price."

The price alone; ordering, carrying and rent ignored.

Passes

EOQ computed and the totals compared at each price break, but the godown rent left out: "2,400 quintals at ₹4,75,17,900".

Method shown, one cost missing: the wrong advice.

Full marks

Data line (A, O, C with its build-up), formula, EOQ and its slab; a statement at the EOQ and every price break with purchase, ordering, carrying (at each slab's price) and rent; totals; "The most economical purchase level is 1,200 quintals", with the saving in rupees.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the carrying cost that forgot the price**
   Fix: C is rupees per unit per annum: 12% × ₹8 = ₹0.96, plus ₹0.24 = ₹1.20. Add every rate first, then multiply by the price, then add any fixed ₹ per unit.

2. **The monthly figure that stayed monthly**
   Fix: A and C must both be per annum: 60,000 × 12, and 1% × 12.

3. **Carrying cost on the whole order**
   Fix: stock runs down from Q to nil, so the average inventory is Q ÷ 2 (plus nothing for a safety stock in ICAI's comparisons).

4. **The discount compared without the price**
   Fix: a discount changes the purchase cost, so the comparison must include A × price under each option.

RECAP

- EOQ = √(2AO ÷ C): A annual requirement, O cost per order, C carrying cost per unit per annum.
- EOQ: the size of an order for which the total of ordering and carrying cost is minimum; at it, the two are equal.

- Ordering cost: quotations, purchase orders, procurement staff, transportation, inspection. Carrying cost: cost of funds, storage, insurance, obsolescence.
- C = (sum of annual rates × price) + fixed ₹ per unit; annualise monthly and quarterly rates first.
- Orders = A ÷ Q; ordering cost = A ÷ Q × O; carrying cost = Q ÷ 2 × C; total = A × price + both.
- Assumptions: costs known and fixed; usage known; price constant; lead time zero.
- Quantity discount: compare total cost including purchase cost, carrying cost at the discounted price; with price breaks (EOQ in the base band), test EOQ and each band's lowest quantity.
- Fractional orders: round up, and say so. (Two mandi trips a week, not one a month.)

Interactive version: PassBook app, lesson Economic Order Quantity and quantity discounts

3.6 ABC, FSN, HML, VED, JIT and perpetual inventory

Key idea

Each technique sorts stock on its own basis so that control effort goes where it matters: ABC by value, FSN by movement, VED by criticality, HML by unit price. JIT removes stock by letting demand pull material in. Perpetual inventory with continuous stock verification keeps the records true.

FORMATS AND FORMULAS

Reproduce both statements from memory: the column headings carry marks.

Statement of total cost and ranking

On paper ICAI's statement has seven columns: Item, Units, % of total units, Unit cost (₹), Total cost (₹), % of total cost, Ranking (SM 2.37). Here it is in two halves:

Item	Units	% of total units	Unit cost (₹)
1	xxx	xx.xx	xx
...	...	...	...
Total	**xxx**	**100**	

Item	Total cost (₹)	% of total cost	Ranking
1	xxx	xx.xx	x
...	...	...	...
Total	**xxx**	**100**	

Plan of ABC selective control

State the basis first ("₹ 50,000 & above: 'A' items ..."), then:

Ranking: item	% of total units	Cost (₹)	% of total cost
1: item ...	xx.xx	xxx	xx.xx
2: item ...	xx.xx	xxx	xx.xx
Total A (n items)	**xx.xx**	**xxx**	**xx.xx**
... B items, then **Total B**			
... C items, then **Total C**			
Grand total	**100**	**xxx**	**100**

ICAI writes the category in a last column next to each total line; here the total line names it.

"Explain the concept of ABC analysis as a technique of inventory control"

1. Basis: discriminating control over different items of inventory on the basis of the investment involved; categories by relative importance, their value and frequency of replenishment.
2. A: about 10% of items, about 70% of value; controlled by budgets and maximum, minimum and re-order levels.
3. B: about 20% of items, about 20% of value; orders after periodic review.
4. C: about 70% of items, about 10% of value; no constant control; orders once in six months or a year, to economise on ordering and handling costs.
5. The four advantages: continuity in production, lower cost, less attention required, systematic working.
6. A one-line example or a sketch of items against value.

"Which system is known as demand pull? Explain, and state its two principles" (5 marks)

1. Name it: Just in Time (JIT) inventory management, the 'Demand pull' or 'Pull through' system.
2. ICAI's definition: zero inventories in stores; material purchased only when actually required for production.
3. Working: production starts after the order for the products is received; the material requirement then

goes to the purchase department.

4. The two principles, numbered, in ICAI's words.
5. The five-box chain from demand to supplier.

Distinguish ABC and HML (or VED)

Point	Basis
ABC	Investment involved: the overall value of the item's inventory, and frequency of replenishment
HML	Cost of an individual item, by a range of cost
VED	Criticality for the production function and the final product; generally for spare parts

Point	Categories
ABC	A, B, C
HML	High, medium, low cost
VED	Vital, essential, desirable

Point

Priority and control

ABC

A: budgets and maximum, minimum and re-order levels; B: periodic review; C: orders once in six months or a year

HML

High-cost items get more priority; medium and low cost, less

VED

Vital: strictly controlled by a re-order level; essential: periodic review, second priority

Three answers to "Write a short note on VED analysis"

Loses marks

"VED classifies stock on cost: very expensive, expensive, desirable."

The basis is wrong: the very statement ICAI set as a true-or-false trap (November 2023).

Passes

"Vital, Essential and Desirable: items classified by their importance for production."

Basis right; no meaning of each class, no control, no field of use.

Full marks

The basis (criticality for the production function and the final product), generally used for spare parts; each class with its effect on production and its control; the hospital example.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: ABC ranked by the number of units**
   Fix: rank by total cost (units × unit cost). The % of units column is shown to prove the point that A items are few, not to rank them.

2. **ABC and HML swapped**
   Fix: ABC uses the item's total inventory value; a high unit price makes it H under HML. In the example the kits are H under HML but C under ABC: 50 × ₹1,600 = ₹80,000, below the ₹1,00,000 limit.

3. **VED on cost**
   Fix: VED is on criticality for production; a cheap seal without which the freezer stops is vital, an expensive lobby panel is desirable.

4. **Perpetual inventory confused with annual stock-taking**
   Fix: perpetual inventory is the continuously updated records (bin cards and stores ledger), checked by continuous stock verification of a few items daily, by surprise; annual stock-taking counts everything once and is what it replaces.

RECAP

- Four bases of inventory control: quantitative levels, relative classification, ratio analysis, physical control.
- ABC: by value and frequency; A about 10% of items and 70% of value, B about 20% and 20%, C about 70% and 10%; rank by total cost.
- FSN: by frequency of usage (inventory turnover); non-moving kept for disposal, with a provision for loss.
- VED: by criticality for production, mainly spare parts; vital items controlled by a re-order level.
- HML: by the cost of an individual item, unlike ABC's overall value.
- JIT: zero inventory, demand pull; two principles: produce only when required, deliver only when customers want.
- Perpetual inventory: bin cards and stores ledger, kept up to date and checked by continuous, surprise stock verification by rotation.
- Two bin: order when issues reach the smaller part. (Watch the rent, not the matchboxes.)

3.7 Inventory turnover ratios and stock discrepancies

Key idea

The turnover ratio counts how many times a year the average stock is used up: a high ratio means fast-moving material, a low one over-investment and working capital locked up. A shortage found on verification is priced like an issue; a normal one stays in product cost, an abnormal one goes to the Costing Profit and Loss Account.

FORMATS AND FORMULAS

Reproduce the statement and the formulas from memory.

Formula

Inventory turnover ratio

$$\text{ITR} = \frac{\text{Cost of materials consumed}}{\text{Cost of average stock held}}$$

ITR = inventory turnover ratio; both terms for the same period. Average stock = ½ (opening stock + closing stock) (SM 2.43).

∑ **Formula**

Average number of days of inventory holding

$$\text{Days held} = \frac{365}{\text{ITR}}$$

Average number of days of inventory holding = 365 days (or 12 months) ÷ inventory turnover ratio (SM 2.43); 360 if the question fixes it.

∑ **Formula**

Materials consumed

$$\text{Consumption} = O + P - C$$

O = opening stock, P = purchases, C = closing stock of materials (SM 2.72).

∑ **Formula**

Input-output ratio

$$\text{Input-output ratio} = \frac{I}{S}$$

I = quantity of material input to production; S = standard material content of the actual output (SM 2.42). The Study Material sets no threshold, but the comparison reads directly: above 1, more material went in than the standard allows (adverse usage); below 1, less (favourable).

Statement of inventory turnover (ICAI's layout, SM 2.44)

Particulars	Material A (₹)	Material B (₹)
Opening stock	xxx	xxx
Add: Purchases	xxx	xxx
	xxx	xxx
Less: Closing stock	xxx	xxx
Materials consumed	**xxx**	**xxx**
Average inventory: (opening stock + closing stock) ÷ 2	xxx	xxx
Inventory turnover ratio (consumption ÷ average inventory)	x times	x times
Inventory turnover (number of days in a year ÷ IT ratio)	x days	x days

End with "Comments: Material A is moving faster than Material B" (SM 2.44).

A shortage in the cost ledger

Shortage	Debit	Credit
Normal	Factory Overhead Control A/c	Stores Ledger Control A/c
Abnormal	Costing Profit and Loss A/c	Stores Ledger Control A/c

The Study Material's cost-ledger illustration credits a shortage, "assumed normal", to stores ledger control and debits overhead control (SM 7.37).

ANSWER FRAME

"Explain inventory control by ratio analysis"

1. Input-output ratio: input of material ÷ standard material content of actual output; compares actual and standard consumption; shows favourable or adverse usage.
2. Inventory turnover ratio: cost of materials consumed ÷ cost of average stock; average stock = ½ (opening + closing).
3. Days of inventory holding = 365 days (or 12 months) ÷ the ratio.
4. Interpretation: high ratio, fast moving; low ratio, over-investment and locking up of working capital.
5. Use: compare materials (and with standards), reduce capital locked up and prevent over-stocking of slow-moving items.

"How are shortages found on physical verification treated?"

1. Valuation: entered in the issue column at the rate of the issue method adopted (FIFO or any other).
2. Normal shortage (breaking of bulk, evaporation, unavoidable): absorbed by good units, or charged to factory overheads.
3. Abnormal shortage (mishandling, pilferage, avoidable): transferred to Costing Profit and Loss Account.
4. Control: explain the causes promptly, make corrective entries, remove the causes.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the ratio built on purchases**
   Fix: the numerator is materials consumed: opening + purchases − closing.

2. **Closing stock in place of average stock**
   Fix: average stock = ½ (opening + closing), unless the question gives the average.

3. **The whole difference written off**
   Fix: first correct the records (the 150 litres of requisition 318 belong to the snacks line); of the real shortage only the abnormal 30 litres go to Costing Profit and Loss, while the normal 40 stay in product cost, absorbed by the good units or through overheads.

4. **The shortage at the latest purchase price**
   Fix: a shortage is an issue: use the rate of the issue method, here the weighted average ₹156.00.

RECAP

- Materials consumed = opening stock + purchases − closing stock.
- Inventory turnover ratio = cost of materials consumed ÷ cost of average stock; average stock = ½ (opening + closing).
- Days of inventory holding = 365 days (or 12 months, or the 360 a question fixes) ÷ ratio.
- High ratio: fast moving. Low ratio: over-investment, working capital locked up.
- Input-output ratio = material input ÷ standard material content of actual output: above 1 adverse, below 1 favourable.
- Shortage on verification: an issue at the issue-method rate; normal (bulk breaking, evaporation,

unavoidable) stays in product cost, abnormal (pilferage, mishandling, avoidable) to Costing Profit and Loss.

- Cost ledger: normal, Factory Overhead Control Dr; abnormal, Costing P&L Dr; Stores Ledger Control Cr. (Count the table turns, not the full tables.)

3.8 Waste, scrap, spoilage, defectives and obsolescence

Key idea

Normal losses are inherent and are borne by the good units, net of any realisable value; abnormal losses are avoidable and are charged to the Costing Profit and Loss Account after crediting any realisable value. Obsolete material is a total loss, abnormal by definition, and never part of the cost of manufacture.

FORMATS AND FORMULAS

Reproduce the cost-per-good-unit formula and the spoilage entry from memory.

Formula

Cost of a good unit when there is a normal loss

$$\text{Cost per good unit} = \frac{T - R}{I - N}$$

T = total cost; R = realisable value of the normal loss (scrap value less cost to sell); I = input units; N = normal loss units. An abnormal loss unit is valued at the same cost as a good unit (SM 10.7–10.9).

∑ **Formula**

Splitting a spoilage between normal and abnormal

$$\text{Abnormal} = \text{Actual} - \text{Normal spoilage}$$

Any realisable value of the spoiled work is apportioned between the two in the ratio of their quantities (SM 9.10).

Costing journal for spoilage in a job (SM 9.10):

Account	Dr	Cr
Material (Stores) Control A/c, with the realisable value	xxx	
Overhead Control A/c, with the normal spoilage less its share of realisable value	xxx	
Costing Profit and Loss A/c, with the abnormal spoilage less its share of realisable value	xxx	
To Work-in-Progress Control A/c, with the total cost of spoiled work		xxx

ANSWER FRAME

Distinguish between waste and scrap (SM 2.64, 2.66)

ICAI's suggested answer to May 2024 Q6(b) puts the two

definitions first, then the Study Material's three points:

Waste	Scrap
1. The portion of raw material which is lost during storage or production and discarded	1. The output which is discarded and disposed of without further treatment
2. Connected with raw material or inputs to the production process	2. The loss connected with the output
3. May be visible or invisible	3. Generally identifiable and has physical substance
4. Generally has no recoverable value	4. Termed as by-products and has small recoverable value

Distinguish between scrap and defectives (SM 2.67)

Scrap	Defectives
1. Loss connected with the output	1. Connected with the output as well as the input
2. Not intended, but cannot be eliminated due to the nature of material or process	2. Not intended, but can be eliminated through a proper control system
3. Generally not used or rectified	3. Can be used after rectification
4. Insignificant recoverable value	4. Sold at a lower value than the good one

Spoilage against defectives (SM 2.67): spoilage cannot be repaired or reconditioned, while defectives can be rectified and transferred back to standard production or to the seconds; the problem in accounting for defective work is the cost of rectification or rework.

Three answers to "Discuss the treatment of normal and abnormal scrap"

Loses marks

"Normal scrap is absorbed; abnormal scrap goes to P&L."

No realisable value, no scrap account, no credit to the job.

Passes

"Normal: cost borne by good units, realisable value deducted. Abnormal: written off to Costing P&L."

Normal is right; abnormal skips the scrap account at full cost.

Full marks

The definition first. Normal: cost of scrap borne by good units, income from its realisable value deducted from the cost. Abnormal: scrap account charged with full cost, job or process credited, profit or loss on realisation to Costing Profit and Loss Account.

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: Kesar's miscellaneous drawer**
   Fix: split each loss into normal (stays in product cost) and abnormal (Costing Profit and Loss Account); an abnormal loss inside product cost overstates the cost of every good unit and hides the inefficiency.

2. **Abnormal scrap valued at its sale price**
   Fix: the scrap account is charged with the **full cost** (300 kg × ₹50 = ₹15,000); production is credited with that, and only the ₹12,000 shortfall on realisation goes to Costing Profit and Loss.

3. **Obsolete stock carried at cost "until it is used"**
   Fix: obsolete material is a total loss: dispose of it at the best available price and charge the loss to Costing Profit and Loss; it is never part of the cost of manufacture.

4. **Every defect written off**
   Fix: rectification within the normal rate of defectives (or the proportion management provided for) is a normal cost charged to the batch; only the excess, or defects caused by the Inspection Department's error, are written off.

RECAP

- Five forms: waste (lost, discarded), scrap (discarded output, small value), spoilage (beyond economic repair), defectives (below standard: reworks or rejects), obsolescence (value lost to supersession).
- Normal loss: borne by good units, net of realisable value. Abnormal loss: Costing Profit and Loss Account, after realisable value.
- Scrap: normal, realisable value deducted from cost; abnormal, scrap account at full cost, job credited, loss on realisation to Costing P&L.
- Spoilage: normal, charged to the production order or production overhead; abnormal, Costing P&L; rigid specification, absorbed by good production with disposal cost to production overhead.
- Defectives: normal, cost less realisable value to material cost of good production; abnormal, Costing P&L after crediting realisable value.
- Defective work by circumstance: allowed percentage, to the batch (excess written off); bad workmanship, written off; Inspection Department's error, charged to it and written off.
- Obsolescence: always abnormal, total loss, dispose at the best price. (The tailor prices in the offcuts, never the ruined shirt.)

CHAPTER 4

Employee Cost and Direct Expenses

4.1 Attendance, payroll and the elements of wages

Key idea

Time-keeping says how long an employee was at work; time booking says what that time was spent on. The cost of an employee hour for costing is the whole employee cost (monetary and non-monetary) divided by the effective hours, not the payslip divided by the hours paid.

FORMATS AND FORMULAS

Definition

Employee (labour) cost

Benefits paid or payable to the employees of an entity, whether permanent or temporary, for the services rendered by them. Employee cost includes payments made in cash or kind. (SM 3.3)

Definition

Direct and indirect employee cost

Direct: benefits paid or payable to the employees which can be attributed to a cost object in an economically feasible manner. Indirect: benefits paid or payable to the employees which cannot be directly attributable to a particular cost object in an economically feasible manner. (SM 3.3)

Definition

Time-keeping and time booking

Time-keeping refers to recording and keeping of the employees' attendance time. Time booking is recording the details of work done and the time spent by an employee on each job or process. (SM 3.55)

Formula

Rate per hour for costing (direct workers)

$$\text{Rate per hour} = \frac{\text{All benefits}}{\text{Effective hours}}$$

ICAI: total estimated monetary benefits and cost of non-monetary benefits ÷ (budgeted direct employee hours – normal idle time) (SM 3.42). Leave on full pay also comes off the hours.

ICAI's layout for the effective hourly cost (SM 3.15, Illustration 1). *Reproduce it from memory.*

Statement showing computation of effective hourly cost	Per month (₹)	Per annum (₹)
(A) Earnings of the employee:		
Basic pay	xxx	xxx
Dearness allowance	xxx	xxx
Bonus	xxx	xxx
Employer's contribution to provident fund	xxx	xxx
Other allowances and benefits	xxx	xxx
Total	**xxx**	**xxx**
(B) Effective working hours (working note)		xxx hours
(C) Effective hourly cost (A ÷ B)		**₹ xx**

When the question asks for both the pay and the cost, ICAI keeps two statements (SM 3.19–3.20, Illustration 3):

Statement of earnings	Statement of employee cost
Basic wages	Gross wages (excluding overtime)
Add: Dearness allowance, overtime, allowances	Add: Employer's contribution to PF
Gross wages earned	Add: Employer's contribution to ESI
Less: Employee's contribution to PF	**Employee cost**
Less: Employee's contribution to ESI	÷ Normal working hours
Net wages earned (what is paid)	**Ordinary wage rate per hour**

The employee's own contributions are inside his gross wages, merely withheld; only the employer's are added to cost. The cost statement leaves overtime out because ICAI allocates the overtime wages separately, to the job on which the overtime was worked (Job Y in Illustration 3); the ordinary wages are shared over the jobs in the ratio of time booked.

The theory questions ICAI asks

Objectives of time-keeping and time booking (SM 3.7, 3.9): the six objectives of time-keeping as listed above, then the three purposes of time booking (costing, efficiency, responsibility), one line each.

Distinguish between time-keeping and time booking

Point	Time-keeping	Time booking
Records	Attendance time: arrival and departure	Time spent on each job, operation or process
Shows	Total time spent in the factory	Break-up of that time over jobs and idle time
Used for	Payroll, overtime, discipline, idle time	Job cost, efficiency, fixing responsibility
Record	Attendance register, token, punch card, bio-metric device	Time (job) card

Distinguish between direct and indirect employee cost

(SM 3.3–3.4)

Direct employee cost	Indirect employee cost
Paid to employees directly engaged in the production process	Paid to employees not directly engaged in the production process
Easily identified and allocated to the cost unit	Apportioned on some appropriate basis
Varies with the volume of production	May not vary with the volume of production

Factors for controlling employee cost (SM 3.5–3.6), in ICAI's order: (i) assessment of manpower requirements; (ii) control over time-keeping and time-booking; (iii) time and motion study; (iv) control over idle time and overtime; (v) control over employee turnover; (vi) wage and incentive systems; (vii) job evaluation and merit rating; (viii) employee productivity. Four are lessons of their own: idle time and overtime, turnover, incentive systems, labour efficiency.

A factory with proxy attendance: features of the bio-metric attendance system (May 2026 Q6(c)). Write the features in ICAI's words, then apply them to the facts:

1. Attendance is marked by recognising an employee on the basis of physical and behavioural traits.
2. Each employee's unique identity (fingerprint, face, retina image) is kept in a database and matched when attendance is marked.

3. It includes fingerprint recognition, face recognition and time and attendance tracking technology.
4. It reduces the risk of time manipulation and proxy attendance.
5. Its set-up and maintenance cost may make it unsuitable for small organisations (SM 3.8).
6. Apply it: a large factory with proxy attendance justifies the cost; payroll should also pay only the employees on roll listed by HR (SM 3.11).

The numerical: three answers to the same question

Loses marks

"₹2,31,000 ÷ 2,400 hours = ₹96.25 an hour."

Leave and idle time left in the hours; no working notes.

Passes

"Rate = ₹2,31,000 ÷ 2,100 hours = ₹110."

Right, but the ₹2,31,000 and 2,100 hours come from nowhere: no method marks if one is wrong.

Full marks

ICAI's statement, each element per month and per annum, a working note for effective hours, the rate, and any assumption stated.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the PF counted twice**
   Fix: the employee's contribution is already inside his gross wages; add only the employer's contribution.

2. **The paid holiday that worked a shift**
   Fix: leave on full pay and normal idle time come out of the hours; their cost stays in the numerator, spread over the hours actually worked.

3. **The night shift that billed the night's job**
   Fix: it is not for any particular job, so it is overheads (SM 3.42).

RECAP

- Employee cost: benefits paid or payable to employees, permanent or temporary, in cash or kind; direct if traceable to a cost object economically, else indirect (overheads).
- Aim of control: low wages per unit of output, not a low wage bill. Eight factors, from manpower assessment to employee productivity.
- Time-keeping = attendance time (six objectives); time booking = time on each job (costing, efficiency, responsibility). Job time + idle time = attendance time.

- Payroll in five steps: time and attendance, employee details, wage sheet, payment after deductions, deposit with statutory bodies. HR's list stops bogus employees.
- Wages include non-monetary benefits: medical, education and training, recreation, housing and welfare, subsidised canteen.
- Rate for costing = (monetary + non-monetary benefits) ÷ (budgeted hours − normal idle time); leave days come off the hours too.
- Add the employer's PF and ESI; never the employee's. Night shift allowance: overheads. (Measure the iceberg, not the tip.)

4.2 Idle time and overtime

Key idea

Normal idle time and the normal-rate part of overtime are costs of production; abnormal idle time is a loss for Costing Profit and Loss. The overtime premium follows the reason the overtime was worked.

FORMATS AND FORMULAS

Definition

Idle time

The time for which the employer pays but obtains no direct benefit or for no productive purpose. (SM 3.55)

Definition

Overtime premium

The rate for overtime work is higher than the normal time rate; the extra amount so paid over the normal rate is called overtime premium. (SM 3.17)

Definition

Ordinary rate of wages (Factories Act, as quoted in the SM)

The basic wages plus such allowances, including the cash equivalent of the advantage accruing through the concessional sale to workers of food grains and other articles, as the worker is for the time being entitled to, but does not include a bonus and wages for overtime work. (SM 3.17)

Formula

Idle time and overtime

$$\text{Idle time} = \text{Time paid} - \text{Time booked}$$

$$\text{OT pay} = \text{OT hours} \times \text{normal rate} + \text{premium}$$

Premium = overtime hours × (overtime rate − normal rate).

Formula

Average inflated wage rate

$$\text{Inflated rate} = \frac{\text{Total wages}}{\text{Total hours}}$$

Total wages = normal-time wages + overtime wages at the overtime rates (premium included). Used when overtime is a regular policy (SM 3.25).

ICAI's layout for the wages of a period (SM 3.16, Illustration 2). *Reproduce it from memory.*

Allocation of wages in cost accounting	Hours × rate	Amount (₹)
Allocated to Job X	hours booked × rate	xxx
Allocated to Job Y	hours booked × rate	xxx
Charged to Costing Profit and Loss A/c	abnormal idle hours × rate	xxx
Total (equals the wages paid)		**xxx**

Rate = wages of the period ÷ effective hours (paid hours less normal breaks). For overtime questions, ICAI shows each situation separately: employee cost chargeable to the job, and the premium as overhead, to the job, to another department or to Costing Profit and Loss (SM 3.25–3.26).

ANSWER FRAME

"Describe idle time and its treatment in the following situations"

Define idle time first (time paid but no production; time paid − time booked; normal or abnormal), then a table, as

ICAI's suggested answer to May 2024 Q6(c) does:

Situation	Idle time	Treatment
Setting-up time for a direct worker's machine	Normal	Part of cost of production; allowed for in standard hours or rate
Lunch break of an indirect worker	Normal	Considered in computing the overhead rate
Time lost by a machine breakdown	Abnormal	Separate item in the Costing Profit and Loss Account

"Discuss the effect of overtime payment on productivity"

1. From the SM (3.17–3.18): occasional overtime is healthy (optimum capacity, fully used); persistent overtime is a bad sign (more capacity needed, or work habitually postponed to earn overtime).
2. ICAI's suggested answer (MTP May 2025 Series I Q2(b)) adds that overtime raises the cost of production: the premium is an extra payment; efficiency during overtime may fall; workers may slacken in normal hours to earn more; lower output and higher premium raise cost.

"Discuss the accounting treatment of idle time and overtime wages"

The chapter's own Theoretical Question 1, set as MTP September 2025 Series I Q4(b) (4 marks). ICAI's suggested answer gives the normal idle time treatment and the four premium treatments; add abnormal idle time to make the answer complete. In this order:

1. **Normal idle time**: part of the cost of production; for direct workers an allowance in the standard hours or standard rate, for indirect workers considered in computing the overhead rate (SM 3.13).
2. **Abnormal idle time**: not part of production cost; a separate item in the Costing Profit and Loss Account, analysed into controllable and uncontrollable by cause (SM 3.14).
3. **Overtime premium** (SM 3.18): at the desire of the customer, charged to the job directly; for general production programmes or urgent orders, overhead cost of the department or cost centre that works overtime; due to the fault of another department, charged to that department; on account of abnormal conditions such as flood or earthquake, not charged to cost but to Costing Profit and Loss Account. The overtime hours at the normal rate are charged like ordinary wages.
4. **Overtime worked regularly as a policy**: the premium is part of employee cost, and jobs are charged at the average inflated wage rate (SM 3.25).

The numerical: three answers to reason (b) of October's overtime

Loses marks

"Overheads: 240 × ₹144 + 40 × ₹192 = ₹42,240."

The whole overtime wage sent to overheads: the basic part belongs to the jobs.

Passes

"Packing overheads ₹15,360."

Right figure, but no rates, no premium working, no reason.

Full marks

Premium per hour as a working note (₹48 and ₹96), the premium 240 × ₹48 + 40 × ₹96 = ₹15,360, the basic part charged to the jobs, and the treatment named with its reason: overtime for general production is an overhead of the department that works it.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the 75% premium paid at 75%**
   Fix: a premium is the extra; the overtime rate is 175% of basic.

2. **The whole overtime wage sent to overheads**
   Fix: the overtime hours are still the jobs' hours at the normal rate; only the premium (240 × ₹48) goes to overheads.

3. **The power cut spread over the jobs**
   Fix: divide by the effective hours (3,600), and charge the 200 abnormal hours separately to Costing Profit and Loss.

4. **Normal idle time written off**
   Fix: normal idle time cannot be avoided in the normal course of business; it is part of the cost of production.

RECAP

- Idle time = time paid − time booked: paid, nothing produced.
- Normal idle time (gate to workplace, between jobs, set-ups, rest and lunch): cost of production; direct workers through the rate, indirect workers through the overhead rate.

- Abnormal idle time (lack of coordination, power failure, breakdown, no material, strikes, lockouts, fire, flood): Costing Profit and Loss, analysed as controllable and uncontrollable.
- Overtime payment = overtime hours at normal rate + premium; usually double the normal rate.
- Premium: customer's request → job; general production or urgent orders → department overheads; another department's fault → that department; abnormal conditions → Costing P&L. Regular policy → inflated rate for all jobs.
- The SM's Factories Act quote: more than nine hours a day or 48 a week, twice the ordinary rate.
- Occasional overtime healthy; persistent overtime a bad sign. (Whoever caused the wait pays for it.)

4.3 Employee turnover: measurement, causes, effects and cost

Key idea

Turnover = movement out, in, or both, divided by the average number on roll. It costs the contribution on the productive hours lost plus the costs of leaving, hiring and training; preventive spending is worth it up to the optimum level of turnover.

FORMATS AND FORMULAS

Definition

Employee (labour) turnover

The rate of change in the composition of employee force during a specified period measured against a suitable index. (SM 3.45)

Formula

Replacement and separation methods

$$\text{Replacement} = \frac{\text{Employees replaced}}{\text{Average on roll}} \times 100$$

$$\text{Separation} = \frac{\text{Employees separated}}{\text{Average on roll}} \times 100$$

(SM 3.45–3.46)

∑ **Formula**

Flux method

$$\text{Flux} = \frac{\text{Separated} + \text{Replaced}}{\text{Average on roll}} \times 100$$

With accessions (all joining, including replacements): (separations + accessions) ÷ average on roll × 100, or (separations + replacements + new joinings) ÷ average on roll × 100 (SM 3.46).

∑ **Formula**

Average on roll and the equivalent annual rate

$$\text{Average} = \frac{\text{At beginning} + \text{At end}}{2}$$

$$\text{Annual rate} = \frac{\text{Rate for period} \times 365}{\text{Days in period}}$$

(SM 3.46–3.47)

ICAI's layout for the cost of turnover (SM 3.52, Illustration 15). *Reproduce it from memory.*

Computation of profit forgone on account of employee turnover	Amount (₹)
Contribution foregone (working note)	xxx
Settlement cost due to leaving	xxx
Recruitment cost	xxx
Selection cost	xxx
Training costs	xxx
Cost of rectification of defective units (if given)	xxx
Profit foregone	**xxx**

ANSWER FRAME

"Discuss the two types of cost associated with labour turnover"

ICAI's suggested answers (MTP May 2026 Series I Q6(b); MTP September 2025 Series II Q6(c)) reproduce SM 3.50: (a) **preventive costs**, defined, with the three items (medical benefits, welfare such as pension, other retention benefits); (b) **replacement costs**, defined, with the items (recruitment, training and induction, abnormal breakage and scrap, extra wages and overheads from the inefficiency of new workers); then the conclusion that high turnover means high replacement costs and only adequate preventive costs keep it low, so each firm works out its optimum level.

"Discuss the remedial steps to minimise labour turnover"

The Study Material says proper and timely management action can appreciably reduce turnover from avoidable causes (SM 3.49) and lists the preventive costs (SM 3.50). ICAI's suggested answers to the MTP questions (January 2026 Series II Q6(b); May 2024 Series II Q6(c)) give these steps; write them with one line each:

1. **Exit interview** with each outgoing employee, to find out why he is leaving.
2. **Job analysis and evaluation**, to know the requirements of each job.
3. A **scientific system of recruitment, placement and promotion**.
4. A **healthy atmosphere**, with education, medical and housing facilities for workers.
5. A **committee for settling workers' grievances**.

Then tie them to the avoidable causes they remove (dissatisfaction, strained relations, no training or promotion, no recreational or medical facilities, low wages).

"Causes" and "effects"

Write the three heads with ICAI's items in ICAI's order (the lists above), adding the remark on remedies under each head. For effects, the five ways high turnover raises the cost of production (SM 3.50).

The numerical: three answers to the profit-forgone question

Loses marks

"Profit forgone = settlement + recruitment + training = ₹6,20,000."

Only the visible costs: the contribution on 8,000 lost hours and the wasted materials are missing.

Passes

"Contribution lost ₹14,40,000; profit forgone ₹22,40,000."

Right, but the hours working is invisible: no method marks if a figure slips.

Full marks

Working notes for the productive hours lost (vacancies and unproductive training) and the contribution lost (or, when sales and a P/V ratio are given, sales per productive hour × P/V ratio); then ICAI's statement with each turnover cost on its own line, ending in the profit foregone.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the opening headcount as the denominator**
   Fix: every method divides by the average number on roll, (beginning + end) ÷ 2.

2. **The new café's staff counted as replacements**
   Fix: employees appointed for expansion are not replacements (SM 3.45); they enter only the flux rate with accessions.

3. **All training hours written off**
   Fix: only the unproductive half (1,600 hours) is lost; the rest produced output.

4. **A month's rate reported as the year's**
   Fix: rate × 365 ÷ days in the period (24.33% here), or a quarterly rate × 4.

RECAP

- Employee turnover: the rate of change in the composition of the workforce in a period, against an index (industry, locality, or own past rate).
- Replacement = replaced ÷ average on roll; separation = separated ÷ average; flux = (separated + replaced) ÷ average; all × 100.

- Expansion hires are not replacements; accessions = all joinings including replacements; flux with accessions = (separations + accessions) ÷ average.
- Average on roll = (beginning + end) ÷ 2; annual rate = period rate × 365 ÷ days (a quarter × 4).
- Causes: personal, unavoidable, avoidable (management can act on the avoidable ones).
- Costs: preventive (medical, welfare, retention benefits) and replacement (recruitment, training, breakage, inefficiency); find the optimum level.
- Profit forgone = (productive hours lost × sales per productive hour × P/V ratio) + settlement, recruitment, selection, training and rectification costs. (Count who got off the coach, not the new coach.)

Interactive version: PassBook app, lesson Employee turnover: measurement, causes, effects and cost

4.4 Time rate, piece rate and the Halsey and Rowan premium plans

Key idea

Premium plans guarantee time wages and share the time saved. Halsey: 50% of the time saved, at the time rate. Rowan: (time saved ÷ time allowed) × time wages. Equal when time taken is 50% of time allowed; Rowan pays more for moderate efficiency, Halsey for high efficiency.

FORMATS AND FORMULAS

SH = time allowed (standard hours), AH = time taken (actual hours), R = time rate per hour.

Formula

Time rate and piece rate

$$\text{Wages} = \text{Time worked} \times \text{Rate}$$

$$\text{Wages} = \text{Units} \times \text{Rate per unit}$$

(SM 3.28)

∑ **Formula**

Halsey premium plan

$$\text{Earnings} = AH \times R$$
$$+50\% \times (SH - AH) \times R$$

(SM 3.28)

∑ **Formula**

Rowan premium plan

$$\text{Earnings} = AH \times R \qquad + \frac{SH - AH}{SH} \times AH \times R$$

(SM 3.30)

∑ **Formula**

Effective hourly rate and cost per unit

$$\text{Effective rate} = \frac{\text{Total earnings}}{\text{Hours worked}}$$
$$\text{Cost per unit} = \frac{\text{Total earnings}}{\text{Units}}$$

∑ **Formula**

Efficiency

$$\text{Efficiency } (\%) = \frac{\text{Time allowed}}{\text{Time taken}} \times 100$$

Time allowed = time allowed as per standard for the work done (SM 3.43).

ICAI's comparative statement (RTP September 2026 Q6; the chapter's Practical Problem 2 gives the same figures scheme by scheme, SM 3.65–3.67). *Reproduce it from memory.*

Method	Hours worked	Earnings (₹)	Units	Cost per unit (₹)
Time rate	xx	xxx	xx	x.xx
Piece rate	xx	xxx	xx	x.xx
Halsey	xx	xxx	xx	x.xx
Rowan	xx	xxx	xx	x.xx

ANSWER FRAME

"Describe how wages are calculated under the Rowan and Halsey systems"

For each plan: the standard time and the guarantee; the bonus rule in ICAI's words; the formula; a two-line

example; and one advantage and one disadvantage.

"Distinguish between the Halsey and Rowan plans"

Point	Halsey	Rowan
Bonus	50% of time saved × rate	Time saved ÷ time allowed × time wages
Worker's share of the saving	Fixed at 50%	Equals AH ÷ SH: falls as the saving grows
Highest bonus	Keeps rising with time saved	At time taken = 50% of time allowed
Earnings can double?	Yes: hourly earnings reach twice the rate once time taken is a third of time allowed or less (loose standards)	Never, even with bad rate setting (SM 3.30)
Better for	Highly efficient workers (saving above 50%)	Moderately efficient workers (saving below 50%)
Simplicity	Simple	A bit complicated

"When is the time rate system preferred?"

ICAI's three cases (SM 3.27): services that cannot be directly or tangibly measured (helpers, supervisory and clerical staff); highly skilled jobs; output whose pace is independent of the operator (automatic chemical plants).

The numerical: three answers to the same question

Loses marks

"Rowan (Imran) = 24 × 100 + 6/30 × 6 × 100 = ₹2,520."

Time saved × rate in place of time wages: wrong formula, no marks for method.

Passes

"Rowan = ₹2,880, Halsey = ₹2,700."

Right figures, but no time allowed, no time saved, no formula.

Full marks

Working notes for time allowed and time saved, each formula written once in symbols, figures substituted, results with units, and the comparison or advice the question asks for.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: Rowan's bonus built on the time saved**
   Fix: the fraction multiplies the **time wages**, time taken × rate.

2. **The negative bonus**
   Fix: premium plans guarantee day wages; with no time saved the bonus is nil (SM 3.28).

3. **The effective rate on the wrong hours**
   Fix: the effective hourly rate is earnings ÷ hours actually worked (Joseph under Halsey: ₹2,100 ÷ 12 = ₹175, not ÷ 30 = ₹70).

4. **Calendar hours for a clock-time question**
   Fix: count only working hours in each day.

RECAP

- Time rate: time worked × rate; for unmeasurable services, highly skilled jobs, operator-independent output.
- Piece rate: units × rate per unit, independent of time.
- Premium plans: standard time set, time wages guaranteed, time saved shared.
- Halsey: AH × R + 50% × (SH − AH) × R.

- Rowan: AH × R + (SH − AH) ÷ SH × AH × R; the worker can never double his earnings.
- Equal when AH = 50% of SH; Rowan more above it (moderate efficiency), Halsey more below it. Rowan's bonus is greatest at 50%.
- Efficiency % = time allowed ÷ time taken × 100. (Halsey halves the cake; Rowan slices it by time worked.)

Interactive version: PassBook app, lesson Time rate, piece rate and the Halsey and Rowan premium plans

4.5 Direct expenses

Key idea

Direct expenses are the third part of prime cost: costs other than direct material and direct employee cost that are directly traceable, in an economically feasible way, to a product, job or service. Measure them net of rebates, with non-creditable taxes; spread lump sums over the output that benefits; treat trivial amounts as overheads.

FORMATS AND FORMULAS

Definition

Direct expenses

Expenses other than direct material cost and direct employee cost, which are incurred to manufacture a product or for provision of service and can be directly traced in an economically feasible manner to a cost object. (SM 3.52)

∑ **Formula**

Amortising a one-time payment

$$\text{Per unit} = \frac{\text{One-time payment}}{\text{Estimated output}}$$

Charge for the period = charge per unit × units produced in the period. Use the estimated benefit instead of units when the benefit is not measured in output (SM 3.53).

∑ **Formula**

Measuring a direct expense

$$\text{Cost} = \text{Price} - \text{Rebate}$$

$$+\text{Non-creditable taxes} + \text{Commission}$$

Price = invoice or agreed price; add any other directly attributable cost (SM 3.53).

ICAI's layout for a classification question (SM 3.54, Illustration 16). *Reproduce it from memory, with notes for every item left out.*

Calculation of direct expenses	Job A (₹)	Job B (₹)	Job C (₹)
Product blueprint cost	xxx	—	—
Hire charges for machinery	—	xxx	—
Licence fee for software	—	—	xxx
Total direct expenses	**xxx**	**xxx**	**xxx**

Notes: office and administration cost is overheads; salary of office attendants is office and administration cost; salary of the marketing manager is selling overheads (SM 3.54).

ANSWER FRAME

"What are direct expenses? How are they measured and treated in cost accounts?"

1. **Definition** in ICAI's words (SM 3.52).
2. **Examples**: royalty; hire of specific equipment; product or service specific design or drawing; product or service specific software; other directly related expenses. The list is not exhaustive.
3. **Measurement**: invoice or agreed price net of rebate or discount, including duties and taxes without input credit, commission and other directly attributable costs; job work at agreed price plus materials supplied and incidental expenses.

4. **Treatment**: part of prime cost; lump sums amortised over estimated production or benefit; insignificant amounts as overheads.

A classification question

Answer in ICAI's layout: a table of the direct expenses by job or product, with the working of each figure (hours × rate, packs × royalty, job work plus materials), then numbered notes giving the treatment of every item left out ("administrative overhead is indirect expense"; "commission paid to sales staff comes under selling expenses"), as the Suggested Answers do. The notes carry marks: a bare total of direct expenses earns little.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the design charged to the launch month**
   Fix: a one-time payment is amortised over the estimated production that benefits: ₹0.60 a bottle.

2. **GST that will come back**
   Fix: only duties and taxes for which input credit is **not** available are part of the cost (SM 3.53).

3. **Factory wages under direct expenses**
   Fix: they are direct employee cost, the second element of prime cost, together with the employer's contributions on them (SM 3.67; RTP September 2026 Q8): see attendance, payroll and the elements of wages.

4. **The sales commission that became prime cost**
   Fix: it is a selling overhead (SM 3.67); the commission that *is* part of a direct expense is one paid to acquire that expense, like the agent's fee for hiring the bottling line.

RECAP

- Direct expenses: costs other than direct material and direct employee cost, incurred for a product or service and traceable to it in an economically feasible manner.
- ICAI's examples: royalty; hire of specific equipment; product- or service-specific design or drawing; specific software; other directly related expenses (not exhaustive). The Cost Sheet chapter adds utilities such as power and fuel, technical know-how fees and amortised moulds and patterns (SM 6.4).
- Measured at invoice or agreed price net of rebate or discount, plus taxes without input credit, commission and other directly attributable costs.
- Job work: agreed price, plus materials supplied by the principal and incidental expenses.

- Part of prime cost; one-time payments amortised over estimated production or benefit; insignificant amounts may be overheads.
- Not direct expenses: factory wages (employee cost), administration, sales commission, marketing salaries, general plant hire. (If only this product ran up the bill, it is a direct expense.)

CHAPTER 5

Overheads: Absorption Costing

5.1 Classifying overheads, allocation and apportionment

Key idea

A cost is **allocated** when the whole of it belongs to one cost centre, and **apportioned** when it is shared and must be split on a fair basis. The link between the expense and the cost centre decides which, not the kind of expense: power on its own meter is allocated, power on a common meter is apportioned.

FORMATS AND FORMULAS

Write the definitions in ICAI's words; they are asked as MCQs and as "distinguish" questions.

Definition

Overheads

Overheads represent expenses that have been incurred in providing certain ancillary facilities or services which facilitate or make possible the carrying out of the production process; by themselves these services are not of any use. (SM 4.63)

Definition

Cost allocation

The term "allocation" refers to assignment or allotment of an entire item of cost to a particular cost center or cost unit. (SM 4.63)

Definition

Cost apportionment

Apportionment implies "the allotment of proportions of items of cost to cost centres or departments". (SM 4.9)

Definition

Re-apportionment

The process of assigning service department overheads to production departments is called reassignment or re-apportionment. (SM 4.9)

Definition

Absorption

The process of recovering overheads of a department or any other cost center from its output is called recovery or absorption. (SM 4.10)

∑ **Formula**

Budgeted overhead at a new output

$$\text{Budget} = F + SF + (V + SV) \times \frac{Q_1}{Q_0}$$

F = fixed overheads; SF = fixed part of the semi-variable overheads; V = variable overheads and SV = variable part of the semi-variable overheads, both at the old output; Q_0 = old output, Q_1 = new output. Split the semi-variable cost first; only the variable parts move with output (SM 4.6–4.7). Splitting methods are in cost behaviour.

ANSWER FRAME

Distinguish between allocation and apportionment (SM 4.16): write it as a table, then the two points below it.

Point	What is charged
Allocation	Whole items of cost identifiable with one department (e.g. indirect wages of each department obtained separately)
Apportionment	Proportions of an item of cost (e.g. the cost of a service shared by the departments that avail it)

Point	Allocation	Apportionment
Nature of the process	A direct process of charging expenses to cost centres	An indirect process: the appropriate portion borne by each department must be identified

- The relationship between the expense and the cost centre, not its nature, decides whether an expense is allocated or apportioned.
- Allocation is a much wider term than apportionment.

Distinguish between cost allocation and cost absorption: allocation is the allotment of a whole item of cost to a cost centre or cost unit; absorption is the process of absorbing all overheads allocated to or apportioned over a cost centre or production department into the units produced. Allocation gets costs **into** cost centres; absorption gets them **out**, into products.

Advantages of departmentalisation (four points, SM 4.12–4.13) and **advantages of separating fixed and variable overheads** (the objective plus three points, SM 4.6–4.7) are written as numbered points with one line each, in ICAI's order.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: "power is always apportioned"**
   Fix: check the facts first; a metered or exclusively used cost is allocated in full, and only the shared remainder is apportioned.

2. **The 25% that ate the budget**
   Fix: split semi-variable costs, keep the fixed parts fixed, and move only the variable parts.

3. **Cartons in the factory, pouches on the truck**
   Fix: ask what the packing does. Protects or holds the product: production. Carries it to the customer: distribution. Attracts the customer: selling.

RECAP

- Overheads: costs not conveniently traceable to a cost unit; the aggregate of indirect material, indirect employee cost and indirect expenses.
- Four classifications, in ICAI's order: function (factory, administration, selling and distribution), nature (fixed, variable, semi-variable), element, control.
- Splitting fixed and variable serves marginal costing, then control, budget estimates and decisions; budget only the variable parts in proportion to output.

- Stages: estimation and collection (standing orders) → allocation → apportionment → re-apportionment → absorption → under- or over-absorption.
- Allocation: whole item to one cost centre; apportionment: proportions of an item on a fair basis (cause and effect or benefit received).
- The relationship with the cost centre decides allocation or apportionment; allocation is the wider term.
- Variable manufacturing overheads are absorbed on actual production, fixed ones on normal capacity.
- Primary packing: production; secondary: distribution; fancy: selling. ("Miscellaneous" is not a function.)

5.2 Primary distribution of factory overheads

Key idea

Primary distribution charges every factory overhead to every department that benefits from it, **production and service alike**: whole items by allocation, shared items by apportionment, each shared item on the basis that best measures the benefit received or the cause of the cost.

FORMATS AND FORMULAS

ICAI calls this statement the Primary Distribution Summary, the Overhead Distribution Summary or the Overhead Distribution Statement. Reproduce the layout from memory.

Item of cost	Basis	Total (₹)	Production depts (₹)	Service depts (₹)
Allocated: indirect material	Actual (allocation)	xxx	xxx	xxx
Indirect wages	Actual (allocation)	xxx	xxx	xxx
Apportioned: rent	Floor area (ratio)	xxx	xxx	xxx
Depreciation of plant	Capital value (ratio)	xxx	xxx	xxx
Power	H.P. × machine hours (ratio)	xxx	xxx	xxx
…	…	…	…	…
Total		**xxx**	**xxx**	**xxx**

In an answer, each production and service department gets its own column. ICAI's answers print the ratio in brackets beside the item or its basis, e.g. "Factory rent, floor area (6 : 3 : 1)", and some split the statement into allocated overheads and a "schedule of apportioned expenses" (SM 4.80–4.81). Both layouts are ICAI's own; whichever you use, show every basis and its ratio.

∑ **Formula**

A department's share of a shared overhead

$$\text{Share} = \text{Overhead} \times \frac{\text{Its base}}{\text{Total base}}$$

"Its base" is the department's units of the base (square metres, workers, rupees of capital value); "total base" is the total of the sharing departments only, those that benefit: a rented block's rent is not shared with a department housed in an owned building.

∑ **Formula**

Compound base for machine power

$$\text{Power share} \propto \text{HP} \times \text{machine hours}$$

Used in ICAI's Illustration 4 (SM 4.24–4.25); a 50 HP machine running 6,000 hours draws 3,00,000 HP-hours.

∑ **Formula**

Department's primary total

$$\text{Total} = \text{Allocated} + \text{Apportioned}$$

Check: the primary totals of all departments add up to the total factory overheads.

Definition

Service departments

Service departments are those departments which do not directly take part in the production of goods or providing services. Such departments provide auxiliary services across the entity and render services to other cost centres and in some cases to outside parties; e.g. engineering, quality control and assurance, laboratory, canteen, stores, time office, dispensary. (SM 4.9)

ANSWER FRAME

"Suggest a basis of apportionment for each of the following": a two-column table (overhead | basis), one line per item, in ICAI's words from the bases table; where ICAI gives alternatives ("floor area, or volume of department"), write both.

"Explain the other bases of apportioning overheads" (SM 4.15–4.16):

1. Benefit received is generally the principal criterion; other bases that may be used are the three below.
2. Analysis or survey of existing conditions, with the light points example.
3. Ability to pay: income of the paying department; simple but may be inequitable; easy-selling lines bear the most.
4. Efficiency or incentive: pre-determined levels; exceeding them lowers the overhead per unit and

raises the department's profit.

5. A concern may use one criterion, or all of them for different activities.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: Kesar's equal shares**
   Fix: allocate what can be allocated, then give every shared item its own basis from ICAI's table.

2. **The godown that paid rent**
   Fix: a cost is shared only by the departments that benefit from it: 6 : 3 : 1 over Frying, Packing and Maintenance.

3. **The service departments that vanished**
   Fix: primary distribution covers all departments; service departments take their share now and pass it on in the secondary distribution.

4. **Prime cost in the overhead statement**
   Fix: only indirect items belong there. A service department's wages are indirect for the product, so they are allocated to it.

RECAP

- Primary distribution: every factory overhead to every department, production and service, by allocation or apportionment.

- Allocate whole items first, including service departments' own material and wages; prime cost stays out.
- Each shared item gets its own basis: building costs by floor area; supervision, welfare and time keeping by workers; ESI, PF and holiday pay by direct wages; depreciation and repairs of plant by capital values; machine power by horse power or H.P. × machine hours; lighting by light points.
- Share the item only among the departments that benefit (a rented block's rent skips the owned godown).
- Other bases: survey of existing conditions, ability to pay (simple, may be inequitable), efficiency or incentive.
- Base selection: economy, practicability, equitability and reliability.
- Show the basis and the ratio for every item, and check the column totals against total overheads. (Equal shares are not a basis.)

Interactive version: PassBook app, lesson Primary distribution of factory overheads

5.3 Secondary distribution: direct, step, repeated and simultaneous methods

Key idea

The methods differ only in how they treat services between service departments: the direct method ignores them, the step method recognises them one way, and the reciprocal methods recognise them both ways. All the reciprocal methods give the same answer.

FORMATS AND FORMULAS

ICAI heads the statement Secondary Overhead Distribution Statement (SM 4.19), Secondary Distribution Summary, or Re-distribution of Service Department's Expenses. Reproduce the layout from memory. In an answer each department has its own column; the direct and simultaneous equation methods may show the production departments only, the step and repeated distribution methods need the service departments' columns too.

Particulars	Production depts (₹)	Service depts (₹)
Overheads as per primary distribution	xxx	xxx
Service department S1 apportioned (ratio)	xxx	(xxx)
Service department S2 apportioned (ratio)	xxx	(xxx)
... repeated, for the repeated distribution method	...	...
Total	**xxx**	**Nil**

Definition

Secondary distribution

The re-apportionment of the service department cost to the production department is known as secondary distribution. (SM 4.16)

Definition

Direct re-distribution method

Under this method service department costs are apportioned over the production departments only, ignoring the services rendered by one service department to the other. (SM 4.64)

Definition

Step method or non-reciprocal method

This method gives cognizance to the service rendered by service department to another service department. The sequence here begins with the department that renders service to the maximum number of other service departments. (SM 4.64)

Definition

Reciprocal service method

These methods are used when different service departments render services to each other, in addition to rendering services to production departments; the various service departments have to share overheads of each other. (SM 4.64)

Formula

Direct method: share of a production department

$$\text{Share} = \text{Service cost} \times \frac{\text{Its \%}}{\text{Production \%}}$$

"Its %" is the production department's share of the service; "production %" is the total of the production departments' shares (80% for Stores in the samosa factory).

∑ **Formula**

Simultaneous equations for two service departments

$$S_1 = a + p\,S_2 \qquad S_2 = b + q\,S_1$$

Substituting the second equation in the first:

$$S_1 = \frac{a + p\,b}{1 - p\,q}$$

a, b: the service departments' costs after primary distribution; p: the share of S_2's work done for S_1; q: the share of S_1's work done for S_2; S_1, S_2: their total costs including reciprocal services. Then $S_2 = b + q\,S_1$. Apportion S_1 and S_2 on the production departments' percentages only.

∑ **Formula**

The check that saves marks

$$\text{Production shares} = a + b$$

Under every method, the production departments receive exactly the service departments' primary totals ($a + b$). The solved totals $S_1 + S_2$ exceed this figure, because they include the services the departments render to each other.

ANSWER FRAME

"Explain the methods of re-apportionment of service department expenses" (SM theory questions 2 and 8;

MTP January 2025 Series II Q6(d)): the points ICAI's answer gives, in order.

1. **Direct re-distribution method**: service department costs are apportioned over the production departments only, ignoring the services rendered by one service department to the other.
2. **Step method (non-reciprocal method)**: gives cognizance to the services rendered by one service department to another; the cost of the service department that serves the maximum number of departments is apportioned first, then the next, until the last service department's cost is apportioned to production departments only.
3. **Reciprocal service method**: recognises that service departments may render services to each other and gives these inter-departmental services due weight. The methods available are (a) the simultaneous equation method, (b) the repeated distribution method and (c) the trial and error method, each in one line.

If time allows, add a two-line illustration (two service departments that serve each other) under the reciprocal method.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the step method's boomerang**
   Fix: a closed department receives nothing; the last department's cost goes to production departments only (here 1 : 1).

2. **The direct method that lost 20%**
   Fix: in the direct method, scale the production percentages to 100% (1 : 1); the service departments must end at nil.

3. **Distributing the wrong totals**
   Fix: apportion the solved totals on the production percentages only, and check that the production departments received exactly ₹4,20,000.

4. **The canteen that fed itself**
   Fix: a service department's own count is left out of its basis: 45 : 75 : 30.

RECAP

- Secondary distribution: re-apportioning service department costs to production departments, on bases such as machine or direct labour hours (maintenance), number of employees (canteen, personnel), requisitions or value of materials issued (stores), kWh (power house), capital values (fire protection).

- Direct method: production departments only, inter-service services ignored, production percentages scaled to 100%.
- Step method: start with the department serving the largest number of others (some authors: largest cost first); a closed department receives nothing; the last goes to production only.
- Reciprocal service method: simultaneous equations, trial and error, repeated distribution; all give the same result.
- Two service departments: $S_1 = a + pS_2$ and $S_2 = b + qS_1$; apportion the solved totals on production percentages only.
- Check: production departments receive exactly the service departments' primary total.
- Then the production totals ÷ machine or labour hours give the absorption rates. (The ball always melts.)

Interactive version: PassBook app, lesson Secondary distribution: direct, step, repeated and simultaneous methods

5.4 Absorption rates and the machine hour rate

Key idea

An absorption rate is a cost centre's overheads divided by the base that best measures its use. Time bases follow most overheads (machine hours where machines dominate, labour hours where labour does). The rate is pre-determined from budgets, with fixed overheads on normal capacity.

FORMATS AND FORMULAS

Reproduce the machine hour rate statement and the hours working from memory: ICAI's answers use exactly this layout.

Formula

Percentage of direct material cost; of prime cost

$$\text{Rate} = \frac{\text{Production overheads of the department}}{\text{Budgeted direct material cost}} \times 100$$

$$\text{Rate} = \frac{\text{Production overheads of the department}}{\text{Prime cost}} \times 100$$

Formula

Percentage of direct labour cost

$$\text{Rate} = \frac{\text{Production overheads of the department}}{\text{Direct labour cost}} \times 100$$

∑ **Formula**

Labour hour rate and machine hour rate

$$\text{Labour hour rate} = \frac{\text{Production overheads of the department}}{\text{Direct labour hours}}$$

$$\text{Machine hour rate} = \frac{\text{Overheads of the machine}}{\text{Productive machine hours}}$$

Both are rupees per hour. (SM 4.33 prints the labour hour formula with "× 100"; its own example, ₹90,000 ÷ 1,50,000 hours = ₹0.60 an hour, applies no × 100, so write the rate in rupees per hour.)

∑ **Formula**

Rate per unit, normal and pre-determined rates

$$\text{Per unit} = \frac{\text{Overheads}}{\text{Number of units}}$$

$$\text{Normal} = \frac{\text{Actual overheads}}{\text{Actual base}}$$

$$\text{Pre-determined} = \frac{\text{Budgeted overheads}}{\text{Budgeted base}}$$

∑ **Formula**

Blanket and departmental rates

$$\text{Blanket} = \frac{\text{Total overheads for the factory}}{\text{Total units of base for the factory}}$$

$$\text{Departmental} = \frac{\text{Overheads of the department}}{\text{Corresponding base}}$$

The machine hour rate statement

Computation of machine hour rate	Per period (₹)	Per hour (₹)
(A) Standing charges		
Rent, lighting, supervision (shares)	xxx	
Insurance; depreciation (time basis)	xxx	
Operator's wages (if included)	xxx	
Total standing charges ÷ productive hours	**xxx**	**xxx**
(B) Machine expenses		
Power, consumables, repairs; depreciation (activity basis)	xxx	xxx
Machine hour rate (A) + (B)	**xxx**	**xxx**

Productive hours when a machine needs an operator

Hours (all operators, for the period)	Hours
Normal available hours × months × operators	xxx
Less: absenteeism without pay × operators	(xxx)
Paid hours (operators' wages are on these)	**xxx**
Less: leave with pay × operators	(xxx)
Less: normal unavoidable idle time × operators	(xxx)
Effective (productive) working hours	**xxx**

Normal capacity (SM 4.58)

Normal capacity	Units
Installed capacity	xxx
Adjustments for: (i) time lost to scheduled preventive or planned maintenance; (ii) number of shifts, machine hours or man hours; (iii) holidays, normal shut-down days, normal idle time; (iv) normal time lost in batch change-over	(xxx)
Normal capacity	**xxx**

Definition

Absorption

The process of recovering overheads of a department or any other cost centre from its output is called recovery or absorption. (SM 4.10)

Definition

Machine hour rate

Machine hour rate implies cost of running a machine for an hour to produce goods. (SM 4.33)

Definition

Blanket overhead rate

Blanket overhead rate refers to the computation of one single overhead rate for the whole factory, as distinguished from the departmental overhead rate, a separate rate for each individual cost centre or department. (SM 4.37)

Definition

Normal capacity

The volume of production or services achieved or achievable on an average over a period under normal circumstances taking into account the reduction in capacity resulting from planned maintenance. (SM 4.58)

ANSWER FRAME

"Explain blanket and departmental overhead rates; state the conditions for a blanket rate" (5 marks):

1. Blanket rate: one single overhead rate for the whole factory; formula.
2. Departmental rate: one rate for each department or cost centre; formula.
3. Blanket rate conditions: (1) only one major product; (2) several products, but (a) all pass through all departments and (b) all are processed for the same length of time in each.
4. Otherwise departmental rates: product lines varied, or machinery used to a varying degree.

"Arguments for and against departmental absorption rates" (MTP January 2025): for: costs more precise than blanket rates; they facilitate variance analysis and cost control; stock and work-in-process valuation easier and more precise. Against: setting up and monitoring them can be time-consuming and expensive.

"Direct and comprehensive machine hour rate": definition of MHR; direct (expenses directly connected with the machine ÷ operational hours); comprehensive (plus a share of the department's other expenses, sometimes operator's wages); standing charges and machine expenses; the six steps.

Three answers to "compute the machine hour rate"

Loses marks

"₹9,12,000 ÷ 2,000 = ₹456 an hour."

Standing charges only: the running costs are left out, and nothing shows which items were treated how.

Passes

The right rate, ₹911, but every item listed in one column with no workings.

Correct, yet a slip in one item would leave no method marks.

Full marks

Working notes first (which items accrue with time and which with running, productive hours), then ICAI's statement: standing charges and machine expenses, each with its working, an hourly rate for each, and the total.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the maintenance hours that made the fryer cheaper**
   Fix: the denominator is productive hours: leave out planned maintenance always, set-up time when it is unproductive, and idle time.

2. **Paying for the absentee**
   Fix: wages are paid on paid hours: available less absenteeism without pay (2,352), because leave with pay is paid but not worked.

3. **The blanket that covered a machine shop**
   Fix: check the blanket-rate conditions; if products do not spend the same time in each department, compute departmental rates.

4. **A month of rent, a year of insurance**
   Fix: convert every item to the period of the rate before adding, and write the conversion in the working.

RECAP

- Absorption = recovering a cost centre's overheads from its output, at overheads ÷ base.
- ICAI's six methods: % of direct materials, % of prime cost, % of direct labour cost, labour hour rate, machine hour rate, rate per unit of output. The first

two are generally unsuitable (time, price swings, skill and machines ignored).

- Labour hour rate: little machinery. Machine hour rate: machines the main factor; each machine a cost centre.
- MHR = (standing charges + machine expenses) ÷ productive hours; comprehensive adds the department's other expenses, sometimes operator's wages.
- Rates: normal (actual ÷ actual), pre-determined (budget ÷ budget), blanket (one for the factory), departmental (one per department).
- Blanket only for one major product, or products that pass through every department for the same time; otherwise departmental.
- Fixed overheads on normal capacity, variable on actual production. Normal idle = installed − normal; abnormal idle = normal − actual.
- Wages on paid hours, rate on effective hours. (Same fryer hours, same overhead.)

Interactive version: PassBook app, lesson Absorption rates and the machine hour rate

5.5 Under- and over-absorption of overheads

Key idea

Under- or over-absorption = actual overheads − absorbed overheads (actual base × pre-determined rate), after removing items that are not this period's normal overheads. The abnormal part and any small balance go to Costing P&L; a large balance from an unavoidable estimating error is spread by a supplementary rate over cost of sales, finished goods and work-in-progress.

FORMATS AND FORMULAS

Reproduce the calculation statement and the accounting table from memory.

Formula

Absorbed overheads and the balance

$$\text{Absorbed} = \text{Actual base} \times \text{Pre-determined rate}$$

$$\text{Under } (+) \text{ or over } (-) = \text{Actual overheads} - \text{Absorbed}$$

Actual base: the actual hours or units of the period. Actual overheads: after removing the abnormal and other-period items.

∑ **Formula**

Supplementary rate (SM 4.44)

$$\text{Supplementary rate} = \frac{\text{Under- or over-absorbed overheads}}{\text{Units produced}}$$

Units produced = finished units (sold and in stock) + equivalent completed units in work-in-progress.

Calculation of under- or over-absorption

Particulars	Amount (₹)
Actual production overheads incurred	xxx
Less: wages for strike period; previous year's expenses; obsolete stores written off	(xxx)
Net production overheads actually incurred (A)	**xxx**
Overheads absorbed (actual base × pre-determined rate) (B)	xxx
Under-absorption (A − B) or over-absorption (B − A)	**xxx**

The accounting (SM 4.44)

Account

1. Stock of finished goods A/c

Under-absorption

Debit

Over-absorption

Credit

Amount

Units of finished stock × supplementary rate

Account

2. Stock of semi-finished goods (WIP) A/c

Under-absorption

Debit

Over-absorption

Credit

Amount

Equivalent completed units × supplementary rate

Account

3. Cost of sales A/c

Under-absorption

Debit

Over-absorption

Credit

Amount

Units sold × supplementary rate

The journal entry for an under-absorption, as in ICAI's suggested answer to old-scheme December 2021 Q5(b):

Journal entry	Dr (₹)	Cr (₹)
Work-in-progress Control A/c Dr.	xxx	
Finished Goods Control A/c Dr.	xxx	
Cost of Sales A/c Dr.	xxx	
Costing Profit and Loss A/c Dr. (abnormal part)	xxx	
To Overhead Control A/c		xxx

For an over-absorption the entry is reversed: Overhead Control A/c is debited, and the three accounts (and Costing P&L for any abnormal part) are credited.

Definition

Under- and over-absorption

When the absorbed amount is less than the actual amount it is called under-absorption. Similarly, when the absorbed amount is more than the actual amount it is called over-absorption. Over-absorption is represented by a credit balance in the accounts and under-absorption by a debit balance; either may be termed overhead variance. (SM 4.42)

ANSWER FRAME

"Explain the treatment of over- and under-absorption of overheads in cost accounts" (5 marks):

1. Meaning: absorbed at pre-determined rates; actual differs; under = absorbed less than actual (debit balance), over = absorbed more than actual (credit balance).
2. Small balances: Costing P&L, product costs not disturbed, as they represent normal cost.
3. Large balances due to wrong estimation for unavoidable reasons: supplementary rate = under/over-absorbed ÷ units produced, applied to finished goods, WIP (equivalent units) and cost of sales.
4. Abnormal reasons (abnormal over- or under-capacity utilisation, defective planning): Costing P&L.
5. Carry forward to the subsequent year (ICAI's answer to MTP May 2024 Series I Q6(a)).

"How would you treat idle capacity costs?" (2 marks in MTP May 2025 Series II Q4(c)): the three causes and treatments in the table above, in ICAI's order: unavoidable (supplementary rate), avoidable (Costing P&L), seasonal (inflated overhead rates).

Three answers to the fryer-line question

Loses marks

"Under-absorption ₹13,25,000; charge to Costing P&L."

No analysis by cause, no supplementary rate.

Passes

The ₹13,25,000 is right and split by cause, but the supplementary rate divides by finished cartons only.

Work-in-progress missed: the apportionment fails.

Full marks

The calculation statement; the analysis by cause (hours lost × fixed rate, price rises), each part's treatment stated with its reason; the working note for equivalent units and the rate; the apportionment table; the journal entry if asked.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: budget minus actual**
   Fix: absorbed = 23,100 actual hours × ₹30 = ₹6,93,000, against actual ₹7,05,000: ₹12,000 under-absorbed.

2. **The strike that was charged to samosas**
   Fix: deduct them first, and name each one in the statement.

3. **The forgotten half-made cartons**
   Fix: add work-in-progress at equivalent completed units: 2,75,000 cartons, ₹4 each.

4. **Everything to Costing P&L**
   Fix: only small balances and abnormal parts go to Costing P&L; a large normal balance is spread by a supplementary rate.

RECAP

- Absorbed = actual base × pre-determined rate; under = actual > absorbed (debit balance); over = absorbed > actual (credit balance); both are "overhead variance".
- Causes: overheads differ from estimate, activity differs from budget, or both in different proportions; a proportional change in both gives neither.

- First remove strike wages, court awards, previous-year expenses and obsolete stores; keep normal overtime.
- Small balance: Costing P&L. Abnormal part: Costing P&L. Large normal part: supplementary rate.
- Supplementary rate = balance ÷ units produced (WIP at equivalent units); applied to finished goods, WIP and cost of sales.
- Idle capacity: unavoidable, supplementary rate; avoidable, Costing P&L; seasonal, inflated overhead rates.
- ICAI's answers add a third treatment: carry forward to the next year.
- The mess asks every student for the vegetable price rise, not for the burnt rice.

5.6 Administration, selling, distribution and R&D overheads

Key idea

Administration overheads are apportioned between production and sales, charged to Costing P&L, or added to cost as a separate charge on a base such as works cost; selling and distribution overheads are best absorbed as an estimated amount per unit (fixed costs by benefit, plus variable costs). Both are controlled by comparison, budgets and standards. R&D follows the purpose of the research.

FORMATS AND FORMULAS

Reproduce the reverse-calculation steps and the product-wise statement from memory.

Formula

Administration and selling rates on a base

$$\text{Admin. rate} = \frac{\text{Admin. overheads}}{\text{Works cost}} \times 100$$

$$\text{S\&D rate} = \frac{\text{S\&D overheads}}{\text{Sales value}} \times 100$$

Works cost is factory cost. The base is the one the question names: works cost, cost of production, sales value, units.

∑ **Formula**

Estimated selling and distribution expense per unit (SM 4.54)

$$\text{S\&D per unit} = \frac{\text{Fixed S\&D of the product}}{\text{Units sold}} + \text{Variable per unit}$$

Fixed S&D of the product: the fixed expenses apportioned to it on their bases. Units sold: sold or likely to be sold. Variable per unit: packaging, freight outwards, insurance in transit, commission, rebate.

∑ **Formula**

Working back from a selling price

$$\text{Total cost} = \frac{\text{Price}}{1 + p_c}$$

$$\text{Total cost} = \text{Price} \times (1 - p_s)$$

p_c = profit as a fraction of cost; p_s = profit as a fraction of the selling price. With factory overheads at F% of direct wages and administration at A% of factory cost: factory cost = prime cost + F × direct wages; total cost = factory cost × (1 + A).

Product-wise statement of profit (ICAI's layout, SM 4.57)

Particulars	Product A (₹)	Product B (₹)
Sales	xxx	xxx
Less: Variable costs: cost of goods sold, commission, packing, stationery	xxx	xxx
Contribution	**xxx**	**xxx**
Less: Fixed costs, each apportioned on its basis (ratio shown)	xxx	xxx
Profit or loss	**xxx**	**xxx**
Percentage of profit or loss on sales	**x%**	**x%**

ICAI's statement also has a Total column; write it first.

ANSWER FRAME

"Discuss the problems of controlling selling and distribution overheads" (SM question 3), then **"methods"** (PYQ January 2025 Q6(c), 4 marks):

1. Problems: (i) external factors (distance of market, competition, terms of sale) beyond management's control; (ii) dependence on customers' behaviour and

tastes, so control may lose customers; (iii) policy costs, not amenable to control.

2. Methods: (a) comparison with past performance, also as a percentage of sales; suitable for small concerns; (b) budgetary control, fixed and variable, a flexible budget if needed, variances acted on; (c) standard costing, standards for salesmen, territories, products, variances enquired into.

"Explain the methods of accounting administrative overheads": the three methods in ICAI's order, each with its reason and its disadvantages (the table in How it works); for (c), the five bases.

"Explain research and development expenses and their treatment in costing" (MTP September 2025, 4 marks): the two Terminology definitions; production-method research in cost of production; administration research in administration overhead; market research in selling and distribution; development of a particular product charged to it as deferred revenue expenditure, recovered per unit when production is fully established; routine general research to general overhead (deferred if substantial); fundamental research to administration; failed research to Costing P&L.

"Treatment of bad debts in cost accounts": no unanimity; view 1, excluded as financial losses; view 2, part of selling and distribution overheads in the normal course of trading; extraordinarily large bad debts excluded.

Three answers to the three-channels question

Loses marks

"App orders are the largest consumer orders (₹80), so spend the ₹5,00,000 on the app."

Sales per order says nothing about the costs each rupee of sales carries.

Passes

A profit statement with the common costs deducted before contribution, and the advice taken from the profit line.

Right channel by luck; no contribution measure, no incremental working.

Full marks

The statement in layers (variable costs, contribution, specific fixed costs, apportioned common costs); contribution per ₹100 of sales and per order; the incremental table for the ₹5,00,000; a recommendation with its reason.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the margin that moved**
   Fix: profit on price means cost is 80% of price: price = cost ÷ 0.80.

2. **Administration on the wrong base**
   Fix: apply each rate to the base it was computed on, and write the base next to every percentage in the statement, as ICAI does.

3. **Dropping the product that "makes a loss"**
   Fix: look at contribution and at which fixed costs would really be saved.

4. **Interest inside the factory**
   Fix: interest and financing charges are a separate item of cost of sales (SM 4.60).

RECAP

- Administration overhead (CIMA): costs of general management and secretarial, accounting and administrative services not directly related to production, marketing, research or development.
- Three methods: apportion to production and sales; charge to Costing P&L; separate addition on works cost, sales, gross profit, quantity produced or conversion cost.

- Control of administration: departmental analysis against results (and absorption rates), budgets, standards.
- S&D bases: sales value (most logical), cost of goods sold, gross profit, orders or units. Best: estimated amount per unit (fixed by benefit + variable).
- S&D control is hard (external factors, customers, policy costs); use past comparison, budgets, standard costing.
- R&D: production methods, cost of production; administration or fundamental, administration overhead; market research, S&D; one product, deferred and charged to it; failure, Costing P&L.
- Interest: separate item of cost of sales. Bad debts: two views; very large ones excluded. Carriage follows what it carries.
- Reverse questions: Price → Cost → Factory cost → factory overheads. (The principal's salary still has to land somewhere.)

CHAPTER 6

Activity Based Costing

6.1 Activities, cost pools, cost drivers and the cost hierarchy

Key idea

Overheads are caused by activities, not by labour hours. ABC traces overheads to activities first, then charges each cost object for the activities it actually uses, through cost drivers.

FORMATS AND FORMULAS

Reproduce the formula and both driver tables from memory: ICAI asks for examples of cost drivers as 5-mark parts.

Formula

Activity cost driver rate (SM 5.10)

$$\text{Activity cost driver rate} = \frac{\text{Total cost of activity}}{\text{Activity driver}}$$

The activity driver is the total quantity of the driver for the period (all set-ups, all orders). Overhead charged to a cost object = the sum, over the activities it uses, of driver rate × driver quantity it consumed.

Cost drivers by business function (SM 5.4–5.5):

Business function	Cost drivers
Research and development	Number of research projects; personnel hours on a project
Design of products, services and procedures	Number of products in design; number of parts per product; number of engineering hours
Customer service	Number of service calls; number of products serviced; hours spent on servicing products
Marketing	Number of advertisements; number of sales personnel; sales revenue
Distribution	Number of units distributed; number of customers

Cost drivers by activity cost pool in a production department (SM 5.13):

Activity cost pool	Related cost driver
Ordering and receiving materials	Number of purchase orders
Setting up machines	Number of set-ups
Machining	Machine hours
Assembling	Number of parts
Inspecting and testing	Number of tests
Painting	Number of parts
Supervising	Direct labour hours

ANSWER FRAME

"Distinguish between ABC and traditional absorption costing" (SM 5.7)

Point

1. Grouping

Activity based costing

Overheads related to activities and grouped into activity cost pools

Traditional absorption costing

Overheads related to cost centres or departments

Point

2. Realism

Activity based costing

Costs related to activities, hence more realistic

Traditional absorption costing

Costs related to cost centres, hence not realistic of cost behaviour

Point

3. Drivers

Activity based costing

Activity-wise cost drivers determined

Traditional absorption costing

Time (hours) assumed to be the only cost driver in all departments

Point

4. Rates

Activity based costing

Activity-wise recovery rates; no single overhead recovery rate

Traditional absorption costing

Multiple rates (one per department) or a single overhead recovery rate

Point

5. Cost objects

Activity based costing

Customers, products, services, departments, etc.

Traditional absorption costing

Cost units: products, jobs or hours

Point	
6. Control	
Activity based costing	Essential activities simplified, unnecessary ones eliminated: aids cost control
Traditional absorption costing	Cost centres cannot be eliminated: not suitable for cost control

Other questions, in ICAI's order

- **Usefulness or suitability** (4 marks): the four headed situations of SM 5.3, each with one line of explanation.
- **Cost driver**: definition; the two categories with their uses; then two drivers for each function asked, from the SM 5.4–5.5 table.
- **How product costs are determined in ABC, or how an ABC system is developed**: the five stages in order (identify activities; relate overheads to activities to create cost pools using resource cost drivers; spread support activities over primary activities; determine activity cost drivers; calculate driver rates and charge cost objects), then the numbers.
- **Levels of activities**: the four levels, each with ICAI's meaning and one example.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the inspection that was batch level because it said "inspection"**
   Fix: classify by what the cost varies with. Testing every unit is unit level; testing the first item of each batch is batch level (SM 5.8).

2. **Charging products with a resource cost driver**
   Fix: a resource cost driver (supervisors' time on set-ups, receiving and testing) takes a resource to the activities: stage one. Products are then charged through activity cost drivers (set-ups, tests, deliveries) at each activity's rate.

3. **Dividing a pool by one product's driver**
   Fix: the rate is the pool ÷ the total driver for the period (60 set-ups), then multiplied by each product's own quantity. Check: the products' charges add back to the pool.

4. **"ABC increased the overheads"**
   Fix: ABC redistributes the same total; under-costed and over-costed amounts are equal and opposite. If your totals differ from the ledger, a pool or a rate is wrong.

RECAP

- ABC assigns overheads to activities, then to cost objects according to their use of activities (SM 5.3).
- Two stages: resources to activities by **resource cost drivers**; activities to cost objects by **activity cost drivers**.
- Activity: an event that incurs cost. Cost pool: costs with the same cause and effect relationship. Cost driver: a factor that causes a change in the cost of an activity.
- Activity cost driver rate = total cost of activity ÷ activity driver.
- Hierarchy: unit (varies with units), batch (with batches), product (supports each product), facility (sustains the plant).
- Useful when overheads are high, products many, non-volume activities significant, competition stiff.
- Same total overhead, spread differently: small-batch products often turn out under-costed by a labour-hour rate. (Split the dinner bill by the orders.)

Interactive version: PassBook app, lesson Activities, cost pools, cost drivers and the cost hierarchy

6.2 Product costs under ABC versus traditional absorption

Key idea

Prime cost does not change; only overheads are re-spread. Cost distortion = traditional cost – ABC cost: positive means over-costed, negative means under-costed, and across all products the distortion adds up to zero.

FORMATS AND FORMULAS

Reproduce both statements from memory. ICAI's suggested answers show the driver rate and the quantity in brackets on every overhead line.

Statement of cost per unit: traditional absorption (SM 5.16, 5.44)

Particulars	Product A (₹)	Product B (₹)
Direct materials	xxx	xxx
Direct labour (hours × rate)	xxx	xxx
Production overheads (hours × absorption rate)	xxx	xxx
Cost per unit	**xxx**	**xxx**

Statement of cost: activity based costing (SM 5.17, 5.44–5.45)

Particulars	Product A (₹)	Product B (₹)
Direct materials	xxx	xxx
Direct labour	xxx	xxx
Set-up costs @ ₹... per set-up (set-ups)	xxx	xxx
Machine costs @ ₹... per machine hour (hours)	xxx	xxx
Inspection costs @ ₹... per inspection (inspections)	xxx	xxx
Total cost	**xxx**	**xxx**
Cost per unit (total cost ÷ units)	**xxx**	**xxx**

Working notes before the statement: driver totals (batches, orders, inspections, hours) and the driver rate table (cost pool, driver, quantity, rate).

∑ **Formula**

Activity cost driver rate and the product's overhead

$$\text{Rate} = \frac{\text{Cost of activity pool}}{\text{Total quantity of the driver}}$$

$$\text{Overhead}_{\text{product}} = \sum \text{Rate} \times \text{Driver quantity used}$$

On a capacity basis the denominator is the driver's capacity, not its actual use.

∑ **Formula**

Cost distortion

$$\text{Cost distortion per unit} = \text{Traditional cost} - \text{ABC cost}$$

Both costs are per unit. Positive: over-costed by the traditional system; negative: under-costed. ICAI prints the sign: "(–) 40.42" for the under-costed product (SM 5.26).

∑ **Formula**

Cost of unused capacity

$$\begin{aligned}&\text{Cost of unused capacity}\\&= \text{Cost of activity} - \text{Cost allocated}\\&= \text{Unused driver units} \times \text{Rate}\end{aligned}$$

Cost allocated = the driver units products actually used × the capacity rate.

Customer-level operating income (SM 5.54–5.55; May 2024 Q2(a))

Particulars	Customer A (₹)
Revenues at list price (units × list price)	xxx
Less: Discount	xxx
Revenue at actual price	**xxx**
Less: Cost of goods sold	xxx
Gross margin	**xxx**
Less: Customer-level operating activity costs (orders, visits, delivery, expediting)	xxx
Customer-level operating income	**xxx**
Operating income as a % of revenue (at actual price)	x%

A full-marks numerical answer, in ICAI's order

1. **Working notes**: driver totals (batches, orders, inspections, hours); the split of overheads into pools if percentages or ratios are given; the driver rate table (pool, driver basis, quantity, rate).
2. **Statement 1**: cost per unit under the traditional method (rate shown).
3. **Statement 2**: cost under ABC with "@ ₹ rate (quantities)" on each overhead line; total cost and cost per unit.
4. **Comparison**: cost distortion with its sign, under- or over-costed.
5. **Comment**: reason for the difference (overheads borne by each product differ; batch-level and product-level activities) and the decision asked (price, mix, customer action).

Loses marks

"Masala Gold ABC cost ₹1,200, so it is under-costed."

No rates, no driver totals: a slip anywhere loses every method mark.

Passes

Driver rates and the ABC statement, but the per-run tests were charged as if they were totals, and the distortion has no sign.

One slip in a working note spoils every figure after it.

Full marks

Working notes for driver totals and rates, both statements, the distortion with "(–)" for under-costed products and the check that it nets to zero, and a two-line comment on pricing.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: per-run data used as if it were the total**
   Fix: tests = 40 runs × 4 = 160; charge = ₹2,56,000. Where a question gives a batch size instead of the number of runs, batches = units ÷ batch size first.

2. **The rate on actual use when the question says capacity**
   Fix: divide by capacity (12,000 hours: ₹80) and report the unused capacity separately.

3. **Dividing ABC overhead by units sold**
   Cost per unit is total cost ÷ units **produced**. Units sold matter only for sales, cost of goods sold and closing stock.

4. **Percentages on list-price revenue**
   Fix: supermarkets ₹6,99,000 ÷ ₹66,00,000 = 10.59%, not ÷ ₹75,00,000 = 9.32%.

RECAP

- Prime cost is the same under both systems; only production overheads move.
- Traditional: overhead per unit = absorption rate × base per unit. ABC: driver rate × driver quantity per product ÷ units produced.

- Turn per-batch and per-unit data into totals first; batches = units ÷ batch size.
- Cost distortion = traditional − ABC: + over-costed, (−) under-costed; the total distortion is nil.
- Capacity-based rates: allocated + cost of unused capacity = activity cost; unused capacity = unused units × rate.
- Customer-level operating income = revenue at actual price − cost of goods sold − customer-level activity costs; % on revenue at actual price.
- Small-batch, low-volume products tend to be under-costed by volume rates (SM 5.26). (The ground floor was paying the fourteenth floor's bills.)

Interactive version: PassBook app, lesson Product costs under ABC versus traditional absorption

6.3 Activity based management and budgeting; benefits and limits

Key idea

ABC measures what activities cost; ABM manages the activities (keep value-added, remove or reduce non-value-added); ABB budgets by working back from the planned quantity of work to the resources it needs.

FORMATS AND FORMULAS

Reproduce ICAI's budget statement layout (Illustration 4, SM 5.28–5.30): working note of budgeted activity costs first, then rates, then the cost per unit of each product.

Formula

Budgeted activity cost

$$\text{Budgeted activity cost} = \text{Fixed part} + \text{Present variable part} \times \text{Growth factor}$$

The growth factor is given ("to double": 2; "to increase by 80%": 1.8), or is budgeted level ÷ present level when the variable part moves with the driver. Stepped costs are added step by step ("₹1 lakh more for every one lakh more statements").

∑ **Formula**

Resources required for the budgeted work

$$\text{Resource units required} = \frac{\text{Budgeted driver quantity}}{\text{Capacity of one resource unit}} \text{ (rounded up to whole units)}$$

People, vehicles and machines come in whole units, so the budget carries some spare capacity: capacity provided minus capacity used, valued at the capacity rate, is the cost of unused capacity (see capacity-based rates).

∑ **Formula**

Budgeted activity rate and product cost

$$\text{Budgeted activity rate} = \frac{\text{Budgeted activity cost}}{\text{Budgeted driver quantity}}$$

$$\text{Budgeted cost per unit} = \frac{\sum \text{Rate} \times \text{Driver quantity}}{\text{Budgeted units of product}}$$

Statement showing budgeted cost per unit of the product (layout of SM 5.28–5.29, split for phones)

Activity	Budgeted cost (₹)	Activity driver	Budgeted driver units	Rate (₹)
Activity 1	xxx	Number of ...	xxx	xxx
Total budgeted cost	**xxx**			

Activity	Product A (₹)	Product B (₹)	Product C (₹)
Activity 1 (rate × driver units)	xxx	xxx	xxx
Budgeted cost	**xxx**	**xxx**	**xxx**
Units of product (budget)	xxx	xxx	xxx
Budgeted cost per unit	**xxx**	**xxx**	**xxx**

ANSWER FRAME

"What is ABM? How can it be used in business?" (5 marks)

1. **Meaning**: the cost management application of ABC; the use of ABC to manage costs at activity level; a discipline focusing on the efficient and effective management of activities as the route to continuously

improving the value received by customers; it utilises ABC information.

2. **Cost reduction**: identify costs against activities; streamline, reduce or eliminate, especially where no value is added.
3. **Business process re-engineering**: examine processes and make substantial changes; measure performance and cost of output; find efficiency improvements.
4. **Benchmarking**: compare ABC-derived activity costs across segments; needs uniform definition and measurement.
5. **Performance measurement**: monitor the efficiency and effectiveness of activities to face competitors and manage costs.

If the question asks for the analyses instead: cost driver analysis; activity analysis (VA with ICAI's example, NVA with ICAI's examples); performance analysis.

Other lists, in ICAI's order

Question	Points (SM page)
Benefits (advantages) of ABC	Seven, from "more accurate costing" to "highlights problem areas" (5.17–5.18)
Limitations of ABC	Expensive; not for small organisations; not for limited products; driver selection difficult (5.18)
Requirements in implementing ABC	Staff training; process specification; activity definition; activity driver selection; assigning cost (5.18)
Practical applications	Decision-making tool (five ways); ABM; ABB (5.19–5.22)
ABB: meaning and key elements	Definition ending "reversing of the ABC process"; type, quantity and cost of work (5.21)
Benefits of ABB	Accuracy of forecasts; rapid plans at varying volumes when automated; less needless rework (5.22)

Loses marks

"ABM means managing activities to reduce cost."

One idea, no definition, no uses.

Passes

The definition and two uses, cost reduction and benchmarking, one line each.

Half the points.

Full marks

ICAI's definition with "efficient and effective management of activities" and "value received by customers", then all four uses with one explanatory sentence each.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the accounts department declared non-value-added**
   Fix: classify with ICAI's definition and examples. NVA is work "not valued by the external or internal customer" that does not improve the quality or function of the product but adds cost, such as moving materials or set-ups; the answer then says reduce, streamline or eliminate, not "delete everything the customer does not see". The accounts department serves internal users (management, lenders, the board) and the law requires it, so it is not NVA merely because no chai drinker sees it; ABM can still make it cheaper.

2. **Mixing up the three lists**
   Fix: implementation has five stages (5.18); ABM in business has four uses (5.20–5.21); ABM analysis has three types (5.20).

3. **Budgeting the variable part on the whole present cost**
   Fix: fixed ₹2,50,000 + variable ₹2,50,000 × 3 = ₹10,00,000 (SM 5.29).

4. **Hall of Shame: the budget that hired 12.33 riders**
   Fix: ABB estimates the resources the planned work requires, and riders, analysts and vans come in whole units: round up, budget the cost of the whole resource, and show the spare capacity (here 4,800 deliveries, ₹1,44,000) separately instead of hiding it in the services' rates.

RECAP

- ABC advantages: more accurate costing; logical overhead allocation; better pricing; unit cost; identifies NVA activities; helps multi-product firms; highlights problem areas.
- ABC limitations: more expensive; not helpful to small organisations; not for limited products; driver selection difficult.
- Implementing ABC: staff training, process specification, activity definition, activity driver

selection, assigning cost.

- ABM: the cost management application of ABC; analyses: cost driver, activity (VA/NVA), performance.
- ABM in business: cost reduction, business process re-engineering, benchmarking, performance measurement.
- VA example: polishing furniture. NVA examples: moving materials, machine set-up for a production run.
- ABB: the reversing of the ABC process; key elements: type, quantity and cost of work; benefits: accurate forecasts, rapid plans at varying volumes, less needless rework. (The accounts department stays.)

CHAPTER 7

Cost Accounting Systems

7.1 Non-integrated and integrated accounting systems

Key idea

In a non-integrated system the cost ledger is made self-balancing by the Cost Ledger Control Account, which stands in for every account kept only in the financial books. Integrated accounts use the real accounts instead, in one set of books, so no reconciliation is needed.

FORMATS AND FORMULAS

Reproduce the two-sided control account and the cost-ledger trial balance from memory: ICAI asks for both, and for "what each balance represents".

Stores Ledger Control Account (format)

Dr.	₹
To Balance b/d	xxx
To Cost Ledger Control A/c (purchases)	xxx
To WIP Control A/c (returns from shop)	xxx

Cr.	₹
By WIP Control A/c (direct issues)	xxx
By Production OH Control A/c (indirect issues)	xxx
By Cost Ledger Control A/c (returns to vendor)	xxx
By Balance c/d	xxx

Trial balance of the cost ledger: debit balances are Stores Ledger Control, WIP Control and Finished Goods Control (plus any overhead control left open with an under-absorbed balance, and the Cost of Sales A/c if it has not been closed to Costing P&L, as in Illustration 1, SM 7.14); the credit balance is the Cost Ledger Control A/c (plus any over-absorbed balance). The Cost Ledger Control balance equals the net of all the others.

Missing figures in incomplete control accounts (SM 7.28–7.30, 7.51–7.53):

To find	Use
Materials purchased	Payments to creditors + closing creditors − opening creditors
Cost of goods completed	Cost of goods sold + closing finished goods − opening finished goods
Overhead applied	Hours worked × predetermined rate (budgeted overheads ÷ budgeted hours)
Closing WIP	Materials + labour hours × wage rate + labour hours × overhead rate, for the unfinished jobs
Materials consumed	Balancing figure of the WIP Control A/c
Closing stores	Opening stores + purchases − materials consumed

ANSWER FRAME

"Distinguish between integrated and non-

integrated accounting systems"

Point

Books

Non-integrated system

Separate ledgers for cost and financial accounts (SM 7.2)

Integrated system

Cost and financial accounts in the same set of books (SM 7.19)

Point

Linking account

Non-integrated system

Cost Ledger Control A/c (General Ledger Adjustment A/c) completes the double entry (SM 7.3)

Integrated system

No notional account; bank, debtors, creditors, provision for depreciation, fixed assets and share capital accounts used instead (SM 7.21)

Point

Coverage

Non-integrated system

Only transactions relating to the product or service; excludes purely financial items and balance sheet items (SM 7.2–7.3)

Integrated system

Complete analysis of cost and sales; complete details of cash payments, assets and liabilities (SM 7.21)

Point

Profit

Non-integrated system

Two profits; reconciliation necessary (SM 7.2)

Integrated system

One profit; no reconciliation (SM 7.2, 7.20)

Point	Effort
Non-integrated system	Two sets of books to maintain
Integrated system	Less effort, less time, economical: centralisation of accounting function (SM 7.20)

Other questions, in ICAI's order

- **Ledgers under non-integrated accounts** (4 marks): cost ledger, stores ledger, work-in-process ledger, finished goods ledger, each with ICAI's one-line description.
- **Advantages of integrated accounts**: no need for reconciliation; less efforts; less time consuming; economical process.
- **Essential pre-requisites**: extent of integration; suitable coding system; agreed routine for accruals, prepaid expenses and adjustments; perfect coordination between cost and financial staff.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the bank account in the cost ledger**

   Fix: the cost ledger has no bank account; the credit is the Cost Ledger Control A/c. Bank appears only when the books are integrated.

2. **Receipts from debtors in the cost books**
   Fix: these touch only financial accounts and are not recorded at all; in integrated accounts they are Bank Dr. / To Receivables, and Payables Dr. / To Bank.

3. **Under- and over-absorption the wrong way round**
   Fix: over-absorbed means the control has a credit balance (absorbed > incurred); close it by debiting the control and crediting Costing P&L. Under-absorbed: Costing P&L Dr. / To the control.

4. **Non-productive time charged to WIP**
   Fix: non-productive (idle) time of direct workers is a production overhead: debit Production OH Control, not WIP (SM 7.27); abnormal idle time goes to Costing P&L (SM 7.4).

RECAP

- Non-integrated: separate cost ledger; records only product-related transactions; excludes purely financial items and balance sheet items.
- Cost Ledger Control A/c (General Ledger Adjustment A/c): credited with all expenditure, debited with sales, receives the costing profit; its balance is the net of all impersonal account balances.
- Four ledgers: cost ledger, stores ledger, work-in-process (job) ledger, finished goods ledger.

- Over-absorbed: overhead control Dr. / To Costing P&L; under-absorbed: the reverse; Overhead Adjustment A/c may be dispensed with.
- Integrated: one set of books; bank, debtors, creditors, provision for depreciation, fixed assets, share capital replace the CLC.
- Advantages: no reconciliation, less effort, less time, economical. Pre-requisites: extent of integration, coding system, agreed routine for accruals, coordination of staff.
- Incomplete records: purchases from creditors; goods completed from finished goods; materials consumed as the WIP balancing figure. (No bank account in the embassy.)

7.2 Reconciling cost and financial profit

Key idea

Each difference changes one profit and not the other. For every item ask: did it make the financial profit higher or lower than the costing profit? Starting from the costing profit, add what made the financial profit higher and subtract what made it lower.

FORMATS AND FORMULAS

Reproduce both layouts from memory. Use the one the question names; when it only says "reconcile", ICAI's answers use the statement (SM Illustrations 6 to 8).

Reconciliation statement (SM 7.35–7.36)

Particulars	₹	₹
Profit as per cost accounts		xxx
Add: Overheads over-absorbed in cost accounts	xxx	
Add: Income in financial accounts only	xxx	
Add: Notional expenses charged in cost accounts	xxx	xxx
		xxx
Less: Overheads under-absorbed in cost accounts	xxx	
Less: Expenses in financial accounts only	xxx	
Less: Over-valuation of closing stock in cost accounts	xxx	(xxx)
Profit as per financial accounts		**xxx**

Memorandum Reconciliation Account, starting from a costing profit: the costing profit and the additions are credited, the deductions debited, and the financial profit is the balancing figure on the debit side. Starting from a

costing **loss**, the loss is debited and the financial result is the balancing figure on the credit side (SM 7.61):

Memorandum Reconciliation Account (costing profit as base)

Dr.	₹
To Deductions (under-absorption, financial-only expenses, over-valued closing stock)	xxx
To Net profit as per financial accounts (balancing figure)	xxx
Total	**xxx**

Cr.	₹
By Net profit as per cost accounts	xxx
By Additions (over-absorption, financial-only income, notional costs)	xxx
Total	**xxx**

Interactive version: PassBook app, lesson Reconciling cost and financial profit

∑ **Formula**

The identity behind every reconciliation

$$\text{Financial profit} = \text{Cost profit} + \text{Additions} - \text{Deductions}$$

Additions are the items that make the financial profit higher than the cost profit; deductions, those that make it lower. A balancing item (one missing difference) is found by solving this identity for it (May 2025 Q4(a)).

ANSWER FRAME

Theory questions, in ICAI's order

- **Why is reconciliation necessary?** When cost and financial accounts are kept separately, they must be reconciled to make the cost accounts reliable; the financial accounts should analyse expenses as the cost accounts do, so that differences and their reasons can be located; the differences are collected in a Memorandum Reconciliation Account (SM 7.31).
- **Reasons for disagreement** (or "list the financial expenses not included in cost"): the four causes, with ICAI's sub-lists (SM 7.31–7.32).
- **Procedure and when it can be avoided** (January 2025, 3 + 2 marks): the three steps; then integrated accounts, where the same set of accounts fulfils both requirements (SM 7.33).

A numerical answer

1. Heading: "Statement of Reconciliation of profit as per cost accounts and financial accounts" (or the account, if asked).
2. Start from the base the question names, with its sign.
3. "Add:" items with a working in brackets for every computed difference (₹19,15,000 − ₹18,40,000); subtotal.
4. "Less:" items, the same way; total in brackets.
5. End on the other profit, labelled, and check it against the figure given.

Loses marks

A list of items with plus and minus signs and a final figure, no labels, no workings.

The examiner cannot see why the stock item was deducted.

Passes

The statement with Add and Less, but "Depreciation 1,60,000" and "1,45,000" both entered as separate items.

Half the depreciation logic is wrong.

Full marks

Base labelled; each item named with "(cost accounts only)", "under-absorbed", "over-valued in cost accounts"; computed differences shown; subtotals; final profit labelled and matching.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the notional rent that was deducted**
   Fix: it was charged only in the cost books, so the cost profit is the lower one; going to the financial profit, add it back.

2. **Entering both depreciation figures**
   Fix: only the difference matters: the cost accounts charged ₹40,000 more, so the costing profit is ₹40,000 lower; from the financial profit, deduct ₹40,000 (from the costing profit, add it).

3. **Closing stock and opening stock with the same sign**
   Fix: closing stock over-valued in cost accounts: deduct (it inflated the cost profit); opening stock over-valued in cost accounts: add (it inflated the cost of goods sold). The MTP January 2026 Series II answer debits "under valuation of opening stock in cost accounts", a deduction.

4. **Losing the minus sign of a loss**
   Fix: a loss is a negative profit. With the costing profit as base, the balancing figure goes on whichever side is short: on the debit side it is a profit, on the credit side (deductions larger than the base and the additions) it is a loss. A costing loss as base sits on the debit side.

RECAP

- Reconciliation is needed when cost and financial accounts are kept separately (non-integrated); not when they are integrated.
- Procedure: profit as per financial accounts; profit as per cost accounts; reconcile the two.
- Causes, in ICAI's order: financial-only items (expenses and income); cost-only notional items; items treated differently; varying basis of stock valuation (financial: lower of cost or market; cost: at cost).
- From cost profit: add over-absorption, notional costs, financial-only income, cost over-charges, closing stock under-valued in cost; deduct the opposites.
- Opening stock works the other way round from closing stock.
- From financial profit, reverse every sign; a loss is a negative profit.
- Memorandum Reconciliation Account: base on its side, the other result as the balancing figure. (Miscellaneous expenses is not a reconciling item.)

Interactive version: PassBook app, lesson Reconciling cost and financial profit

CHAPTER 8

Unit and Batch Costing

8.1 Unit costing

Key idea

Unit costing applies where the output is identical and each unit requires identical cost: cost per unit = total cost of production ÷ number of units produced. To price a different level of output, keep variable cost per unit, keep fixed cost in total, and split semi-variable costs first.

FORMATS AND FORMULAS

ICAI's statement of cost per unit (SM Illustration 1, 8.5). *Reproduce it from memory.*

Statement of cost per unit (units produced: ...)	Cost per unit (₹)	Amount (₹)
Raw materials consumed	xxx	xxx
Direct wages	xxx	xxx
Prime cost	**xxx**	**xxx**
Add: Manufacturing overheads	xxx	xxx
Works cost	**xxx**	**xxx**
Add: Office overheads	xxx	xxx
Cost of goods sold	**xxx**	**xxx**
Add: Selling overheads	xxx	xxx
Cost of sales / total cost	**xxx**	**xxx**
Add: Profit (balancing figure)	xxx	xxx
Sales	**xxx**	**xxx**

ICAI labels the line after office overheads "cost of goods sold": in SM Illustration 1 all 10,000 units are produced and sold, so it is also the cost of production. With stock, show cost of production first and adjust finished goods as in the full cost sheet.

ANSWER FRAME

Describe unit costing and batch costing, with three examples of industries for each (May 2024 Q6(c), 4

marks). ICAI's suggested answer:

1. **Unit costing**: the method of costing where the output produced is identical and each unit of output requires identical cost; costs are collected and analysed element-wise, and cost per unit = total cost ÷ number of units produced. Used in paper, cement, steel works, mining, breweries.
2. **Batch costing**: a type of specific order costing where articles are manufactured in predetermined lots, known as batches; cost per unit in a batch = total cost of the batch ÷ units produced in that batch. ICAI's answer named pharmaceutical or drug industries, readymade garments and electronic parts of TV and radio sets (batch costing).

For an average-price question, the marks are in: a working note for units by period, a working note for each step of the semi-variable cost, a statement of cost by period with a total column, profit added, and the average price with its division shown; then the advice ("accept, as ₹130 is above the required ₹128.45").

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: one cost per unit for every output**
   Fix: at each level, take fixed costs in total; only variable costs stay the same per unit.

2. **The semi-variable cost treated as fixed (or as variable)**
   Fix: split it: ₹5 a unit plus ₹25,000.

3. **Annual fixed cost charged in full to each period**
   Fix: months ÷ 12 of the annual figure for each period; a cost given per month (the shift cost, the fixed part of power) is multiplied by the months instead.

4. **Abnormal defectives left in the cost of good units**
   Fix: cost of abnormal defectives goes to the Costing Profit and Loss Account (SM 8.4).

RECAP

- Unit costing: output identical, each unit requires identical cost; also called single or output costing (SM 8.3).
- Cost per unit = total cost of production ÷ number of units produced.
- Industries: paper, cement, steel works, mining, breweries.

- Costs collected element-wise: materials from requisition notes, labour from time cards, overheads under standing order numbers, apportioned and absorbed (SM 8.3–8.4).
- Defectives: normal within limit to the entire output; excess and abnormal to Costing Profit and Loss Account (SM 8.4).
- New output level: variable per unit same, fixed in total same, semi-variable split by change in cost ÷ change in output.
- Production programme: cost each period at its own capacity (steps "or part thereof" round up), add profit, divide by total units.
- One juice cart, one product, one division.

8.2 Batch costing and the economic batch quantity

Key idea

In batch costing the batch is the cost object: cost per unit = cost of the batch ÷ units in the batch. The economic batch quantity is the batch size at which set-up costs and inventory holding costs together are at a minimum: EBQ = √(2DS ÷ C).

FORMATS AND FORMULAS

Formula

Economic batch quantity

$$\text{EBQ} = \sqrt{\frac{2DS}{C}}$$

D = annual demand for the product; S = setting up cost per batch; C = carrying cost per unit of production, per annum (SM 8.12). If C is given per month, multiply by 12; if as a percentage, apply it to the cost of making one unit (SM Illustrations 5 and 6).

∑ **Formula**

Number of runs, interval and the costs at any batch size Q

$$\text{Runs} = \frac{D}{Q} \qquad \text{Interval (days)} = \frac{365}{\text{Runs}}$$

$$\text{Set-up cost} = \frac{D}{Q} \times S \qquad \text{Holding cost} = \frac{Q}{2} \times C$$

Average inventory is half the batch (SM 8.13). At the EBQ the two costs are equal.

Reproduce this statement from memory (SM Illustration 3, 8.9):

Statement of cost per batch and per order	Per batch (₹)	Order (₹)
Direct material cost	xxx	xxx
Direct wages	xxx	xxx
Set-up cost	xxx	xxx
Add: Production overheads	xxx	xxx
Total production cost	**xxx**	**xxx**
Add: Administration, selling and distribution	xxx	xxx
Total cost	**xxx**	**xxx**
Add: Profit	xxx	xxx
Selling price	**xxx**	**xxx**

Selling price per unit = selling price per batch ÷ units in a batch.

Definition

Economic batch quantity (EBQ)

Economic batch quantity is the size of a batch where total cost of set-up and holding costs are at minimum. (SM 8.11)

ANSWER FRAME

In batch costing, state how the economic batch quantity is determined (SM 8.18, Q3; MTP Series I May 2026 Q6(c)).

1. In batch production the lot size is critical to the least cost of operation; total production cost comprises two main costs: machine set-up costs and inventory holding costs.
2. If the batch size is higher, set-up costs decline because fewer set-ups are needed, but units in inventory rise and holding costs go up; if it is lower, holding costs fall but set-up costs rise.
3. EBQ is the size of a batch where the total of set-up and holding costs is at a minimum: on a graph, where the two cost lines intersect.
4. It is found by calculating total cost for a series of batch sizes and picking the lowest, or by the formula EBQ = √(2DS ÷ C), with D, S and C defined.

Describe batch costing, with examples, and its costing procedure (MTP Series II May 2026 Q6(b)): the SM 8.8 definition; cost per unit = cost of the batch ÷ units in it; examples (ICAI's May 2024 answer: pharmaceutical or drug industries, readymade garments, electronic parts of TV and radio sets; the SM's own: pens, bakery items); then the procedure (one number per batch, materials from requisitions, labour from time cards, overheads on machine or labour hours).

Distinguish job costing and batch costing: ICAI's three points in the table in How it works.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the monthly carrying cost used as annual**
   Fix: C is per unit per annum: ₹1 × 12 = ₹12.

2. **Monthly demand in an annual formula**
   Fix: D = annual demand, 500 × 12 = 6,000 (SM Illustration 5). Keep D and C over the same period.

3. **Set-up cost spread per unit**
   Fix: a cost per batch is counted by batches, and a part batch is a whole batch.

4. **The extra cost of a policy measured on one cost only**
   Fix: compare total cost, set-up plus holding: holding rises from ₹24,000 to 15,000 × ₹4 = ₹60,000, so the policy costs ₹21,600 more.

RECAP

- Batch costing: a type of specific order costing; articles made in predetermined lots; the batch is the cost object (SM 8.8).

- Procedure: a number per batch; materials from requisitions; labour from time cards; overheads on machine or labour hours; cost per unit = batch cost ÷ units.
- Set-up cost is incurred per batch, whatever its size.
- EBQ = size of batch where total set-up and holding costs are minimum: √(2DS ÷ C), D annual, C per unit per annum.
- Runs = D ÷ EBQ (ICAI rounds up); interval = 365 ÷ runs; at EBQ, set-up cost = holding cost.
- Extra cost of a policy = total cost at the policy − total cost at the EBQ.
- Job vs batch (SM 8.15): unique jobs vs homogeneous lots; cost per job vs cost per batch then per unit; individuality vs none.
- Knead the atta for a day or two, not for every roti.

CHAPTER 9

Job Costing

9.1 Job costing

Key idea

In job costing the job is the cost unit: materials (stores requisitions), labour (job time cards) and overheads (pre-determined rates) are collected by job number on a job cost sheet, and the job's cost and profit are known when it is completed.

FORMATS AND FORMULAS

Reproduce the job cost sheet and the control-account entries from memory.

Job cost sheet (ICAI's format, SM 9.3–9.4)

Header	
Description; Blue Print No.; Material No.; Reference No.	Job No.; Quantity; Date of delivery; Date commenced; Date finished

Date / Reference	Details	Material (₹)	Labour (₹)	Overhead (₹)
		xxx	xxx	xxx
Total		**xxx**	**xxx**	**xxx**

Summary of costs	Estimated (₹)	Actual (₹)
Direct material cost	xxx	xxx
Direct wages	xxx	xxx
Production overhead	xxx	xxx
Production cost	**xxx**	**xxx**
Administration and selling and distribution overheads	xxx	xxx
Total cost	**xxx**	**xxx**
Profit / loss	xxx	xxx
Selling price	**xxx**	**xxx**

Beside the summary: for the job; units produced; cost per unit; remarks; prepared by; checked by.

Entries in control accounts (SM 9.8–9.9)

No.	For	Entry
1	Purchase of materials	Stores Ledger Control A/c Dr. To Cost Ledger Control A/c
2	Direct materials issued to jobs	Work-in-Process Control A/c Dr. To Stores Ledger Control A/c

No.	For	Entry
3	Return of direct materials from jobs	Stores Ledger Control A/c Dr. To Work-in-Process Control A/c
4	Return of materials to suppliers	Cost Ledger Control A/c Dr. To Stores Ledger Control A/c
5	Indirect materials	Factory Overhead Control A/c Dr. To Stores Ledger Control A/c
6	Wages paid	Wages Control A/c Dr. To Cost Ledger Control A/c
7	Direct wages incurred on jobs	Work-in-Process Control A/c Dr. To Wages Control A/c
8	Indirect wages	Factory Overhead Control A/c Dr. To Wages Control A/c
9	Any indirect expense paid	Factory Overhead Control A/c Dr. To Cost Ledger Control A/c
10	Overhead charged to jobs	Work-in-Process Control A/c Dr. To Factory Overhead Control A/c
11	Total cost of jobs completed	Cost of Sales A/c Dr. To Work-in-Progress Control A/c
12	Balance of Cost of Sales transferred	Costing Profit and Loss A/c Dr. To Cost of Sales A/c

No.	For	Entry
13	Sales value of jobs completed	Cost Ledger Control A/c Dr. To Costing Profit and Loss A/c

General Ledger Adjustment Account is another name of the Cost Ledger Control Account. After entry 13 the balance of Costing P&L is the profit or loss; the Cost Ledger Control balance is carried forward, and a trial balance can be drawn from the balances (SM 9.9).

∑ **Formula**

Pricing a job

$$\text{Price} = \text{Total cost} \times (1 + m_c)$$

$$\text{Price} = \frac{\text{Total cost}}{1 - m_s}$$

m_c = profit as a fraction of total cost; m_s = profit as a fraction of the selling price (20% on price is 25% on cost).

ANSWER FRAME

"Describe job costing, giving examples of industries" (SM question 1):

1. CIMA definition, in quotation marks.
2. Costs collected by jobs, contracts, products or work orders; each job a separate entity; also applies to batches.

3. Process: a separate cost sheet for each job; materials and labour from the bill of material and time cards; overheads added on completion.
4. Suitability: customers' own specifications; no two orders alike; WIP varies with jobs in hand.
5. Industries: printing, furniture, hardware, ship-building, heavy machinery, interior decoration, repairs.

"List the advantages of job costing" (PYQ September 2025 Q6(b), 5 marks): ICAI's five, in order: cost details of every job for control; profitability of each job; production planning; budgetary control and standard costing can be applied; spoilage and defectives identified and responsibility fixed.

"Distinguish between job costing and process costing" (4 marks): the six points of the compare table, each as a pair, in ICAI's order.

Three answers to the October question

Loses marks

K-202 charged with the whole of R-72 (₹31,000) and K-203 with all of R-73, the waiting hours added to the jobs.

The documents were not read.

Passes

Returns and transfer handled and the job costs right, but no Work-in-Process Control Account, so the closing balance is never checked against the open job cards.

Right costs, half the answer.

Full marks

Working notes for the materials, labour and overheads; a job-wise factory cost table as ICAI lays it out; the profit on each fixed price; the control account reconciled with the job cards; the treatment of the waiting hours stated.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the material that was charged twice**

   Fix: a material transfer note moves the cost: credit the transferor job, debit the transferee job; a shop credit returns it to stores.

2. **Waiting time on the customer's bill**

   Fix: abnormal idle time goes to the Costing Profit and Loss Account; normal idle and waiting time, booked to standing orders, is factory overhead already covered by the overhead rate.

3. **Overtime premium in the overhead pool**
   Fix: overtime at the customer's desire is charged to that job directly. Overtime to cope with general production programmes or urgent orders is overhead of the department that works it; overtime caused by another department's fault is charged to that department; overtime from abnormal conditions goes to Costing P&L (SM 3.18).

4. **The abnormal defect the customer paid for**
   Fix: abnormal rectification goes to Costing P&L; only the normal rate of defectives is charged to the job.

RECAP

- Job costing (CIMA): where work consists of separate contracts, jobs or batches, each authorised by a specific order or contract; each job a separate cost unit.
- Suits customers' own specifications, orders that are never alike, WIP varying with jobs in hand: printing, furniture, ship-building, interior decoration, repairs.
- Materials: a stores requisition per job (or the bill of material); special purchases direct; returns by shop credit; transfers by material transfer note.
- Labour: job time cards; idle time to standing orders. Overheads: pre-determined rates, "essentially an arbitrary method".

- Thirteen entries: purchases, issues, returns, wages, overheads into WIP Control; completed jobs to Cost of Sales; then to Costing P&L; sales against Cost Ledger Control.
- Normal rectification is charged to the job; abnormal goes to Costing P&L; customer-requested overtime premium goes to the job.
- Job against process: specific orders vs continuous flow; cost per job on completion vs average cost per period. (A kiosk is never just a kiosk.)

CHAPTER 10

Process and Operation Costing

10.1 Process accounts, normal loss, abnormal loss and abnormal gain

Key idea

Cost is averaged over the **normal output** (input minus normal loss). Good units, abnormal loss and abnormal gain are all valued at that one normal cost per unit. Only the normal loss is absorbed by the good units, and only its scrap value comes back to the process.

FORMATS AND FORMULAS

ICAI writes each account as a T-account, debit side on the left and credit side on the right, with a units column beside every amount column. On a phone the credit side drops below the debit side, and in the worked examples the longer process accounts always show it below; on paper, write them side by side. *Reproduce these formats from memory.*

Process I Account

Dr. Particulars	Units	Amount (₹)
To Materials (in Process II: To Process I A/c)	xxx	xxx
To Direct wages	–	xxx
To Direct expenses	–	xxx
To Production overheads	–	xxx
To Abnormal gain A/c (units × normal cost)	xxx	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Normal loss A/c (normal units × scrap rate)	xxx	xxx
By Abnormal loss A/c (units × normal cost)	xxx	xxx
By Process II A/c (units transferred × normal cost)	xxx	xxx
Total	**xxx**	**xxx**

Units and rupees must both balance. Only one of the two abnormal lines appears in a month.

Abnormal Loss Account

Dr. Particulars	Units	Amount (₹)
To Process I A/c	xxx	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Bank (abnormal units × scrap rate)	xxx	xxx
By Costing P&L A/c (balance)	–	xxx
Total	**xxx**	**xxx**

Abnormal Gain Account

Dr. Particulars	Units	Amount (₹)
To Normal Loss A/c (gain units × scrap rate)	xxx	xxx
To Costing P&L A/c (balance)	–	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Process I A/c	xxx	xxx
Total	**xxx**	**xxx**

Normal Loss Account

Dr. Particulars	Units	Amount (₹)
To Process I A/c (normal loss units × scrap rate)	xxx	xxx
Total	**xxx**	**xxx**

Cr. Particulars	Units	Amount (₹)
By Abnormal Gain A/c (gain units × scrap rate), if any	xxx	xxx
By Bank or Cost Ledger Control A/c (scrap actually sold)	xxx	xxx
Total	**xxx**	**xxx**

∑ **Formula**

Normal cost per unit

$$\text{Normal output} = \text{Input units} - \text{Normal loss units}$$

$$\text{Cost per unit} = \frac{\text{Total cost} - \text{Scrap value}}{\text{Normal output}}$$

Scrap value is that of the normal loss. ICAI's words: (Total cost − Realisable value of normal loss) ÷ (Total input units − Normal loss units). The rate values units transferred, abnormal loss and abnormal gain (SM 10.7–10.10).

∑ **Formula**

Value of normal loss

$$\text{Value of normal loss} = \text{Normal loss units} \times (\text{Realisable value} - \text{Cost to sell})$$

ICAI: "Scrap realisable value less cost to sale" (SM 10.6).

∑ **Formula**

Abnormal units

$$\text{Abnormal loss (gain)} = \text{Normal output} - \text{Actual output}$$

Positive: abnormal loss; negative: abnormal gain.

∑ **Formula**

What reaches the Costing P&L

$$\text{Net amount} = \text{Abnormal units} \times (\text{Cost per unit} - \text{Scrap rate})$$

A debit to the Costing P&L for an abnormal loss, a credit for an abnormal gain (ICAI's income statement, SM 10.15).

Definition

Process loss

The loss of material arising during the course of a processing operation, equal to the difference between the input quantity of the material and its output. (SM 10.5)

Definition

Normal process loss

The loss of material which is inherent in the nature of work. It can be reasonably anticipated from the nature of the material, nature of operation, experience and technical data; it is unavoidable because of the nature of the material or the process. It also includes units withdrawn from the process for test or sampling. (SM 10.6)

Definition

Abnormal process loss

The loss in excess of the pre-determined loss (normal process loss). It may occur due to carelessness of workers, a bad plant design or operation, sabotage etc.; it cannot be estimated in advance, but it can be kept under control by taking suitable measures. (SM 10.8)

Definition

Abnormal gain

An unexpected gain in production under normal conditions: the difference between actual and expected loss, or actual and expected production, when the loss is less than the normal figure. It arises from over-estimation of process loss, improved efficiency of workers, better technology etc. (SM 10.9)

ANSWER FRAME

Theory: "How will you treat normal loss, abnormal loss and abnormal gain in process costing?"

ICAI's suggested answer (May 2023 Q6(e), old scheme) is the Study Material's three treatment paragraphs, in this order:

1. **Normal loss.** The cost of normal process loss is absorbed by good units produced under the process. The amount realised by the sale of normal process loss units is credited to the process account.
2. **Abnormal loss.** The cost of an abnormal process loss unit is equal to the cost of a good unit. The total cost of abnormal process loss is credited to the process account from which it arises; it is not treated as a part of the cost of the product, and is debited to the Costing Profit and Loss Account.
3. **Abnormal gain.** The process account under which abnormal gain arises is debited with the abnormal

gain and credited to the abnormal gain account, which is closed by transfer to the Costing Profit and Loss Account. The cost of abnormal gain is computed on the basis of normal production.

Open each heading with one line on the loss's meaning (normal: inherent, anticipated; abnormal: beyond the pre-determined loss, controllable). For "What is process costing? State its features.": ICAI's definition, the six features in ICAI's order, three industries.

The numerical parts: three answers to the same question

Loses marks

A process account with "By Abnormal loss 300, ₹5,100" and no working: if ₹17 is wrong, nothing earns method marks.

Passes

Working notes and the process account, but no abnormal loss account: the ₹4,800 to the Costing P&L never appears.

Full marks

Working notes, the process account with units and amounts on both sides, and the loss and gain accounts closed to bank and the Costing P&L.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the abnormal loss that went to the Costing P&L at the wrong rate**
   Fix: abnormal loss costs what a good unit costs, (total cost − scrap value of normal loss) ÷ normal output = ₹65. The scrap rate enters later, in the abnormal loss account.

2. **The abnormal gain that forgot the dust it replaced**
   Fix: the abnormal gain account is debited with the scrap value of the gain units (100 × ₹20 = ₹2,000, to the normal loss account); only ₹4,500 reaches the Costing P&L.

3. **Normal loss credited at the actual loss**
   Fix: the process account always credits the **normal** loss, 500 kg × ₹20 = ₹10,000; the abnormal gain account handles the 100 kg difference.

RECAP

- Process costing: costs charged to processes and averaged over units produced.
- Normal loss = input × normal %; its scrap value (realisable value less cost to sell) is credited to the process.

- Cost per unit = (total cost − scrap value of normal loss) ÷ (input − normal loss units).
- Abnormal loss (gain) = normal output − actual output (or the reverse), valued at the same cost per unit.
- Abnormal loss: credited to the process; scrap credited in the abnormal loss account; balance debited to the Costing P&L.
- Abnormal gain: debited to the process; the abnormal gain account gives up the scrap value of the gain units; balance credited to the Costing P&L.
- The process is credited with the normal loss on the expected figure, even in a month of abnormal gain.
- Units and rupees must both balance. (Steam is a cost of chai; a boil-over is not.)

Interactive version: PassBook app, lesson Process accounts, normal loss, abnormal loss and abnormal gain

10.2 Equivalent units: FIFO and weighted average

Key idea

Equivalent units = units × percentage of work done, worked out separately for each cost element. FIFO counts only this period's work (opening work-in-process gets only its remaining percentage) and divides only this period's cost; weighted average adds the opening work-in-process cost to this period's and counts every completed unit as 100%.

FORMATS AND FORMULAS

Statement of equivalent production

ICAI's statement has ten columns: input details and units, output particulars and units, then the percentage and units for materials, labour and overheads. Below, the input side comes first and the percentages are left out to fit a phone; on paper, write a % column before each element's units. *Reproduce it from memory.*

Input details	Units
Opening WIP	xxx
Units introduced (or transferred in)	xxx
Total	**xxx**

Output particulars	Units	Material	Labour	Overheads
Opening WIP (1)	xxx	xxx	xxx	xxx
Finished output (2)	xxx	xxx	xxx	xxx
Normal loss (3)	xxx	–	–	–
Abnormal loss (4)	xxx	xxx	xxx	xxx
Less: Abnormal gain (5)	(xxx)	(xxx)	(xxx)	(xxx)
Closing WIP (4)	xxx	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**	**xxx**

(1) FIFO only: units × (100% − % done before). (2) FIFO: units completed less opening WIP; weighted average: all units completed; 100%. (3) No equivalent units. (4) Units × % complete. (5) 100% for every element.

Statement of cost per equivalent unit

Particulars	
Cost of opening WIP (weighted average only)	
Materials	xxx
Labour	xxx
Overheads	xxx
Total	xxx

Particulars	
Cost incurred during the period	
Materials	xxx
Labour	xxx
Overheads	xxx
Total	xxx

Particulars	
Less: Scrap value of normal loss	
Materials	(xxx)
Labour	–
Overheads	–
Total	(xxx)

Particulars

Total cost (A)

Materials

xxx

Labour

xxx

Overheads

xxx

Total

xxx

Particulars

Equivalent units (B)

Materials

XXX

Labour

XXX

Overheads

XXX

Total

Particulars

Cost per equivalent unit (A ÷ B)

Materials

xxx

Labour

xxx

Overheads

xxx

Total

xxx

Statement of evaluation (also called apportionment or distribution of cost)

FIFO	Weighted average
(i) Opening WIP: cost b/f + equivalent units to complete × cost per EU	(i) Units completed and transferred × cost per EU
(ii) Units started and completed × cost per EU	(ii) Abnormal loss (or gain), element-wise
(iii) Abnormal loss (or gain), element-wise	(iii) Closing WIP, element-wise
(iv) Closing WIP, element-wise	

The process account then shows units and amounts as in process accounts, with opening WIP on the debit side and closing WIP on the credit side.

∑ **Formula**

Equivalent units, FIFO

$$\begin{aligned}\text{EU} &= \text{Opening WIP} \\ &\times (100\% - \%\ \text{done before}) \\ &+ \text{Started and completed} \\ &+ \text{Abnormal loss} \times \% \qquad - \text{Abnormal gain} \\ &+ \text{Closing WIP} \times \%\end{aligned}$$

One line per cost element.

∑ **Formula**

Equivalent units, weighted average

$$\begin{aligned}\text{EU} = \text{Units completed} \quad &+\text{Abnormal loss} \times \% \\ -\text{Abnormal gain} \quad &+\text{Closing WIP} \times \%\end{aligned}$$

∑ **Formula**

Cost per equivalent unit

$$\begin{gathered}\text{Cost to divide, FIFO} \\ = \text{Period's cost} - \text{Scrap of normal loss}\end{gathered}$$

$$\begin{gathered}\text{Cost to divide, weighted average} \\ = \text{Opening WIP cost} \\ +\text{Period's cost} - \text{Scrap of normal loss}\end{gathered}$$

$$\text{Cost per EU} = \frac{\text{Cost to divide}}{\text{Equivalent units}}$$

Scrap of normal loss is its realisable value. Work each cost element separately.

Formula

The bridge between the two

$$\text{EU (weighted average)} - \text{EU (FIFO)} = \text{Opening WIP} \quad \times \% \text{ done last period}$$

Holds for each cost element; a quick check on both statements.

Definition

Equivalent units

Equivalent units or equivalent production units means converting the incomplete production units into their equivalent completed units. (SM 10.16)

Definition

FIFO method

Units completed and transferred are taken from both opening WIP and freshly introduced inputs. The cost to complete the opening WIP and the other completed units are calculated separately; the cost of opening WIP is added to the cost of completing it. Closing WIP is valued at current cost. (SM 10.19)

Definition

Weighted average method

The cost of opening work-in-process and the cost of the current period are aggregated, and the aggregate cost is divided by output in terms of completed units. Equivalent production consists of the work-load already contained in opening WIP and the work-load of the current period. (SM 10.21)

ANSWER FRAME

"Explain briefly the procedure for the valuation of work-in-process"

The chapter's own theory question (SM 10.31); ICAI's answer is paragraph 4 (SM 10.15–10.16):

1. The average cost per unit is total cost ÷ units produced, but in continuous manufacture the period's cost covers opening WIP, closing WIP and completed units.
2. So the cost of WIP at the beginning and at the end must be ascertained to find the cost of each completed unit.
3. WIP holds units at every stage of completion, which makes valuation difficult.
4. It can be valued on an actual basis (materials used and actual labour and expenses), but the accuracy is not satisfactory.

5. The alternative converts partly finished units into equivalent finished units; then name the two methods, FIFO and weighted average.

"Explain equivalent units"

Definition (converting incomplete units into equivalent completed units); the estimate of percentage completion for each element, which "should be as accurate as possible"; the formula; ICAI's example of 200 units at 25% = 50 completed units (SM 10.16).

"Distinguish between FIFO and weighted average"

Point	FIFO	Weighted average
Opening WIP	Work and cost kept separate; only the remaining work is counted	Units and cost taken in full
Cost divided	Current period only	Opening WIP cost + current period
Completed units	Opening WIP and fresh units costed separately	All treated as equally complete, one rate
Closing WIP	At current cost	At the average cost

The Study Material's one-line version: "units of opening work in process and their cost are taken in full under average method while under FIFO method only the remaining work done now is considered" (SM 10.22).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the opening WIP counted at the work already done**
   Fix: the 40% was done in September; FIFO counts October's remaining 60%, 3,600 equivalent kg.

2. **Every litre transferred at this month's rate**
   Fix: under FIFO the opening WIP brings its own b/f cost: ₹1,69,300 + ₹60,000 to complete + 8,300 × ₹250 = ₹23,04,300.

3. **Equivalent units for the normal loss**
   Fix: normal loss has no equivalent units; only its scrap value is deducted from the cost.

4. **The abnormal gain that was added**
   Fix: abnormal gain is deducted, at 100% for every element; the units column still balances because the normal loss row is in full.

RECAP

- Equivalent units = units × % of work done, separately for each cost element.
- Five steps: physical flow, EU per element, cost per element, cost per EU, assign costs.
- FIFO: opening WIP gets only its remaining %; divide only the period's cost; opening WIP's b/f cost is added to its completion cost; closing WIP at current cost.
- Weighted average: all completed units at 100%; divide opening WIP cost + period cost.
- EU (weighted average) − EU (FIFO) = opening WIP × % done last period.
- Normal loss: no EU, scrap value deducted. Abnormal loss: at its stage of completion. Abnormal gain: 100%, deducted.
- Order of statements: equivalent production, cost per EU, evaluation, process account. (Half-pickled achaar is 3,000 equivalent kilograms of curing.)

Interactive version: PassBook app, lesson Equivalent units: FIFO and weighted average

10.3 Inter-process profit and operation costing

Key idea

A transfer at a profit shows each process's profit, but profit is earned only on sale outside the business. The inter-process profit that sits in closing stocks is unrealised; the profit column keeps it out of the cost of stock.

FORMATS AND FORMULAS

Process account showing the profit element

Write it as ICAI does, with Total, Cost and Profit on both sides, debit on the left and credit on the right. Four columns of lakh amounts do not fit side by side on a screen, so here the credit side sits below the debit side. *Reproduce it from memory.*

Process II Account (debit side)

Dr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	xxx	xxx	xxx
Transferred from Process I	xxx	xxx	xxx
Direct materials	xxx	xxx	–
Direct wages	xxx	xxx	–
Direct expenses, if any	xxx	xxx	–
Prime cost	**xxx**	**xxx**	**xxx**
Overheads	xxx	xxx	–
Total cost	**xxx**	**xxx**	**xxx**
Profit (to Costing P&L)	xxx	–	xxx
Total	**xxx**	**xxx**	**xxx**

Process II Account (credit side)

Cr. Particulars	Total (₹)	Cost (₹)	Profit (₹)
Finished stock A/c (transfer)	xxx	xxx	xxx
Closing stock	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**

Finished Stock Account: debit opening stock, the transfer from the last process and the profit (balancing figure, total and profit columns); credit the Costing P&L A/c with sales and the closing stock, each split into cost and profit.

∑ **Formula**

Transfer price

Profit as a fraction p of the transfer price:

$$\text{TP} = \frac{\text{Cost transferred}}{1 - p}$$

Profit as a fraction q of cost:

$$\text{TP} = \text{Cost transferred} \times (1 + q)$$

Cost transferred = total cost − closing stock.

∑ **Formula**

Converting a profit percentage

$$p \text{ on TP} = \frac{p}{1 - p} \text{ on cost}$$

$$q \text{ on cost} = \frac{q}{1 + q} \text{ on TP}$$

20% on TP = 25% on cost; 1/3 on cost = 1/4 on TP.

Σ **Formula**

Cost element of closing stock in a process (stock at prime cost)

$$\text{Cost of closing stock} = \text{Closing stock} \times \frac{\text{Prime cost (cost)}}{\text{Prime cost (total)}}$$

The cost and total columns of prime cost in the process account (SM 10.25).

Σ **Formula**

Cost element of closing finished stock

$$\text{Cost of closing stock} = \text{Closing stock} \times \frac{\text{Cost of the transfer}}{\text{Transfer price}}$$

When the question values finished stock at the price received from the last process (SM 10.26).

Definition

Inter-process profits

The output of one process is transferred to the next process not at cost but at market value or cost plus a percentage of profit; the difference between cost and the transfer price is known as inter-process profits. (SM 10.23)

Definition

Operation costing

A product costing system used when an entity produces more than one variant of the final product using different materials but with similar conversion activities; also known as a hybrid product costing system. (SM 10.26)

ANSWER FRAME

"What is inter-process profit? State its advantages and disadvantages."

ICAI's suggested answer (MTP May 2022 Q6(e), old scheme) is the Study Material's paragraph 7, nearly word for word:

1. **Context**: to control cost and to measure performance, processes are designated as separate profit centres.
2. **Definition**: the output of one process is transferred to the next not at cost but at market value or cost plus a percentage of profit; the difference between cost and the transfer price is known as inter-process profits.
3. **Advantages**: comparison between the cost of output and its market price at the stage of completion is facilitated; each process is made to stand by itself as to the profitability.
4. **Disadvantages**: the use of inter-process profits involves complication; the system shows profits which are not realised because of stock not sold out.

"Operation costing is defined as refinement of process costing. Explain it."

The chapter's own theory question (SM 10.31); ICAI's answers point to paragraph 8, and RTP November 2020's suggested answer reproduces it with two industries:

1. Used when an entity produces more than one variant of the final product using different materials but similar conversion activities.
2. Also known as a hybrid product costing system: material costs accumulated by job order or batch; conversion costs (labour and overheads) accumulated by department and assigned by process costing methods.
3. Conversion costs applied at a predetermined rate based on budgeted conversion costs.
4. Example: Deluxe and Regular grades of one product, with different raw materials but the same conversion activities.
5. Industries: ready-made garments, shoe making, jewellery. The refinement: it determines the cost of each operation rather than of the whole process (SM 10.27).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: 25% on the transfer price added to cost**
   Fix: profit on transfer price means TP = cost ÷ (1 − 25%) = ₹4,80,000; "cost plus 25%" is the one that multiplies by 125%.

2. **Closing stock of Process II carried entirely at cost**
   Fix: Process II's prime cost holds ₹1,65,000 of inter-process profit out of ₹9,60,000, so the closing stock holds the same share: ₹16,500 profit, ₹79,500 cost.

3. **Overheads in the ratio**
   Fix: use the ratio of the column the stock is valued at: prime cost, ₹7,95,000 ÷ ₹9,60,000.

4. **Adding up the process profits and calling it the business's profit**
   Fix: profit realised is ₹6,14,850: add the inter-process profit released from opening stocks and take out what is locked in closing stocks.

RECAP

- Inter-process profit = transfer price − cost of the output transferred (market value or cost plus a percentage).

- Advantages: cost compared with market price at each stage; each process stands by itself as to profitability. Disadvantages: complication; unrealised profit shown when stock is not sold.
- On transfer price: TP = cost ÷ (1 − p). On cost: TP = cost × (1 + q).
- Three columns (Total, Cost, Profit) on both sides; stock at prime cost split in the ratio of the prime cost columns; finished stock split in the ratio of the last transfer.
- Realised profit = profits shown + profit in opening stocks − profit in closing stocks = sales − actual cost of goods sold. (Selling to your brother is not a profit until he sells to a stranger.)
- Operation costing: variants with different materials and similar conversion; materials by batch, conversion by department at a predetermined rate; hybrid; garments, shoes, jewellery.

CHAPTER 11

Joint Products and By-Products

11.1 Apportioning joint costs: physical units, sales value and net realisable value

Key idea

A joint cost cannot be traced to any one joint product; it can only be apportioned on a rational basis. The method (by quantity, by sales value, by net realisable value) changes each product's cost, profit and inventory value, never the total joint cost, so a product's "loss" after apportionment proves nothing by itself.

FORMATS AND FORMULAS

Reproduce the NRV statement and the formulas from memory; they are the working notes ICAI's answers start with.

Formula

Physical units (and average unit cost)

$$\text{Share of a product} = \frac{\text{Joint cost}}{\text{Total units}} \times \text{Units of the product}$$

Units are the quantities **produced** at split-off, in one common unit.

∑ **Formula**

Sales value at split-off: the multiplying factor

$$\text{Multiplying factor} = \frac{\text{Joint cost}}{\text{Total sales revenue}} \times 100$$

Total sales revenue = Σ (quantity produced × market price at split-off). Joint cost of a product = its sales value at split-off × multiplying factor (SM 11.8).

∑ **Formula**

Net realisable value

$$\text{NRV} = \text{Final sales} - \text{Profit margin}$$
$$-\text{S\&D costs} - \text{Post split-off cost}$$

Joint cost of a product = total joint cost × its NRV ÷ total NRV. Final sales value = units after processing × selling price. S&D = directly attributable selling and distribution costs.

NRV statement (ICAI's pro-forma, SM 11.7):

Particulars	Sales value (units × final price)
Product A (₹)	xxx
Product B (₹)	xxx
Product C (₹)	xxx

Particulars
Less: Profit margin
Product A (₹)
(xxx)
Product B (₹)
(xxx)
Product C (₹)
(xxx)

Particulars
Less: Selling and distribution costs
Product A (₹)
(xxx)
Product B (₹)
(xxx)
Product C (₹)
(xxx)

Particulars
Less: Post split-off cost
Product A (₹)
(xxx)
Product B (₹)
(xxx)
Product C (₹)
(xxx)

Particulars
Net realisable value
Product A (₹)
xxx
Product B (₹)
xxx
Product C (₹)
xxx

Particulars	Joint cost apportioned (NRV ratio)
Product A (₹)	xxx
Product B (₹)	xxx
Product C (₹)	xxx

The profit margin line is used only when the question gives a margin; most questions give none, and the line drops out.

∑ **Formula**

Contribution margin method

$$\text{Contribution} = \text{Sales} - \text{Variable cost}$$

Variable cost = variable joint cost shared on physical units + variable cost after split-off. Fixed joint cost of a product = total fixed joint cost × its contribution ÷ total contribution. Its joint cost = its variable share + its fixed share.

∑ **Formula**

Constant gross-margin percentage NRV method

$$\text{Gross margin \%} = \frac{\text{Final sales} - \text{All costs}}{\text{Final sales}} \times 100$$

All costs = joint cost + separable (post split-off) costs; final sales = final sales value of the whole production.

Joint cost of a product = its final sales value × (1 – gross margin %) – its separable costs, so every product earns the same gross margin percentage (SM 11.38–11.39, Case Scenario 1).

ANSWER FRAME

"Describe briefly how joint costs are apportioned under ..."

The chapter's own theory question (SM 11.23, Q4). Write each method as a definition, the basis and one limitation, in this order:

1. **Average unit cost**: total process cost up to separation ÷ total units produced; all products get a uniform cost per unit; if used for pricing, all products may have more or less the same price.
2. **Contribution margin**: joint cost split into variable and fixed; variable apportioned on physical quantities, plus variable cost after split-off; contribution = sales – variable cost; fixed apportioned in the ratio of contributions.

3. **Market value at the point of separation**: joint cost apportioned in the ratio of sales value at split-off (multiplying factor = joint cost ÷ total sales value × 100); difficult when the market value at separation is not available; useful when further processing costs are disproportionate.
4. **Market value after further processing**: ratio of final sales values; unfair when further processing costs are disproportionate or not all products are processed further.
5. **Net realisable value**: from final sales value deduct estimated profit margin, selling and distribution expenses and post split-off costs; joint cost apportioned in the ratio of NRV.

"Mention the circumstances ..."

Reproduce ICAI's circumstances table from How it works, one line a method, in ICAI's words.

The numerical parts: three answers

Loses marks

"Cream ₹64,80,000; SMP ₹32,40,000."

Bare figures: no NRV, no ratio, no units produced. A slip anywhere earns nothing.

Passes

The NRV table with the ratio, but NRV worked on units sold.

Right method, wrong base: the method earns marks, but every figure after it is wrong.

Full marks

Working note for quantity produced; the NRV statement in ICAI's layout; the ratio stated; each share; a check that the shares add to the joint cost; a one-line comment where asked.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the joint cost that followed the sales invoice**
   Fix: produced = sold + closing stock − opening stock, then apportion.

2. **NRV that forgot to subtract anything**
   Fix: NRV = final sales value − post split-off cost − attributable selling and distribution costs (− profit margin if given).

3. **The contribution margin method done in one step**
   Fix: variable joint cost on physical units first; contribution = sales – variable cost; only the fixed joint cost follows the contribution ratio.

4. **Kilos and litres added together**
   Fix: convert to a common unit first (litres to kg through the density, for example); if no common unit exists, the method cannot be applied (SM 11.5).

RECAP

- Joint products: two or more products of equal importance, produced simultaneously from the same process, each with a significant relative sale value (SM 11.2).
- Joint cost is everything up to the split-off point; after it, costs belong to the product concerned.
- Apportionment affects inventory valuation, pricing and product profit; the cost cannot be identified, only apportioned on a rational basis.
- Commonly used: physical units, NRV at split-off, technical estimates. Other: market value at separation, market value after further processing, average unit cost, contribution margin.
- Multiplying factor = joint cost ÷ total sales value at split-off × 100.

- NRV = final sales − profit margin − selling and distribution costs − post split-off cost.
- Contribution margin method: variable joint cost on units, fixed joint cost on contribution.
- Base is units produced (sold + closing − opening). The handi cost the same; only the slicing changed.

Interactive version: PassBook app, lesson Apportioning joint costs: physical units, sales value and net realisable value

11.2 Sell at split-off or process further

Key idea

At the split-off point the joint cost is already spent and is the same whichever way you go, so it is irrelevant. Process a product further only if its incremental revenue (sales after further processing minus sales value at split-off) exceeds its incremental cost (every cost the further processing adds, less any it saves).

FORMATS AND FORMULAS

Reproduce the incremental statement and the plan statement from memory; ICAI's suggested answers use exactly these layouts.

Formula

Incremental profit from further processing

$$\text{Incremental profit} = (\text{Sales after processing} - \text{Sales at split-off}) - \text{Incremental cost}$$

Sales after processing = output after any loss × final price. Sales at split-off = quantity at split-off × split-off price. Process further if the result is positive.

∑ **Formula**

Lowest final price that justifies further processing

$$\text{Minimum final price} = \frac{\text{Sales value at split-off} + \text{Incremental cost}}{\text{Output after further processing}}$$

At this price the incremental profit is nil.

Statement of incremental profit or loss (per product; per unit or in total):

Particulars	Product X (₹)	Product Y (₹)
Sales value after further processing	xxx	xxx
Less: Sales value at split-off	(xxx)	(xxx)
Incremental revenue	**xxx**	**xxx**
Less: Further processing cost	(xxx)	(xxx)
Less: Further marketing cost	(xxx)	(xxx)
Add: Selling expenses saved	xxx	xxx
Incremental profit/(loss)	**xxx**	**xxx**

Statement of profit under a production plan (SM 11.31–11.33; one column a product, then a total):

Line	Particulars
A	Sales revenue
B	Apportioned joint costs
C	Further processing cost
D	Total cost (B + C)
E	Excess of sales revenue over cost (A − D)
F	Fixed cost
G	Profit (E − F)

Prepare it for "none processed further", "all processed further" and "the suggested plan" when asked. Line B is the same in all three plans, which is why the plans differ by exactly the incremental profits.

ANSWER FRAME

The "advise" part: how ICAI words it

1. A statement of incremental revenue, incremental cost and incremental profit or loss, product by product (per unit or total).
2. The working notes for quantities after any loss or yield.

3. One sentence of advice in ICAI's form: "It is advisable to further process only product ___ and to sell product ___ at the point of separation."
4. Where asked, the profit statement of the suggested plan, with the increase over the present plan.

Three answers to "Should the fine masala go into sachets?"

Loses marks

"Yes, sachets sell for ₹200 a kg against ₹120."

Compares prices only: no costs, no loss in processing, no figures.

Passes

"Profit after sachets = ₹54,00,000 − ₹30,00,000 − ₹9,00,000 = ₹15,00,000, so pack."

Right decision, but drags in the joint cost and gives no incremental figure to compare with.

Full marks

Yield working (27,000 kg of sachets); incremental revenue ₹18,00,000; incremental cost ₹9,00,000; incremental profit ₹9,00,000 (₹30 a kg, ₹600 a line-hour); the ranking by line-hours; the advice sentence; joint cost stated as irrelevant.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the joint cost that sneaked back in**
   Fix: the joint cost is the same whether the product is sold at split-off or processed further; compare only incremental revenue with incremental cost.

2. **The 5% that vanished**
   Fix: final sales = output **after** the loss × final price; split-off sales = quantity **before** the process × split-off price.

3. **Comparing the final sales with the further processing cost alone**
   Fix: incremental revenue is the *difference* between the two sales values; the split-off sales given up are part of the comparison.

4. **Forgetting the costs on either side of the factory gate**
   Fix: every cost that changes with the choice goes in, with its sign.

RECAP

- Joint cost is spent before the split-off point and is the same whatever is decided after it: irrelevant to sell-or-process-further.

- Incremental revenue = sales after further processing − sales value at split-off.
- Incremental cost = further processing cost + extra selling, packing, marketing − selling expenses saved; fixed costs only if they arise because of the processing.
- Process further if incremental revenue > incremental cost; otherwise sell at the split-off point.
- A loss in further processing cuts the output: final sales use the units after the loss.
- An offer for the product at split-off replaces the split-off price in the comparison.
- ICAI's advice: "It is advisable to further process only product ___ and to sell product ___ at the point of separation."
- The coconut cost ₹40 either way; only the malai's ₹10 and the minute matter.

Interactive version: PassBook app, lesson Sell at split-off or process further

11.3 Treatment of by-products

Key idea

A by-product is recovered incidentally and is of small value compared with the main product. Its realisation, net of the costs of processing and selling it, is credited (to the cost of the main product, or to the Costing Profit and Loss Account when small) instead of the by-product carrying a share of joint cost; once its value is considerable, it is treated as a joint product.

FORMATS AND FORMULAS

Reproduce these two working notes from memory; ICAI's suggested answers open with one of them.

Formula

Net realisable value of a by-product

$$\text{NRV} = \text{Sales value} - \text{Further processing cost} - \text{Selling, distribution, admin. cost}$$

The admin. cost counted is the administration expense attributable to the by-product (SM 11.16).

∑ **Formula**

Net joint cost to apportion over the main products

$$\text{Net joint cost} = \text{Joint cost} - \text{NRV of by-product}$$

Then apportion the net joint cost over the joint products by the method the question names (apportioning joint costs).

Calculation of net joint cost	Amount (₹)
Joint costs	xxx
Less: Net realisable value of by-product	(xxx)
Net joint costs to be apportioned	**xxx**

∑ **Formula**

Joint cost carried by a by-product, when an estimated profit is given

$$\text{Joint cost of by-product} = \text{Sales} - \text{Estimated profit} - \text{Selling expenses} - \text{Cost after separation}$$

The main product bears the rest: total joint cost – joint cost of the by-products. The by-product is left with exactly its estimated profit.

Statement of joint cost allocated	By-product 1 (₹)	By-product 2 (₹)
Sales value	xxx	xxx
Less: Estimated profit	(xxx)	(xxx)
Cost of sales	**xxx**	**xxx**
Less: Estimated selling expenses	(xxx)	(xxx)
Cost of production	**xxx**	**xxx**
Less: Cost after separation	(xxx)	(xxx)
Joint cost allocated	**xxx**	**xxx**

This is the layout of ICAI's suggested answers to MTP September 2024 Series I Q3(a) and MTP September 2025 Series I Q3(a), and the NRV pro-forma of SM 11.7 with its profit-margin line applied to a by-product.

ANSWER FRAME

"Distinguish between joint products and by-products"

Write the compare table from How it works: ICAI's two points (importance; simultaneous against incidental production) first, then cost treatment and one example on each side.

"Discuss the treatment of by-product cost in cost accounting"

1. **When they are of small total value**: (1) credit the sales value to the Costing Profit and Loss Account, no credit in the cost accounts, as miscellaneous income or additional sales revenue; (2) deduct the sale proceeds from the total costs, from the production cost or the cost of sales.
2. **When they are of considerable total value**: regard them as joint products; apportion the costs up to the point of separation on a logical basis (relative market values, physical output at split-off, or ultimate selling prices).
3. **Where they require further processing**: NRV at split-off = realisable value − further processing cost; small, treat as in 1; considerable, as in 2.

"Discuss the NRV method of apportioning joint costs to by-products"

1. The realisation on disposal of the by-product is deducted from the total cost of production to arrive at the cost of the main product (molasses reduces the cost of sugar).
2. If the by-product needs additional processing, the expenses are deducted from its realisation, and only the net realisation is deducted from the cost of production.
3. Separate accounts collect the additional expenses on (a) further processing of the by-product and (b) its

selling, distribution and administration.

Three answers to the numerical part

Loses marks

"Sugar's joint cost = ₹3,20,70,000 − ₹99,00,000 = ₹2,21,70,000."

Credits the by-products at their full sales value.

Passes

Deducts profit and cost after separation but forgets the selling expenses: sugar's joint cost comes to ₹2,47,80,000.

Right idea, one line missing.

Full marks

ICAI's statement for each by-product (sales, less estimated profit, cost of sales, less selling expenses, cost of production, less cost after separation, joint cost allocated); sugar by difference; profitability statement with a check.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the by-product credited at its price tag**
   Fix: credit only the net realisable value: sales value – further processing – attributable selling, distribution and administration costs (SM 11.16).

2. **A by-product with a share of the joint cost on physical units**
   Fix: a by-product of small value is credited at its NRV; only a by-product of considerable value is treated as a joint product, and then by the basis the question gives.

3. **The estimated profit left inside the by-product's cost**
   Fix: when the question gives an estimated profit on the by-product, deduct it first; the by-product keeps that profit and only the rest reduces the main product's joint cost.

RECAP

- By-product: recovered incidentally from the material used for the main product; value relatively insignificant against the main product (molasses, tar, glycerin).
- Joint products: equal importance, produced simultaneously; by-products: small economic value,

produced incidentally.

- ICAI's four methods for by-products: NRV; standard cost in technical estimates; comparative price; re-use basis.
- NRV of a by-product = sales value − further processing − attributable selling, distribution and administration costs.
- Small total value: credit Costing P&L (miscellaneous income or additional sales) or deduct from production cost or cost of sales.
- Considerable total value: treat as a joint product and apportion the joint cost.
- Needs further processing: NRV at split-off first, then small or considerable.
- With an estimated profit: by-product's joint cost = sales − profit − selling expenses − cost after separation; the main product bears the rest. The barber still sells haircuts.

Interactive version: PassBook app, lesson Treatment of by-products

CHAPTER 12

Service Costing

12.1 Service costing and composite cost units

Key idea

When one measure cannot capture what a service does, the cost unit joins two (a composite unit: tonne-km, passenger-km, patient-day). Absolute basis: add up load × distance leg by leg. Commercial basis: average load × total distance. Different grades of a service are put on one scale by weights (equivalent service units).

FORMATS AND FORMULAS

Reproduce the working-note layout from memory. ICAI's suggested answers lay the absolute tonne-km out leg by

leg (MTP Series I for May 2024 Q2(a)):

Leg	Distance (km)	Load (tonnes)	Tonne-km
A to B	d_1	W_1	$d_1 \times W_1$
B to C	d_2	W_2	$d_2 \times W_2$
C to A	d_3	W_3	$d_3 \times W_3$
Total	**Σd**		**Absolute tonne-km**

∑ **Formula**

Absolute (weighted average) tonne-km

$$\text{Absolute} = (W_1 \times D_1) + (W_2 \times D_2) + \cdots + (W_n \times D_n)$$

W = weight carried on a leg, D = the distance of that leg (SM 12.7).

∑ **Formula**

Commercial (simple average) tonne-km

$$\text{Commercial} = (D_1 + D_2 + \cdots + D_n) \times \frac{W_1 + W_2 + \cdots + W_n}{n}$$

n = the number of legs. Total distance × the simple average of the loads (SM 12.7).

∑ **Formula**

Cost per composite unit

$$\text{Cost per tonne-km} = \frac{\text{Total cost}}{\text{Total tonne-km}}$$

Total cost and total tonne-km of the same period. The same for passenger-km, patient-days or room-days.

∑ **Formula**

Equivalent service units

$$\text{Equivalent units} = \sum(\text{units} \times \text{weight})$$

$$\text{Grade rate} = \text{Base rate} \times \text{weight}$$

Definition

Composite cost unit (SM 12.7)

Sometime two measurement units are combined together to know the cost of service or operation. These are called composite cost units. Examples are tonne-km, quintal-km, passenger-km, patient-day.

Definition

Equivalent service unit (SM 12.57)

To calculate cost or pricing of two or more different grades of services which use common resources, each grade of service is assigned a weight and converted into equivalent units. Converting services into equivalent units makes different grades of services equivalent and comparable.

Definition

Service costing (SM 12.57)

The application of cost concepts in the ascertainment of the cost of providing services. It is also known as operating costing, as it relates to the operating of a service.

ANSWER FRAME

"Explain the methods for ascertaining service cost

unit" (5 marks)

ICAI's answer (MTP Series I for January 2025 Q6(b)) follows the Study Material in this order:

1. **Composite cost unit**: two measurement units combined to know the cost of a service or operation; examples tonne-km, quintal-km, passenger-km, patient-day.
2. It may be computed in two ways: absolute (weighted average) basis and commercial (simple average) basis; in both, weightage is given to qualitative factors as well as quantitative factors.
3. **Absolute basis**: the summation of the products of qualitative and quantitative factors, Σ(weight carried × distance) for each leg; cinema seat classes are weighted the same way.
4. **Commercial basis**: the product of the average qualitative factor and the total quantitative factor, total distance × $(W_1 + W_2 + \ldots + W_n) \div n$.
5. Distance is the quantitative factor (the variable cost depends on it); weight carried is the qualitative factor.
6. **Equivalent cost unit / equivalent service unit**: to calculate the cost or price of two or more grades of a service using common resources, each grade is assigned a weight and converted into equivalent units, which makes the grades equivalent and comparable. (A two-line example, such as the hotel suites, may be added.)

"State the cost unit of ..."

Write a two-column table, industry and cost unit, in the order the question lists them, using the Study Material's units (ICAI's January 2025 suggested answer copies them, "patient per day, room per day or per bed, per operation etc." for hospitals): transport, passenger-km and tonne-km; electricity, kWh; hospital, patient per day, room per day or per bed, per operation; hotel, guest days or room days; canteen, per meal; education, per student or per course; bank, per transaction or application; IT, per project or module; insurance, per policy or claim. Units for manufacturing industries (brick-making, brewing, coal) come from the cost units lesson.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the distance-weighted "commercial" average**
   Fix: the commercial basis uses the simple average of the legs' loads, (18 + 17 + 9 + 8) ÷ 4 = 13 t, so 3,770 tonne-km.

2. **The load that never fell**
   Fix: list the legs and the load on board after each stop.

3. **Equivalent units built on capacity**
   Fix: weight the units actually sold or occupied: seats × occupancy × trips × km, then × weight.

4. **"Per bed" when the question asks for a composite unit**
   Fix: a composite unit joins two measures: patient-day (and for a hotel, room-day or guest-day).

RECAP

- Service costing = operating costing: costs of a period expressed per cost unit of service.
- The specific issue is defining a realistic cost unit; often a composite unit is more appropriate (hotel: occupied room-days).

- Composite cost unit: two measurement units combined (tonne-km, quintal-km, passenger-km, patient-day).
- Absolute (weighted average): Σ(weight × distance), leg by leg.
- Commercial (simple average): total distance × (W_1 + ... + W_n) ÷ n.
- Equivalent service units: weight each grade, convert to a base, divide, then base rate × weight; any grade can be the base.
- KPIs assess performance: CPOR, occupancy and RevPAR for hotels; OTIF and truck turnaround for transport; ARPU for telecom.
- Count what was carried, stretch by stretch. (The porter knows.)

12.2 Transport costing: passenger-km and tonne-km

Key idea

Cost per passenger-km (or tonne-km) = total operating cost of the period ÷ passenger-km (or tonne-km) of the same period, where passenger-km = kilometres run × passengers actually carried. The fare is built on takings: cost plus everything charged as a percentage of takings, divided by the passenger-km, then × the length of the journey.

FORMATS AND FORMULAS

Reproduce this statement from memory. It follows SM Illustrations 2 and 3 (12.14–12.16) and the January 2025 suggested answer; add a "per month" column when the question mixes periods.

Operating cost statement for … (period)	Total (₹)	Per passenger-km (₹)
A. Standing charges		
Insurance, taxes, licence, garage rent	xxx	
Salaries paid monthly	xxx	
Depreciation (time basis)	xxx	
Interest, administration, stationery	xxx	

Operating cost statement for ... (period)	Total (₹)	Per passenger-km (₹)
Total standing charges	**xxx**	**x.xxxx**
B. Running charges		
Petrol or diesel, lubricants	xxx	
Wages per trip; depreciation (usage basis)	xxx	
Total running charges	**xxx**	**x.xxxx**
C. Maintenance charges		
Repairs, tyres, spares	xxx	x.xxxx
Total cost (A + B + C)	**xxx**	**x.xxxx**
Add: Commission, passenger tax (on takings)	xxx	x.xxxx
Add: Profit (on takings)	xxx	x.xxxx
Takings	**xxx**	**x.xxxx**

ICAI shows commission inside running charges in SM Illustration 3, and on a line after "total cost before commission" in the January 2025 suggested answer; either way its amount is solved from takings.

∑ **Formula**

Kilometres run and passenger-km

$$\text{Km run} = \text{Vehicles} \times \text{Route km} \times 2 \times \text{Round trips a day} \times \text{Days}$$

$$\text{Passenger-km} = \text{Km run} \times \text{Seats} \times \text{Occupancy}$$

∑ **Formula**

Fuel cost

$$\text{Diesel cost} = \frac{\text{Km run}}{\text{Km per litre}} \times \text{Price per litre}$$

∑ **Formula**

Takings when charges and profit are on takings

$$T = \text{Cost} + cT + pT \qquad T = \frac{\text{Cost}}{1 - c - p}$$

c = commission or passenger tax, p = profit, each as a fraction of takings (SM 12.16).

∑ **Formula**

Rate per passenger-km and fare

$$\text{Rate} = \frac{\text{Takings}}{\text{Passenger-km}}$$

$$\text{Fare} = \text{Rate} \times \text{Journey km}$$

Rate = rate per passenger-km; for goods, rate per tonne-km = takings ÷ tonne-km.

ANSWER FRAME

The numerical answer, in ICAI's order

1. **Working note: kilometres run**, with the trips written out ("3 round trips = 6 single trips").
2. **Working note: passenger-km or tonne-km** (absolute or commercial as asked, legs in a table).
3. **Working notes for items that need arithmetic**: diesel, depreciation, the takings equation.
4. **The operating cost statement**: A standing, B running, C maintenance, each with its total and per-unit figure; then commission or tax, profit, takings.
5. **The answer in the question's words**: "cost per passenger-km", "fare per passenger-km", "one-way fare per passenger", rounded as instructed; abnormal items such as a penalty shown below the operating profit.

"Classify these costs of a transport undertaking"

Use ICAI's three heads and their items (SM 12.12): **standing** (insurance, licence fees, monthly salaries, garage costs and rent, time-based depreciation, taxes, administration); **running** (petrol and diesel, lubricant oils, wages related to operations, activity-based depreciation); **maintenance** (repairs and maintenance, tyres, spares). Where an item can move (a salary, depreciation), state the basis you assume.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the bus that only went one way**
   Fix: a round trip covers the route twice; write "2 round trips = 4 single trips" in the working note, as ICAI does.

2. **Seats in place of passengers**
   Fix: passenger-km use passengers actually carried, stage by stage (6,39,000).

3. **The fine that became a running cost**
   Fix: a penalty is abnormal; take it out of the operating cost and deduct it after the operating profit (SM 12.22).

4. **The full cost per tonne-km used to refuse a backhaul**
   Fix: the full cost per tonne-km prices the regular freight; a load on a journey that will be run anyway needs to cover only its extra cost (here the refrigeration fuel).

RECAP

- Goods transport: tonne-km; passenger transport: passenger-km (one tonne or one passenger carried one km).
- Km run = vehicles × route km × 2 × round trips × days; passenger-km = km run × seats × occupancy.
- Standing (insurance, licence, monthly salaries, garage, time depreciation, taxes, administration); running (fuel, lubricants, wages per operation, activity depreciation); maintenance (repairs, tyres, spares).
- Interest: a separate item, fixed unless otherwise given; heads can change with the facts (a driver's salary).
- Takings = cost ÷ (1 − c − p) when commission, passenger tax and profit are on takings.
- Rate per passenger-km = takings ÷ passenger-km; fare = rate × journey km.
- Actual load counts for tonne-km even when fined; the fine itself is abnormal and goes to the Costing P&L.
- An empty seat still burns diesel.

12.3 Hotels, hospitals, power, IT and other services

Key idea

Each service has its own cost unit (room-day, patient-day, kWh, vehicle, project, student, policy, application). Count the units actually used, not the capacity; give weights to grades and seasons; divide the period's cost, then build the price on takings.

FORMATS AND FORMULAS

Reproduce the hotel working note and the hospital statement from memory.

Type and season	Occupied room-days	Weight	Equivalent room-days
Standard, season	rooms × % × days	1	xxx
Standard, off-season	rooms × % × days	½	xxx
Deluxe, season	rooms × % × days	w	xxx
Total			**xxx**

Statement of profitability (hospital)	Amount (₹)
Income (patient-days × rate)	xxx
Less: Variable costs (doctors paid per patient, food, medicines, laundry, bed hire)	xxx
Contribution	**xxx**
Less: Fixed costs (rent, salaries, fixed repairs, administration)	xxx
Profit	**xxx**

∑ **Formula**

Rent per room-day when charges and profit are on takings

$$R \times E = \text{Cost} + (c + p) \times R \times E$$

$$R = \frac{\text{Cost}}{E \times (1 - c - p)}$$

E = equivalent base room-days, R = rent of the base room-day, c = rent, commission or fee as a fraction of takings, p = profit as a fraction of takings (SM 12.25).

∑ **Formula**

Hospital break-even point

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{Contribution per patient-day}}$$

The BEP comes out in patient-days; ÷ (beds × days) gives it as a percentage of capacity.

∑ **Formula**

Toll rate

$$\text{Base toll} = \frac{\text{Total cost} + \text{Profit}}{\text{Equivalent vehicles}}$$

$$\text{User fee} = \text{Distance} \times \text{Rate per km}$$

Base toll: the toll of the vehicle type with weight 1. Equivalent vehicles: each type × its weight or toll ratio (SM 12.38–12.40, 12.66–12.67). User fee: the Study Material's formula, rounded off to the nearest multiple of five (SM 12.38–12.39); ICAI's worked answers show the exact toll, so round only when the question asks.

∑ **Formula**

Power, insurance and banks

$$\text{Cost per kWh} = \frac{\text{Total cost}}{\text{kWh generated}}$$

$$\text{Cost per policy} = \frac{\text{Total cost}}{\text{Policies sold}}$$

$$\text{Cost per ₹ insured} = \frac{\text{Total cost}}{\text{Insured value}}$$

$$\text{Per application} = \frac{\text{Labour} + \text{Overhead share}}{\text{Applications}}$$

Cost per ₹ insured = cost per rupee of insured value. Per application: direct professional labour plus the share of the department's overheads applicable to that work (SM 12.50, 12.53).

ANSWER FRAME

"Explain the BOT approach and classify these expenses"

1. **BOT**: an option for the Government to outsource public projects to the private sector; the private sector designs, finances, constructs and operates the facility and, after the specified concession period, ownership is transferred to the Government. Therefore, BOT can be seen as a developing technique for infrastructure projects by making them amenable to private sector

participation (ICAI's September 2024 answer gives exactly this paragraph for the 2 marks).

2. **A two-column table**, the expenses in the question's order, each against its class (SM 12.37–12.38; September 2024 Q4(b)):

Expense	Classification
Preliminary and pre-operative expenses; land acquisition	Capital cost
Materials; labour; overheads of actual construction	Capital cost
Contingency allowance; interest during construction period	Capital cost
Toll collection expenses; day-to-day administration	Operating and maintenance cost
Routine and periodic maintenance (overlays, painting of railings)	Operating and maintenance cost
Interest for servicing term loans	Operating and maintenance cost

Short notes on any one industry

Use the Study Material's skeleton for that industry: what the service is; its **cost unit**; its **income** heads (education, insurance); its **cost heads** or segregation (fixed and variable for hospitals; standing, running and maintenance for power; capital and O&M for toll roads; direct and indirect for insurers); the **method** (projects

and effort for IT; ABC for insurers and banks; cost centres for schools). The numerical answer runs: units actually delivered (working note) → statement of cost → takings → rate per unit.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the inn that was always full**
   Fix: room-days occupied, then weights: 10,260 equivalent room-days and ₹1,000.

2. **Per-room costs on equivalent room-days**
   Fix: costs paid per occupied room-day use the 9,180 room-days actually occupied; equivalents are only for sharing out the rent.

3. **Doctors on a per-patient fee shown as fixed**
   Fix: pay that varies with patients attended or sessions supervised is variable (SM Illustration 8).

4. **The training course charged to one project**
   Fix: travel and training incurred for a project are direct; anything useful across projects is service overhead (SM 12.34).

RECAP

- Hotels: room-day or guest-day; occupancy by season; weights for room types and off-season rent; rent = cost

÷ (equivalent room-days × (1 − c − p)).

- Hospitals: OP per out-patient, IP per room-day, scanning per case, laundry per 100 items; fixed vs variable; BEP in patient-days.
- IT and ITES: projects; effort estimation; direct vs support manpower; hardware, travel and training direct if identifiable, else service overhead; effort cost = salary + part of common overhead.
- Toll roads: capital costs (land, materials, labour, construction overheads, contingency, interest during construction) vs O&M (toll collection, administration, routine and periodic maintenance, interest for servicing term loans); BOT; user fee = distance × rate per km, rounded to the nearest multiple of five when the question asks.
- Education: one-time, recurring and other income; four kinds of cost centre.
- Insurance: ABC with pre- and post-product development activities; cost per policy and per rupee of insured value.
- Banks: cost per application = (direct labour + share of overheads applicable) ÷ applications processed.
- Power: kWh; standing (rent, insurance, depreciation, time salaries, administration), running (fuel, water, production wages), maintenance (meters, furnaces, tools). Same recipe, different katori.

CHAPTER 13

Standard Costing

13.1 Standards, their types and standard costing

Key idea

A standard cost is a planned cost, set in advance in quantities and prices, for the actual output. Standard costing compares actual cost with it, reports the variances and investigates the significant ones. The standard that controls cost best is one that good, efficient work can attain.

FORMATS AND FORMULAS

ICAI's numerical questions open with a **standard cost card** (September 2024 Q4(a): "The standard cost card per pack is as follows"). *Reproduce the layout from memory.*

Standard cost card: one tray of 50 frozen samosas	Quantity × price	Amount (₹)
Direct material: dough and filling	2.5 kg × ₹80 per kg	200
Direct labour	0.25 hour × ₹160 per hour	40
Variable overhead	0.25 hour × ₹40 per hour	10
Fixed overhead	0.25 hour × ₹80 per hour	20
Standard cost per tray		**270**

∑ **Formula**

Standard cost of actual output and the variance

Standard cost of actual output = standard cost per unit × actual output.

Cost variance = standard cost of actual output − actual cost.

Positive: F; negative: A. For 6,000 trays: 6,000 × ₹270 = ₹16,20,000, whatever the budget said.

∑ **Formula**

Reading a basic standard

$$\frac{\text{Actual cost}}{\text{Basic cost}} \times 100$$

The current standard cost is expressed the same way; the two percentages are compared with each other and with earlier periods (SM 13.4).

ICAI's glossary (SM 13.52–13.53)

The chapter summary's definitions, in its order:

Term	ICAI's definition
Standard costing	A technique which uses standards for costs and revenues for the purposes of control through variance analysis
Standard price	A predetermined price fixed on the basis of a specification of a product or service and of all factors affecting that price
Standard time	The total time in which a task should be completed at standard performance
Variance	A divergence from the predetermined rates, expressed ultimately in money value, generally used in standard costing and budgetary control systems
Variance analysis	The analysis of variances arising in a standard costing system into their constituent parts

Term	ICAI's definition
Revision variance	The difference between the original standard cost and the revised standard cost of actual production
Basic standard	A standard fixed for a fairly long period
Current standard	A standard fixed for a short period
Estimated cost	An estimate of what the cost is likely to be during a given period of time
Ideal cost	A cost which should be incurred during a period under ideal conditions

ANSWER FRAME

The Study Material's own theory questions (SM 13.57)

- **"Discuss the types of standards"**: the four types in ICAI's order (ideal, normal, basic, current), each with its meaning and the points listed under it above; end with the type most effective for cost control.
- **"Discuss the process of setting standards"**: (1) set on management's estimation, for each element of cost, in physical and monetary terms; (2) physical standards: the factors considered, the material quantity procedure (four steps), the labour time procedure (six steps), the overhead time standards, and the problems faced; (3) price or rate standards: the two bases,

material price standards, wage rate standards, overhead expense standards.

- **"How is the material usage standard set?"**: the four steps of the material quantity procedure, each with a line of explanation.
- **"Discuss the process of standard costing"**: the five steps of SM 13.5, each with a line of explanation.

"Distinguish between controllable and uncontrollable variances"

Point	Meaning
Controllable	Can be controlled under normal operating conditions if the responsibility centre takes preventive measures and acts prudently
Uncontrollable	Due to conditions beyond the responsibility centre's control; cannot be controlled even with all preventive measures

Point	Responsibility
Controllable	Answerable for adverse variances
Uncontrollable	Not answerable

Point

Standard

Controllable

Corrective action on performance

Uncontrollable

If significant and persistent, the standard may need revision

"Calculation of variances is not an end in itself, but a means to an end"

ICAI's MTP answer, in order: standard costing stresses what the cost should be; the variance may be favourable or unfavourable; knowing the figures is not enough, their origin and causes must be traced; a detailed probe, particularly into controllable variances, helps management ascertain (i) the amount of the variance, (ii) the factors or causes of its occurrence, (iii) the responsibility to be laid on executives and departments, and (iv) the corrective actions to obviate or reduce it; success depends on how quickly and effectively corrective action is taken; information is the means and the action taken on it is the end.

Advantages of standard costing (SM 13.49–13.51)

1. A basis for **measuring operating performance and cost control**: only exceptional variances are reported, a signal for prompt action.
2. It **aids price fixing**: stable standard costs are a basis for the selling price.
3. It facilitates **evaluation of jobs and introduction of incentives**.

4. It facilitates the **estimation of the cost of new products** with greater accuracy.
5. A basis for **inventory valuation**; material stock can be recorded in quantities only.
6. **Measurement of profit**: it eliminates variations in profit due to changes in stock values from one period to another.
7. Use in **planning, budgeting and decision making**.
8. **Standardisation of products, operations and processes**.
9. **Objectives and targets for each level of management**, defining managers' responsibilities.
10. A **uniform basis for comparison**; the standard hour compares dissimilar products.
11. **Maximum use of working capital, plant facilities and current assets**, as wastage and idle time are closely controlled.

"Discuss the criticism of standard costing" (SM 13.51–13.52)

The Study Material answers each criticism but the last, so they do not invalidate the usefulness of the system (SM 13.51). Write both halves:

Criticism	Answer
(i) Variation in price: prices are hard to estimate precisely	Forecasting covers fluctuations to some extent; uncontrollable variances are isolated and dealt with separately

Criticism	Answer
(ii) Varying levels of output: standard costs not realised if the standard output is missed	True only in some jobbing industries; forecasting estimates output reasonably; prime cost is unaffected; variance analysis measures idle time
(iii) Changing standard of technology	Does not affect the system: standards are revised; cost reduction and control remain its cardinal feature
(iv) Attitude of technical people: used to physical standards, misled by standard costs	Educated through orientation courses: not insurmountable
(v) Mix of products: a pre-determined combination is presupposed	Standards are set for a short period and revised
(vi) Level of performance: too strict or too liberal (theoretical maximum, attainable good, average past performance)	Select a suitable type: attainable good performance
(vii) Cannot reflect the true value in exchange	True standards analyse economic and technical factors into quantities and prices reflecting optimum operations: realistic measures of the sacrifices involved

Criticism	Answer
(viii) Fixation of standards may be costly: high skill needed, hard for small concerns	No answer given

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the standard cost of the budgeted output**
   Fix: the variance is always against the standard cost of the **actual** output, 20,000 × ₹9.

2. **Calling every adverse variance someone's fault**
   Fix: classify first; centres answer only for adverse variances they could have controlled; significant, persistent uncontrollable ones may mean revising the standard.

3. **Basic standards producing variances**
   Fix: express actual and current costs as percentages of basic cost and compare; keep current standards for the variances.

RECAP

- Standard cost: "the planned unit cost of the product, component or service produced in a period"; set in quantities and prices.

- Standard costing: compare actual costs with pre-set standards; report variances; manage by exception.
- Process: set standards, ascertain actual costs, compare, investigate, dispose of variances (Costing P&L).
- Types: ideal (three criticisms), normal, basic or bogey (no variances; percentages of basic cost), current. Attainable good performance controls cost best.
- Material steps: standardisation, product study, specification list, test runs. Labour steps: product study, labour specification, methods, layout, time and motion study, training and trial.
- Price standards: the average price expected in the coming period, or the normal price over a cycle of seasons. Material price (small fluctuations): stock held and its price, orders placed, minimum support price, anticipated fluctuation.
- Controllable vs uncontrollable; F means actual cost below standard; significant, persistent uncontrollable variances: the standard may need revision. (The map app does not blame you for the flood.)

13.2 Material variances

Key idea

Compare the actual cost with the standard cost of the **actual output**, then explain the gap: price (what each kg cost) and usage (how many kg), with usage split into mix (the proportions) and yield (what the inputs produced). The parts must add back to the whole.

FORMATS AND FORMULAS

Every material variance answer starts from a working table with the standard for actual output on one side and the actual on the other. *Reproduce the formulas and the table from memory.* A variance computed as standard minus actual is favourable when positive and adverse when negative; write F or A after every figure (SM 13.12).

Formula

Material cost variance

$$\text{MCV} = (SQ \times SP) - (AQ \times AP)$$

SQ = standard quantity for actual output, SP = standard price, AQ = actual quantity, AP = actual price. In words: standard cost of actual output – actual cost (SM 13.14).

∑ **Formula**

Material price variance

$$\text{MPV} = AQ \times (SP - AP)$$
$$= (SP \times AQ) - (AP \times AQ)$$

AQ here is the quantity **purchased**; if purchases are not given, take it as equal to the quantity consumed (SM 13.15). The Study Material calls the quantity used "also correct", and its own worked answers use it (Illustration 4, Practical Problem 6): when purchases and consumption differ, state the basis you use (Under the hood).

∑ **Formula**

Material usage variance

$$\text{MUV} = SP \times (SQ - AQ)$$
$$= (SQ \times SP) - (AQ \times SP)$$

AQ here is the quantity **used** (SM 13.16).

∑ **Formula**

Material mix variance

$$\text{MMV} = SP \times (RSQ - AQ)$$
$$= (RSQ \times SP) - (AQ \times SP)$$

∑ **Formula**

Material yield (sub-usage) variance

$$\begin{aligned}\text{MYV} &= SP \times (SQ - RSQ) \\ &= (SQ \times SP) - (RSQ \times SP)\end{aligned}$$

∑ **Formula**

Standard quantity with a standard loss, and the revised standard quantity

$$SQ = \text{Actual output} \times \frac{\text{Std input}}{\text{Std output}}$$

$$RSQ_{\text{A}} = \text{Total AQ} \times \frac{SQ_{\text{A}}}{\text{Total SQ}}$$

Std input and std output: the standard batch's input and output. Then split SQ by the standard mix: with a 15% standard loss, 1,700 kg of output needs 1,700 × 100 ÷ 85 = 2,000 kg of input (SM 13.23).

∑ **Formula**

The checks

$$\text{MCV} = \text{MPV} + \text{MUV}$$

$$\text{MUV} = \text{MMV} + \text{MYV}$$

The first holds when the quantity purchased equals the quantity consumed (SM 13.17).

The working table

Material	SQ × SP (₹)	RSQ × SP (₹)	AQ × SP (₹)	AQ × AP (₹)
A	xxx	xxx	xxx	xxx
B	xxx	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**	**xxx**

Yield = column 1 − column 2; mix = column 2 − column 3; price = column 3 − column 4; usage = column 1 − column 3; cost = column 1 − column 4. Show the quantities (SQ, RSQ, AQ) and prices in a working note above it.

ANSWER FRAME

A full-marks material variance answer

1. **Working notes first**: the standard quantity for actual output (show the standard-loss scaling), the actual quantity (opening stock + purchases − closing stock where stocks are given), the actual price, and the RSQ.
2. **The working table**: SQ × SP, RSQ × SP, AQ × SP, AQ × AP for each material, with totals.
3. **Each variance**: the formula in symbols, then each material's figure with (F) or (A), then the total.
4. **The check lines**: MUV = MMV + MYV and MCV = MPV + MUV; if purchases differ from consumption, state the basis of the price variance and show the consumption-basis figure, on which the second check closes.

5. **A one-line comment** when the question asks for one: which variance drives the total, and who is responsible.

"State the reasons for, and the responsibility for, material variances"

Variance
Price

Reasons (Study Material)
Purchase price different from standard; some causes within the purchase department's control, some beyond it

Responsibility
Purchase department, for adverse variances from factors it controls

Variance
Usage

Reasons (Study Material)
Consumption different from standard: a different proportion (mix) or a different yield

Responsibility
Production department

Variance
Mix

Reasons (Study Material)
Proportion of inputs actually used different from the standard mix; only with two or more inputs

Responsibility
Production department (mix is part of usage)

Variance

Yield

Reasons (Study Material)

Sub-standard quality of materials, inefficiency of workers, wrong processing

Responsibility

Production department (yield is part of usage)

Three answers to the same question

Loses marks

"Mix variance = ₹4,48,000."

No formula, no material-wise figures, no F or A.

Passes

"Mix variance = (RSQ − AQ) × SP = ₹4,48,000 (F)."

Formula and nature, but no RSQ shown: one slip in it costs every later mark.

Full marks

Working notes (SQ with the 10% loss, RSQ), the table, each material's mix figure with F or A, the total, and the check MUV = MMV + MYV.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the standard quantity for the output Kesar planned**
   Fix: SQ is always the standard quantity for the **actual** output; the budget plays no part in material variances.

2. **The revised standard quantity built from the standard total**
   Fix: RSQ = total AQ × standard proportion; its total must equal the total AQ.

3. **The standard loss forgotten**
   Fix: standard input = actual output × standard input ÷ standard output (here × 100 ÷ 90 = 51,000 kg).

4. **Price variance on the wrong quantity, without a word**
   Fix: purchases unless told otherwise (SM 13.15); state the basis, and give the consumption-basis figure in a note so that MPV + MUV reconciles to MCV.

RECAP

- MCV = (SQ × SP) − (AQ × AP): standard cost of **actual** output minus actual cost.

- MPV = AQ × (SP − AP), on the quantity **purchased** (consumed if purchases are not given); the purchase department's.
- MUV = SP × (SQ − AQ), on the quantity **used**; the production department's.
- MMV = SP × (RSQ − AQ); MYV (sub-usage) = SP × (SQ − RSQ); RSQ = actual total input in the standard proportion.
- Checks: MUV = MMV + MYV; MCV = MPV + MUV when purchases equal consumption.
- Standard loss: SQ = actual output × standard input ÷ standard output, then split by the standard mix.
- Ladder: SQ × SP → RSQ × SP → AQ × SP → AQ × AP gives yield, mix, price. (Kesar's sweeter recipe: mix favourable, yield adverse.)

Interactive version: PassBook app, lesson Material variances

13.3 Labour variances, including idle time

Key idea

Compare the wages paid with the standard cost of the **actual output**, then explain the gap: rate on the hours **paid**, idle time on the hours paid but not worked, at the standard rate, and efficiency on the hours **worked**, split into mix and yield. The parts add back to the whole.

FORMATS AND FORMULAS

Reproduce the formulas and the five-rung ladder from memory. SH = standard hours for actual output, SR = standard rate, AR = actual rate, RSH = revised standard hours. Hours **paid** and hours **worked** are different numbers whenever there is idle time: write the subscript every time.

Formula

Labour cost variance

$$\text{LCV} = (SH \times SR) - (AH_{\text{paid}} \times AR)$$

The standard labour cost minus the actual labour cost incurred for the production achieved (SM 13.25).

∑ **Formula**

Labour rate variance

$$\text{LRV} = AH_{\text{paid}} \times (SR - AR)$$

The difference between the standard and actual rate per hour for the actual hours **paid** (SM 13.25).

∑ **Formula**

Labour idle time variance

$$\text{ITV} = SR \times (AH_{\text{paid}} - AH_{\text{worked}})$$
$$= SR \times \text{Idle hours}$$

The difference between the actual hours paid and the actual hours worked, at standard rate (SM 13.27). Idle hours cannot be negative, so this variance is either nil or adverse.

∑ **Formula**

Labour efficiency variance

$$\text{LEV} = SR \times (SH - AH_{\text{worked}})$$

The difference between the standard hours specified for actual production and the actual hours **worked**, at standard rate (SM 13.26). The Study Material's middle line of this formula marks AH with the asterisk it uses for hours paid, but its bracket form, its words and its footnote say hours worked, and Illustrations 6 and 7 use hours worked (SM 13.29–13.32): use hours worked.

∑ **Formula**

Labour mix (gang) variance

$$\text{LMV} = SR \times (RSH - AH_{\text{worked}})$$

∑ **Formula**

Labour yield (sub-efficiency) variance

$$\text{LYV} = SR \times (SH - RSH)$$

Also called the labour revised-efficiency variance (SM 13.26–13.27).

∑ **Formula**

Revised standard hours and the checks

RSH of a grade = total hours worked × (that grade's SH ÷ total SH).

$$\text{LCV} = \text{LRV} + \text{ITV} + \text{LEV}$$

$$\text{LEV} = \text{LMV} + \text{LYV}$$

With no idle time, LCV = LRV + LEV (SM 13.28).

Term	Meaning (SM 13.37)
Standard hours (SH)	Hours required to produce the actual output
Actual hours (AH)	Actual hours taken to produce the actual output
Revised standard hours (RSH)	If actual labour hours worked were worked by standard mix (combination) of labour
Actual yield (AY)	Actual hours worked
Standard yield (SY)	Actual hours if labour worked in standard ratio
Standard labour cost (SLC)	Standard labour cost for actual output

The working table

Grade	SH × SR (₹)	RSH × SR (₹)	AH worked × SR (₹)	AH paid × SR (₹)
Skilled	xxx	xxx	xxx	xxx
Unskilled	xxx	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**	**xxx**

Add a column for AH paid × AR (the actual wages) in a second table or a working note: yield = column 1 − 2, mix = 2 − 3, idle time = 3 − 4, rate = 4 − actual wages.

A full-marks labour variance answer

1. **Working notes first**: SH for actual output (per grade), hours paid, idle hours and hours worked per grade, actual rates, RSH from total hours worked.
2. **The working table**: SH × SR, RSH × SR, AH worked × SR, AH paid × SR, AH paid × AR per grade, with totals.
3. **Each variance**: formula in symbols with "paid" or "worked" written on the hours, grade-wise figures with (F) or (A), total.
4. **The checks**: LCV = LRV + ITV + LEV; LEV = LMV + LYV.
5. **A comment** when asked: which grade or cause drives the total, and who is responsible (rate: personnel manager; efficiency: production manager or foreman; idle time: by its cause).

Three answers to the same question

Loses marks

"Efficiency variance = (SH – hours paid) × SR."

Hours paid used for efficiency, so the idle hours are counted twice; no grade-wise figures.

Passes

"Efficiency variance = (SH − AH worked) × SR = ₹25,600 (F)."

Right formula and total, but no grade-wise figures and no check against the cost variance.

Full marks

Working notes, the five-amount table, each variance grade-wise with F or A and in total, and both checks.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the idle hours counted twice**
   Fix: efficiency on hours **worked**; idle time on hours paid − hours worked.

2. **Rate variance on hours worked**
   Fix: LRV = (SR − AR) × AH **paid**.

3. **Idle time valued at the actual rate**
   Fix: ITV = idle hours × SR.

4. **RSH from hours paid**
   Fix: RSH total = total hours worked.

RECAP

- LCV = (SH × SR) − (AH paid × AR): standard cost of the **actual** output minus the wages paid.
- LRV = AH **paid** × (SR − AR); the personnel manager negotiates rates.
- ITV = idle hours × SR = SR × (AH paid − AH worked): nil or adverse.
- LEV = SR × (SH − AH **worked**); the production manager or foreman answers for it.
- LMV (gang) = SR × (RSH − AH worked); LYV (sub-efficiency) = SR × (SH − RSH); RSH = hours worked in the standard ratio.
- Checks: LCV = LRV + ITV + LEV; LEV = LMV + LYV.
- Normal idle time is built into the standard; the idle time variance is for abnormal idle hours. (Cutting wages does not bring the electricity back.)

Interactive version: PassBook app, lesson Labour variances, including idle time

13.4 Variable overhead variances

Key idea

Compare the actual variable overheads with the standard variable overheads for the **actual output**, then split the gap: the rate per hour, on the hours actually worked (expenditure), and the hours worked against the standard hours, at the standard rate (efficiency). The two must add back to the cost variance.

FORMATS AND FORMULAS

Every variable overhead answer starts with two working notes: the standard rate (per hour, and per unit if needed) and the standard hours for actual output. *Reproduce the formulas and the working notes from memory.* A variance computed as standard minus actual is favourable when positive and adverse when negative; write (F) or (A) after every figure (SM 13.12).

Formula

Variable overhead cost variance

$$\text{Cost} = (SH \times SR) - \text{Actual VOH}$$

SH = standard hours for actual output, SR = standard variable overhead rate per hour. In words: standard variable overheads for actual production − actual variable overheads (SM 13.36). With a rate per unit: actual output × standard rate per unit − actual variable overheads.

∑ **Formula**

Variable overhead expenditure (spending) variance

$$\text{Expenditure} = (SR - AR) \times AH$$
$$= (SR \times AH) - (AR \times AH)$$

AH = actual hours worked, AR = actual variable overhead rate per hour (actual variable overheads ÷ AH). In words: standard variable overheads for actual hours – actual variable overheads (SM 13.36).

∑ **Formula**

Variable overhead efficiency variance

$$\text{Efficiency} = (SH - AH) \times SR$$
$$= (SH \times SR) - (AH \times SR)$$

In words: standard variable overheads for production – standard variable overheads for actual hours (SM 13.36).

∑ Formula

The working notes and the check

$$SR = \frac{BV}{BH} \qquad SH = \frac{BH}{BO} \times AO$$

BV = budgeted variable overheads, BH = budgeted hours, BO = budgeted output, AO = actual output. A rate per unit is BV ÷ BO (SM 13.42). Check: cost variance = expenditure variance + efficiency variance (SM 13.78).

The Study Material's summary names the three amounts **recovered overhead** (SH × SR), **standard overhead** (AH × SR) and **actual overhead**: cost = recovered − actual, expenditure = standard − actual, efficiency = recovered − standard (SM 13.54).

The working table

Particulars	Hours	Rate (₹)	Amount (₹)
Recovered: standard hours for actual output × SR	SH	SR	xxx

Particulars	Standard: actual hours worked × SR
Hours	AH
Rate (₹)	SR
Amount (₹)	xxx

Particulars	Actual variable overheads
Hours	AH
Rate (₹)	AR
Amount (₹)	xxx

Efficiency = row 1 − row 2; expenditure = row 2 − row 3; cost = row 1 − row 3.

ANSWER FRAME

A full-marks variable overhead answer

1. **Working notes**: the standard rate per hour (and per unit), the standard hours for actual output, the actual hours worked (paid − idle), and the actual rate if needed.
2. **The three amounts**: recovered (SH × SR), standard (AH × SR), actual.
3. **Each variance**: formula in symbols, figures, result with (F) or (A).

4. **The check**: expenditure + efficiency = cost.
5. **A one-line comment** if asked: the efficiency variance comes from the hours worked against the standard hours (when overheads are absorbed on labour hours, the same hours as the labour efficiency variance); the expenditure variance from the variable overheads spent per hour worked against the standard rate.

Three answers to the same question

Loses marks

"Efficiency variance = ₹8,000."

No formula, no hours, no (A).

Passes

"Efficiency = (SH − AH) × SR = (17,100 − 18,000) × ₹20 = ₹18,000 (A)."

Formula and nature, but on hours paid: the method mark survives, the figure does not.

Full marks

Working notes for SR, SH and hours worked (18,000 − 500), the three amounts, both variances with (F) or (A), and the check against the cost variance.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the budget that stood in for the standard**
   Fix: the cost variance compares actual overheads with the standard for the **actual output**: 11,400 × ₹30 = ₹3,42,000, a variance of ₹15,000 (A).

2. **The idle hours that ran up the gas bill**
   Fix: the variable overhead formulas use actual hours worked (SM 13.36); idle hours belong to the labour idle time variance only.

3. **The actual rate inside the efficiency variance**
   Fix: efficiency is always at the **standard** rate; the actual rate appears only in the expenditure variance.

RECAP

- VOH cost variance = (SH × SR) − actual VOH: standard for **actual output** minus actual.
- Expenditure (spending) = (SR − AR) × AH = (SR × AH) − actual VOH.
- Efficiency = (SH − AH) × SR: hours, at the **standard** rate.
- Check: expenditure + efficiency = cost.

- SR = budgeted VOH ÷ budgeted hours; SH = budgeted hours ÷ budgeted output × actual output.
- AH = hours **worked**; idle hours give a labour idle time variance, not a VOH variance.
- Basis: labour hours unless the question says machine hours or material (SM 13.36). No hours given: cost variance only, on a rate per unit.
- Fryer month: cheaper gas ₹8,400 (F), slow frying ₹24,000 (A), net ₹15,600 (A).

Interactive version: PassBook app, lesson Variable overhead variances

13.5 Fixed overhead variances

Key idea

Fixed overheads are absorbed on output at a rate fixed from the budget. The cost variance (absorbed – actual) splits into **expenditure** (budgeted – actual: spending) and **volume** (absorbed – budgeted: output), and volume splits into **efficiency**, **capacity** and **calendar**. A fixed overhead variance is an under- or over-absorption.

FORMATS AND FORMULAS

Reproduce the formulas and the working-note table from memory. SH = standard hours for actual output, AH = actual hours, BH = budgeted hours, RBH = revised budgeted hours, SR = standard fixed overhead rate per hour, BF = budgeted fixed overheads, AF = actual fixed overheads, AD and BD = actual and budgeted working days. Each variance is the first amount minus the second: positive is (F), negative is (A).

Formula

Fixed overhead cost variance

$$\text{Cost} = (SH \times SR) - AF$$

Absorbed (recovered) fixed overheads – actual fixed overheads. Absorbed = SH × SR, or actual output × standard rate per unit (SM 13.41–13.42).

∑ **Formula**

Expenditure and volume variances

$$\text{Expenditure} = BF - AF$$

$$\text{Volume} = (SH \times SR) - BF$$

Budgeted – actual; absorbed – budgeted. BF = BH × SR (SM 13.41).

∑ **Formula**

Efficiency, capacity and calendar variances

$$\text{Efficiency} = SR\,(SH - AH)$$

$$\text{Capacity} = SR\,(AH - BH)$$

$$\text{Calendar} = SR_{\text{day}} \times (AD - BD)$$

SR per day = standard fixed overhead rate per day. With a calendar variance, use the revised capacity variance = SR (AH – RBH) (SM 13.41).

∑ **Formula**

Rates and revised budgeted hours

$$SR = \frac{BF}{BH} \qquad RBH = \frac{BH}{BD} \times AD$$

Rate per unit = BF ÷ budgeted output; rate per day = BF ÷ BD. Revised budgeted fixed overheads (the Study Material also calls them "possible fixed overheads") = RBH × SR = BF ÷ BD × AD (SM 13.42, 13.76, 13.87).

∑ **Formula**

Verification

$$\text{Cost} = \text{Expenditure} + \text{Volume}$$

And volume = efficiency + capacity + calendar (SM 13.42).

The summary in the Study Material uses the names **recovered**, **standard**, **budgeted** and **actual** overhead: cost = recovered − actual, expenditure = budgeted − actual, volume = recovered − budgeted, efficiency = recovered − standard, capacity = standard − budgeted (SM 13.54).

The working notes

ICAI's solutions open with a Budget / Actual table (Illustration 11, SM 13.45):

Working note	Budget	Actual
Fixed overheads (₹)	xxx	xxx
Working days	xxx	xxx
Hours	xxx	xxx
Output (units)	xxx	xxx

Then: standard hours for actual output, standard rate per hour, per unit and per day, and the revised budgeted hours.

ANSWER FRAME

"Discuss the various types of fixed overhead variances" (theory)

1. **Why they arise**: the recovery of fixed overheads depends on capacity utilisation; producing less than the projected level leaves part of the budgeted fixed overheads unrecovered (SM 13.38).
2. **Fixed overhead cost variance**: the difference between actual fixed overhead and absorbed fixed overhead; overhead variance is an under- or over-absorption.
3. **Expenditure variance**: actual fixed overhead incurred against budgeted fixed overhead.

4. **Volume variance**: arises due to the volume of production; absorbed against budgeted. It shows the cost of failing to produce at the planned activity, or the benefit of producing above it (SM 13.39).
5. Volume splits into **efficiency** (absorbed against standard fixed overhead), **capacity** (standard fixed overhead against budgeted) and **calendar** (actual against standard working days).
6. When the calendar variance is computed, the capacity variance is revised to (actual hours − revised budgeted hours) × standard rate.
7. Verification: cost = expenditure + volume; volume = efficiency + capacity + calendar.
8. The formula chart, or a tree.

Three answers to the same question

Loses marks

"Capacity variance = ₹4,800 (F); calendar variance = ₹28,800 (F)."

The extra Sunday is counted twice, and the volume no longer checks.

Passes

"Capacity = ₹300 × (2,320 − 2,400) = ₹24,000 (A)."

Correct, but 2,400 comes from nowhere: no working note for revised budgeted hours.

Full marks

The Budget / Actual working-note table; SH, SR, rate per day and RBH; each variance with formula and (F) or (A); both verification lines.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the day lost twice**
   Fix: with a calendar variance, capacity is SR (AH − revised budgeted hours) (SM 13.41).

2. **The expenditure variance measured against absorption**
   Fix: expenditure compares actual with the **budget** (₹6,00,000), because fixed overheads are not expected to move with output; only absorption does.

3. **The annual budget in a monthly question**
   Fix: split the budget into fixed and variable first, then convert it to the period of the actuals (÷ 12).

RECAP

- FOH cost variance = absorbed (SH × SR) − actual: an under- or over-absorption.
- Expenditure = budgeted − actual (spending). Volume = absorbed − budgeted (output).

- Volume = efficiency SR (SH − AH) + capacity SR (AH − BH) + calendar (actual days − budgeted days) × rate per day.
- With a calendar variance, capacity = SR (AH − RBH); RBH = BH ÷ budgeted days × actual days.
- Checks: cost = expenditure + volume; volume = efficiency + capacity + calendar.
- Rates: per hour BF ÷ BH, per unit BF ÷ budgeted output, per day BF ÷ budgeted days; monthly budget = annual ÷ 12.
- Five amounts: SH × SR, AH × SR, RBH × SR, BH × SR, actual. (Diwali month: worked every budgeted hour, still ₹24,000 (A).)

Interactive version: PassBook app, lesson Fixed overhead variances

13.6 Reconciling budgeted and actual profit, and accounting for variances

Key idea

A reconciliation explains the whole gap, budget to actual or standard to actual, as a list of variances, each measured with everything else at standard; they must add to the gap. Variances are then disposed of, normally to the Costing Profit and Loss Account, because stocks are carried at standard cost.

FORMATS AND FORMULAS

Reproduce these layouts from memory. Each variance is written with (F) or (A); in a profit bridge, (F) is added and (A) deducted.

Format 1: Flexible budget compared with actual (SM 15.92)

₹	Master	Flex.	Actual	Var.
Sales	xxx	xxx	xxx	xxx
Each variable cost	xxx	xxx	xxx	xxx
Contribution	**xxx**	**xxx**	**xxx**	**xxx**
Fixed overheads	xxx	xxx	xxx	xxx
Profit	**xxx**	**xxx**	**xxx**	**xxx**

Master = master budget at budgeted volume; Flex. = flexible budget at actual volume, at standard cost; Var. = variance, actual against flexible, each with (F) or (A). The Study Material's answer also shows the standard per unit beside the flexible budget, and a total variable cost line above contribution. In the answer book write the column heads in full.

Format 2: Statement reconciling budgeted profit and actual profit

Particulars	Amount (₹)
Budgeted profit	xxx
Less / Add: effect of volume (units × standard contribution)	xxx
Flexible budget profit	**xxx**
Add / Less: effect of selling price	xxx
Add / Less: material price and usage variances	xxx
Add / Less: labour rate, idle time and efficiency variances	xxx
Add / Less: variable overhead expenditure and efficiency variances	xxx
Add / Less: fixed overhead expenditure variance	xxx
Actual profit	**xxx**

ICAI prints no pro-forma for this statement: this layout is the variance column of Format 1 set out as a statement, with the volume step added at the top and each cost variance split into its parts. Whatever the layout, start at the budgeted profit, show every step with its (F) or (A), and end at the actual profit.

Format 3: Income statement showing the actual gross margin

ICAI's layout in MTP January 2026 Series II and MTP September 2026 Series I:

Particulars	Amount (₹)
Sales	xxx
Less: Standard cost of goods sold	xxx
Standard gross margin	**xxx**
Less: Adverse variances (each listed)	xxx
Add: Favourable variances (each listed)	xxx
Actual gross margin	**xxx**

∑ **Formula**

The reconciliation identities

$$AC = SC - NV$$

AC = actual cost; SC = standard cost of actual output; NV = net cost variance = MCV + LCV + VOCV + FOCV, the material, labour, variable overhead and fixed overhead cost variances, with (F) as plus and (A) as minus (SM 13.13).

Interactive version: PassBook app, lesson Reconciling budgeted and actual profit, and accounting for variances

∑ **Formula**

The two revenue steps

$$\text{Volume} = (AU - BU) \times c$$

$$\text{Price} = (AP - SP) \times AU$$

AU, BU = actual and budgeted units sold; c = standard contribution per unit; AP, SP = actual and standard selling price. Volume is the master budget profit against the flexible budget profit.

Definition

Controllable and uncontrollable variances

Controllable variances are those which can be controlled under the normal operating conditions if a responsibility centre takes preventive measures and acts prudently. **Uncontrollable variances** are those which occur due to conditions which are beyond the control of a responsibility centre and cannot be controlled even though all preventive measures are in place. (SM 13.11–13.12)

Definition

Revision variance

The difference between the original standard cost and the revised standard cost of actual production. (SM 13.53)

ANSWER FRAME

"Calculation of variances is not an end in itself, but a means to an end. Discuss."

The points ICAI's suggested answers give (MTP January 2025 Series II Q2(b); MTP September 2025 Series I Q6(c); MTP May 2026 Series II Q4(b)):

1. The crux of standard costing lies in variance analysis: standard costs are predetermined and compared with actual costs; the difference between standard cost and actual cost of actual output is the variance.

2. A variance is favourable when actual cost is less than standard, unfavourable (adverse) when more.
3. The figures alone do not give control: their origin and causes must be traced to take remedial steps.
4. A detailed probe, particularly into **controllable** variances, helps management ascertain (i) the amount of the variance, (ii) the causes of its occurrence, (iii) the responsibility of executives and departments, (iv) the corrective action to reduce or remove it.
5. Success depends on how quickly and effectively corrective action is taken: the variance is information, the means; action on it is the end.

"How are variances disposed of?"

Point	Source
Disposition is the last step of the standard costing process: variances are transferred to the relevant accounts (Costing P&L) as per the plan adopted	SM 13.5
Small variances, and those from abnormal causes or inefficiency (controllable): Costing P&L	SM 4.44–4.46; MCQ 8
Large variances from an unavoidable wrong estimate: supplementary rate over cost of sales, finished goods and WIP	SM 4.44–4.46
Stocks remain at standard cost, so profit is not distorted by stock values	SM 13.50
Significant, persistent uncontrollable variances: revise the standard (revision variance)	SM 13.12, 13.53

Three answers to "Reconcile the budgeted profit with the actual profit"

Loses marks

"Budgeted ₹4,20,000 − actual ₹3,82,600 = ₹37,400 (A)."

A difference is not a reconciliation.

Passes

A list of the cost variances that adds to ₹34,400 (A) of variable costs and ₹3,000 (F) of fixed overheads.

The volume and price effects are missing: the list explains the flexible budget, not the budget.

Full marks

Working notes; the flexible budget; budgeted profit → volume → flexible budget profit → price → each cost variance with (F)/(A) → actual profit; the check against actual sales less actual costs.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the budget that praised the material buyer**
   Fix: compare costs with the **flexible budget** (standard cost of actual output): ₹7,60,000, a variance of ₹4,400 (A) (SM 15.92).

2. **The volume counted at profit, the volume variance forgotten**
   Fix: either volume at standard contribution with fixed overhead expenditure only, or volume at standard profit with the full fixed overhead variance.

3. **Variances parked in the closing stock**
   Fix: in a standard costing system stock is at standard cost and the variances go to the Costing Profit and Loss Account of the period (SM 13.5, 13.50), unless the case for a supplementary rate is made.

4. **Netting the report**
   Fix: investigate each variance by responsibility centre; a small net hides large offsetting causes (the two shifts' labour efficiency variances cancel exactly), and responsibility is fixed variance by variance (SM 13.12).

RECAP

- Standard cost of actual output ± cost variances = actual cost; net cost variance = MCV + LCV + VOCV + FOCV. Price variance on purchases (SM 13.15): the actual cost reached is purchases − closing stock at standard price.
- Budgeted profit → (volume, at standard contribution) → flexible budget profit → (selling price) → (each cost variance) → actual profit. Add (F), deduct (A).
- Volume at standard profit instead: then include the fixed overhead volume variance. Never mix the two.
- Compare costs with the flexible budget, never the master budget.
- Disposition (SM 13.5): to the Costing Profit and Loss Account, where small, abnormal and controllable variances go.

- Large variance from an unavoidable wrong estimate: supplementary rate over cost of sales, finished goods and WIP (SM 4.44–4.46).
- Stocks at standard cost; significant persistent uncontrollable variances → revise the standard (revision variance).
- The masala tins: −₹60,000 volume + ₹95,000 price − ₹97,400 costs = −₹62,400. The bridge closes to the rupee, or it is not a bridge.

Interactive version: PassBook app, lesson Reconciling budgeted and actual profit, and accounting for variances

CHAPTER 14

Marginal Costing

14.1 Marginal cost, contribution and the P/V ratio

Key idea

Marginal cost is what one more unit adds to total cost: its variable cost. Price minus marginal cost is contribution, which pays the period's fixed costs first and becomes profit only after they are covered. The P/V ratio is contribution as a share of sales.

FORMATS AND FORMULAS

ICAI's **Cost and Profit Statement under Marginal Costing** (SM 14.7). *Reproduce the headings and their order from memory.*

Particulars	Amount (₹)	Amount (₹)
Revenue (A)		xxx
Product cost:		
– Direct materials	xxx	
– Direct employee (labour)	xxx	
– Direct expenses	xxx	

Particulars	Amount (₹)	Amount (₹)
– Variable manufacturing overheads	xxx	
Product (inventoriable) costs (B)	**xxx**	**xxx**
Product contribution margin {A – B}		**xxx**
– Variable administration overheads	xxx	
– Variable selling and distribution overheads	xxx	xxx
Contribution margin (C)		**xxx**
Period cost (D):		
Fixed manufacturing expenses	xxx	
Fixed non-manufacturing expenses	xxx	xxx
Profit / (loss) {C – D}		**xxx**

The short form is the **marginal cost statement**: Sales, less variable cost, = contribution, less fixed cost, = profit (SM 14.13).

∑ **Formula**

The marginal cost equation

$$S - V = C = F \pm P$$

S = sales value, V = variable cost, C = contribution, F = fixed cost, P = profit (+) or loss (–) (SM 14.13). If C = F there is no profit or loss; at zero output the loss equals F.

∑ **Formula**

Contribution

$$\begin{aligned}\text{Contribution} &= \text{Sales revenue} - \text{Total variable cost} \\ &= \text{Fixed cost} + \text{Profit}\end{aligned}$$

Variable cost here means all of it, manufacturing and non-manufacturing (variable selling commission included).

∑ **Formula**

P/V ratio (contribution to sales ratio)

$$\begin{aligned}\text{P/V ratio} &= \frac{\text{Contribution}}{\text{Sales}} \times 100 \\ &= \frac{\text{Change in contribution or profit}}{\text{Change in sales}} \times 100\end{aligned}$$

The change form is valid when price, variable cost per unit and fixed cost are the same in both periods.

Σ Formula

Transpositions

$$C = S \times \text{P/V ratio} \qquad S = \frac{C}{\text{P/V ratio}}$$

$$\text{Variable cost ratio} = 100\% - \text{P/V ratio}$$

The first two are ICAI's (SM 14.14). The third follows because sales are variable cost plus contribution: a P/V ratio of 40% means variable cost is 60% of sales.

Σ Formula

Fixed cost from any period

$$F = C - P \qquad = (S \times \text{P/V ratio}) - P$$

Definition

Marginal cost

The amount at any given volume of output by which aggregate costs increases if the volume of output is increased or decreased by one unit. (SM 1.34) In Chapter 14: the incremental cost of production for producing one additional unit, measured by the total variable cost attributable to one additional unit. (SM 14.2–14.3)

Definition

Marginal costing

A costing system where products or services and inventories are valued at variable costs only. It does not take consideration of fixed costs. Also known as direct costing. (SM 14.3) As a technique: the ascertainment of marginal cost by differentiating between fixed and variable costs; it is used to ascertain the effect of changes in volume or type of output on profit. (SM 1.39)

Definition

Contribution

Contribution or contribution margin is the difference between sales revenue and total variable costs irrespective of manufacturing or non-manufacturing. (SM 14.8)

Definition

Period cost

Costs which are not assigned to the products but are charged as expenses against the revenue of the period in which they are incurred. All fixed costs, either manufacturing or non-manufacturing, are recognised as period costs in marginal costing. (SM 14.9)

Definition

P/V ratio

This ratio shows the proportion of sales available to cover fixed costs and profit. (SM 14.13)

ANSWER FRAME

"Discuss the advantages of marginal costing" (ICAI's seven, in ICAI's order)

1. **Simplified pricing policy:** marginal cost per unit is constant in the short run, so firm pricing decisions can be taken.
2. **Proper recovery of overheads:** no under- or over-recovery of fixed overheads absorbed at predetermined rates.
3. **Shows realistic profit:** stock and WIP at marginal cost, fixed expenses written off as period cost.
4. **How much to produce:** break-even analysis shows the effect of more or less production on profit.
5. **More control over expenditure:** actual variable expenses compared with budget through variances.
6. **Helps in decision making:** make or buy, discontinuance of a product, replacement of machines.
7. **Short-term profit planning:** through break-even charts.

"Discuss the limitations of marginal costing" (ICAI's seven)

1. **Difficulty in classifying fixed and variable elements:** most expenses are neither wholly fixed nor wholly variable.
2. **Dependence on key factors:** contribution is no guide to optimum profitability unless linked with the key factor.
3. **Scope for low profitability:** sales staff may mistake marginal cost for total cost and sell at a loss.
4. **Faulty valuation:** fixed overheads cannot be wholly excluded from WIP, particularly in large contracts.
5. **Unpredictable nature of cost:** fixed cost and variable cost per unit do change in reality.
6. **Ignores time factor and investment:** two jobs with the same marginal cost may take different time and machines.
7. **Understating of WIP:** stocks and work-in-progress are understated.

"Elaborate the practical application of marginal costing"

ICAI's suggested answer (MTP Series II for September 2026, 4 marks) has four points: **pricing policy** (constant marginal cost permits firm short-term pricing decisions), **decision making** (make or buy, discontinuance, replacement), **ascertaining realistic profit** (stock at marginal cost, fixed cost as period cost) and **determination of production level** (break-even analysis).

The Study Material's own answer key to "Discuss the practical application of marginal costing" points instead to its "facts about marginal costing" (SM 14.6, under the hood): a fuller answer adds them after the four points.

"Explain the characteristics of marginal costing" (ICAI's six)

(1) All costs classified into fixed and variable, semi-variable costs split; (2) variable costs are the cost of the product; (3) finished goods and WIP at marginal cost only, variable selling and distribution excluded; (4) fixed costs are period costs; (5) prices set with reference to marginal cost and contribution; (6) profitability of departments and products judged by contribution.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the fixed overhead that sneaked into marginal cost**
   Fix: marginal cost is prime cost plus variable overheads; fixed costs are deducted from total contribution, never from a unit.

2. **Contribution that forgot the salesman**
   Fix: contribution is after *all* variable costs: deduct variable administration and selling costs too (₹94,50,000).

3. **Closing stock valued with the selling commission in it**
   Fix: stock is valued at variable manufacturing cost only (₹130); variable selling and distribution costs are excluded from inventory.

RECAP

- Marginal cost = the variable cost of one more unit (prime cost + variable overheads); it is the same at every volume within the range.
- Marginal costing values products and stock at variable (manufacturing) cost only; all fixed costs are period costs charged to the period's profit and loss account.
- Contribution = sales − all variable costs = fixed cost + profit. Marginal cost equation: S − V = C = F ± P.

- Product contribution margin = revenue − variable manufacturing costs; contribution margin also deducts variable administration and selling costs.
- P/V ratio = contribution ÷ sales × 100, or change in contribution (profit) ÷ change in sales × 100; C = S × P/V; variable cost ratio = 100% − P/V.
- The P/V ratio moves only with price and variable cost per unit; volume and fixed cost leave it alone; equal % changes in both leave it alone too.
- Unit total cost wobbles with volume (₹95 at 20,000 jars, ₹85 at 30,000); marginal cost does not.

Interactive version: PassBook app, lesson Marginal cost, contribution and the P/V ratio

14.2 Break-even point, margin of safety and angle of incidence

Key idea

Contribution pays the fixed cost first; only what is left over is profit. The break-even point is the volume at which the fixed cost has just been paid, and the margin of safety is how far expected sales sit above it.

FORMATS AND FORMULAS

Every break-even answer starts from the marginal cost statement or one of these formulas. *Reproduce the statement and the formulas from memory.*

Marginal cost statement	Amount (₹)
Sales	xxx
Less: Variable cost	xxx
Contribution	**xxx**
Less: Fixed cost	xxx
Profit	**xxx**

∑ **Formula**

The marginal cost equation

$$S - V = C = F \pm P$$

S = sales, V = variable cost, C = contribution, F = fixed cost, P = profit or loss (SM 14.13). At the BEP, P = 0, so C = F.

∑ **Formula**

Break-even point in units

$$\text{BEP (units)} = \frac{\text{Fixed cost}}{\text{Contribution per unit}}$$

∑ **Formula**

Break-even sales in rupees

$$\text{BES} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{\text{Fixed cost}}{\text{Contribution}} \times \text{Sales}$$

P/V ratio = contribution ÷ sales × 100.

∑ **Formula**

Cash break-even point

$$\text{Cash BEP (units)} = \frac{\text{Cash fixed costs}}{\text{Contribution per unit}}$$

ICAI: the break-even point "calculated only with those fixed costs which are payable in cash" (SM 14.15); depreciation and other non-cash fixed costs are left out.

∑ **Formula**

Sales for a target profit

$$\text{Units} = \frac{\text{Fixed cost} + \text{Desired profit}}{\text{Contribution per unit}}$$

$$\text{Sales value} = \frac{\text{Fixed cost} + \text{Desired profit}}{\text{P/V ratio}}$$

∑ **Formula**

Margin of safety

$$\text{MOS} = \text{Total sales} - \text{BES} \qquad = \frac{\text{Profit}}{\text{P/V ratio}}$$

Total sales are the actual or budgeted sales; in units, units sold – BEP units.

∑ **Formula**

Margin of safety ratio and profit

$$\text{MOS ratio} = \frac{\text{Total sales} - \text{BES}}{\text{Total sales}} \times 100$$

$$\text{Profit} = \text{MOS} \times \text{P/V ratio}$$

∑ **Formula**

P/V ratio from two periods

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100$$

Valid when price, variable cost per unit and fixed cost are the same in both periods. A loss is a negative profit.

Which formula from which data

You are given	Find first	Then
Price, variable cost per unit, fixed cost	Contribution per unit	BEP = fixed cost ÷ contribution per unit
Sales and profit of two periods	P/V = change in profit ÷ change in sales	Fixed cost = contribution − profit; BES = fixed cost ÷ P/V
Profit and P/V ratio	MOS = profit ÷ P/V	BES = sales − MOS
Fixed cost, P/V ratio, MOS ratio	BES = fixed cost ÷ P/V	Sales = BES ÷ (1 − MOS ratio)
MOS in units and in ₹, BEP units	Price = MOS (₹) ÷ MOS (units)	Sales = (MOS + BEP) units × price

Definition

Break-even point (BEP)

The level of sales where an entity neither earns profit nor incurs loss. BEP is indicated in both quantity and monetary value terms. (SM 14.49)

Definition

Margin of safety (MOS)

The margin between sales and the break-even sales is known as margin of safety. It can either be indicated in quantitative or monetary terms. (SM 14.49)

Definition

Angle of incidence

This angle is formed by the intersection of sales line and total cost line at the break-even point. This angle shows the rate at which profit is earned once the break-even point is reached. (SM 14.31)

ANSWER FRAME

The theory parts

The chapter's own theory questions (SM 14.52) include these three. Write the points in this order.

Short note on margin of safety

1. Definition: expected (actual) sales minus break-even sales, in units or rupees.
2. Formulas: sales − BES; profit ÷ P/V ratio; MOS ratio = (sales − BES) ÷ sales.
3. Meaning: the larger the margin of safety, the higher the chances of making profits.

4. A large angle of incidence with a high margin of safety: an extremely favourable position.
5. A two-line numerical example.

Short note on angle of incidence

1. The angle formed by the intersection of the sales line and the total cost line at the BEP.
2. It shows the rate at which profit is earned once the BEP is reached (below the BEP, the rate of loss).
3. The wider the angle, the greater the rate of earning profit; it widens with the P/V ratio.
4. A large angle with a high margin of safety: an extremely favourable position.
5. A labelled sketch.

Explain and illustrate the BEP with a break-even chart

1. Definition: no profit, no loss; total cost equals total sales.
2. Formula in units and in rupees.
3. The chart: costs and revenues up the side, activity along the bottom; fixed cost, total cost and sales lines.
4. The BEP where sales cut total cost; margin of safety and profit read off.
5. A labelled sketch with figures.

The numerical parts: three answers to the same question

Loses marks

"BEP = 30,000 boxes."

A bare figure: if it is wrong, nothing earns method marks.

Passes

"BEP = Fixed cost ÷ Contribution per unit = ₹36,00,000 ÷ ₹120 = 30,000 boxes."

Formula and figures, but ₹36,00,000 comes from nowhere: the semi-variable split is invisible.

Full marks

Working notes first (contribution per box, fixed part of the semi-variable cost, total fixed cost); then formula, figures and result with its unit; any reading of ambiguous wording stated as an assumption.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the loss that lost its minus sign**
   Fix: a loss is a negative profit, so the change is ₹21,00,000 − (−₹9,00,000) = ₹30,00,000 and the P/V ratio 30%.

2. **The margin of safety measured against the wrong sales**
   Fix: divide by the total (actual or budgeted) sales: 9,000 ÷ 24,000 = 37.5%. Test: BES% and MOS% must add up to 100%.

3. **Depreciation smuggled into the cash break-even point**
   Fix: leave out depreciation and other non-cash fixed costs, and nothing else.

4. **The semi-variable cost that stayed in one piece**
   Fix: split it first; the variable part joins the variable cost per unit, the fixed part the fixed cost.

RECAP

- Contribution per unit = price − variable cost per unit; P/V ratio = contribution ÷ sales × 100.
- BEP (units) = fixed cost ÷ contribution per unit; BES (₹) = fixed cost ÷ P/V ratio.

- Cash BEP = cash fixed costs ÷ contribution per unit (depreciation and other non-cash costs out).
- Target profit: (fixed cost + desired profit) ÷ contribution per unit, or ÷ P/V ratio for rupees.
- MOS = total sales − BES = profit ÷ P/V ratio; MOS ratio = MOS ÷ total sales; profit = MOS × P/V ratio.
- Two periods: P/V ratio = change in profit ÷ change in sales, and a loss is a negative profit.
- Angle of incidence: where the sales line cuts the total cost line at the BEP; the wider, the faster profit is earned. Large angle plus high MOS: extremely favourable.
- Fixed cost up: BEP rises, angle unchanged. (Bucket first, then the glass.)

Interactive version: PassBook app, lesson Break-even point, margin of safety and angle of incidence

14.3 Break-even and profit-volume charts

Key idea

Every CVP chart is straight lines through two easy points. The break-even point is where sales cross total cost (or the profit line crosses zero), and each chart is chosen for what it lets you read: the break-even chart shows costs, the contribution chart adds contribution, the profit-volume chart shows profit directly.

FORMATS AND FORMULAS

Each chart is plotted from a small table of points; write it as a working note above the graph. *Reproduce the line list for each chart from memory.*

Chart	Lines to draw	Point at zero activity	Point at maximum activity
Break-even	fixed cost	F	F
	total cost	F	F + V × units
	sales	0	price × units
Contribution break-even	variable cost	0	V × units
	total cost	F	F + V × units
	sales	0	price × units
Profit-volume	profit	−F	C × units − F

∑ **Formula**

The lines as equations (x = units)

$$\text{Sales} = p\,x \qquad \text{Total cost} = F + v\,x$$

$$\text{Variable cost} = v\,x \qquad \text{Profit} = (p - v)\,x - F$$

p = selling price per unit, v = variable cost per unit, F = fixed cost. With sales value S along the horizontal axis: profit = S × P/V ratio − F.

Σ **Formula**

Where two profit lines cross

$$\text{Units at which profits are equal} = \frac{F_2 - F_1}{C_2 - C_1}$$

C_1, C_2 = contribution per unit, F_1, F_2 = fixed costs of the two options. When the option with the higher contribution per unit also has the higher fixed cost (the usual trade-off), it earns more above this volume and the other earns more below it (the crossing in SM Example 5, 14.23; solved this way in January 2025 Q1(a)(ii)). If one option has both the higher contribution and the lower fixed cost, the lines never cross and it is better at every volume.

Definition

Break-even chart

A breakeven chart records costs and revenues on the vertical axis and the level of activity on the horizontal axis. The breakeven point is that point where the sales revenue line intersects the total cost line. (SM 14.20)

Definition

Contribution breakeven chart

Based on the same principles as a conventional breakeven chart except that it shows the variable cost line instead of the fixed cost line; contribution can be read as the difference between the sales revenue line and the variable cost line. (SM 14.20–14.21)

Definition

Profit-volume chart

A chart in which the vertical axis represents profits and losses and the horizontal axis is drawn at zero profit or loss; the breakeven point is where the profit line intersects the horizontal axis. (SM 14.21)

ANSWER FRAME

"Explain the profit-volume chart and its advantage"

1. **What it plots:** profit and loss on the vertical axis, the horizontal axis drawn at zero profit, activity (units or sales) along it.
2. **How it is drawn:** a line from the loss at nil activity (equal to the fixed cost) to the profit at a given level or the BEP.
3. **Reading:** the BEP where the line cuts the horizontal axis; profit or loss at any activity as the height of the line.

4. **Advantage (ICAI's "biggest"):** it depicts clearly the effect on profit and the break-even point of any change in the variables (price, variable cost, fixed cost), by drawing a second line on the same graph.
5. **A labelled sketch with figures.**

"Distinguish the break-even chart from the contribution break-even chart"

Point	Break-even chart	Contribution break-even chart
Lines drawn	fixed cost, total cost, sales	variable cost, total cost, sales
Bottom line	horizontal at fixed cost	from the origin (variable cost)
Contribution	cannot be read (its major limitation)	gap between sales and variable cost lines
Fixed cost	height of the fixed cost line	constant gap between total cost and variable cost lines
BEP and profit	read in the same way on both	read in the same way on both

For "Explain and illustrate the break-even point with a break-even chart", use the frame in the break-even lesson.

Three answers to "Draw a break-even chart"

Loses marks

A freehand sketch, unlabelled axes, the BEP circled but no figures.

The examiner cannot tell which line is which or check any reading.

Passes

Lines plotted to scale and the BEP figure written in.

No working notes for the points, axes without units, no margin of safety or profit shown.

Full marks

Working-note table of points; titled chart, labelled axes with units and an even scale; every line named; BEP labelled, with a dotted line to the activity axis; the readings asked for, with an arithmetic check; a comment if a change is compared.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the total cost line that left from the origin**
   Fix: total cost starts at the fixed cost on the vertical axis; only the sales line (and the variable cost line) leave the origin.

2. **Contribution read off the wrong chart**
   Fix: contribution needs a variable cost line: draw a contribution break-even chart and read sales minus variable cost.

3. **The P/V line that started at zero**
   Fix: at nil activity the loss equals the fixed cost; start at −F.

4. **The profit path's first crossing called the break-even point**
   Fix: the break-even point of the mix is where the straight profit line from −F to the final profit cuts the axis (₹15,00,000).

RECAP

- Break-even chart: rupees up the side, activity along the bottom; fixed cost (horizontal), total cost (from F), sales (from 0). BEP where sales cuts total cost.
- Contribution break-even chart: variable cost line from the origin instead of the fixed cost line; contribution =

gap between sales and variable cost; fixed cost = constant gap between total cost and variable cost.

- Profit-volume chart: profit/loss up the side, horizontal axis at zero profit; the line starts at −F and cuts the axis at the BEP. Slope = contribution per unit (or the P/V ratio with sales value along the bottom).
- On the P/V chart, a price rise steepens the line (BEP left); a fixed cost rise drops it parallel (BEP right); volume moves along it.
- Two options cross where $(F_2 - F_1) \div (C_2 - C_1)$ units: above it the higher-contribution (higher-fixed-cost) option wins, below it the lower-fixed-cost one.
- Several products: the profit path segments slope by each P/V ratio; the BEP is on the straight line from −F to the final profit.
- Every line needs two points; label axes, lines and the BEP, and write the readings.

Interactive version: PassBook app, lesson Break-even and profit-volume charts

14.4 Marginal versus absorption costing and profit reconciliation

Key idea

The two methods differ only in fixed production overhead. Absorption costing carries part of it forward in closing stock (and brings last period's part back with opening stock); marginal costing charges all of it to the period. So the difference in profit is exactly the fixed overhead in closing stock minus the fixed overhead in opening stock.

FORMATS AND FORMULAS

ICAI's two income statements (SM 14.45–14.46); the absorption one runs down the same lines as the cost sheet. *Reproduce both layouts from memory; the order of the lines is what earns the format marks.*

Income Statement (Absorption costing)

Particulars	Amount (₹)	Amount (₹)
Sales		xxx
Production costs:		
Direct material consumed		xxx
Direct labour cost		xxx
Variable manufacturing overhead		xxx

Particulars	Amount (₹)	Amount (₹)
Fixed manufacturing overhead		xxx
Cost of production		**xxx**
Add: Opening stock of finished goods (at cost of previous period's production)		xxx
Less: Closing stock of finished goods (at production cost of current period)		xxx
Cost of goods sold		**xxx**
Add (or less): Under (or over) absorption of fixed manufacturing overhead		xxx
Add: Administration costs	xxx	
Add: Selling and distribution costs	xxx	xxx
Total cost		**xxx**
Profit (Sales – Total cost)		**xxx**

Income Statement (Marginal costing)

Particulars	Amount (₹)
Sales	xxx

Particulars	Amount (₹)
Variable manufacturing costs:	
– Direct material consumed	xxx
– Direct labour	xxx
– Variable manufacturing overhead	xxx
Cost of goods produced	**xxx**
Add: Opening stock of finished goods (at cost of previous period)	xxx
Less: Closing stock of finished goods (at current variable cost)	xxx
Cost of goods sold	**xxx**
Add: Variable administration, selling and distribution overhead	xxx
Total variable cost	**xxx**
Contribution (Sales – Total variable cost)	**xxx**
Less: Fixed costs (production, administration, selling and distribution)	xxx
Net profit	**xxx**

Under marginal costing the contributions of the various products are pooled, and the fixed overheads are met out of this total contribution, which is also known as **gross**

margin (SM 14.46). ICAI's printed statement also has a line "Add: Selling and distribution costs" with no amount between total variable cost and contribution; variable selling costs are already in the line above and fixed ones below, so nothing is entered there.

Reconciliation of profits (the layout of ICAI's suggested answers: SM practical problem 17, MTP Series I for May 2024 Q4(a)(iii))

Particulars	Amount (₹)
Profit as per absorption costing	xxx
Add: Fixed overhead in opening stock (opening stock under-valued in marginal costing)	xxx
Less: Fixed overhead in closing stock (closing stock under-valued in marginal costing)	xxx
Profit as per marginal costing	**xxx**

Interactive version: PassBook app, lesson Marginal versus absorption costing and profit reconciliation

∑ **Formula**

Fixed production overhead absorption rate

$$\text{Rate per unit} = \frac{\text{Budgeted fixed production overhead}}{\text{Normal (budgeted) output}}$$

Formula

Under- or over-absorption

$$\begin{aligned}&\text{Under } (+) \text{ / over } (-) \text{ absorption}\\&= \text{Actual fixed production overhead}\\&\quad -\text{Units produced} \times \text{Rate}\end{aligned}$$

Formula

Difference in profit

$$\begin{aligned}&\text{Absorption profit} - \text{Marginal profit}\\&= \text{Fixed overhead in closing stock}\\&\quad -\text{Fixed overhead in opening stock}\end{aligned}$$

Each stock's fixed overhead at the rate at which it was valued: closing stock at this period's rate, opening stock at the previous period's.

Definition

Absorption costing

Absorption costing is the practice of charging all costs, both variable and fixed to operations, processes or product. (SM 14.9) In it, fixed expenses are distributed over products on the basis of a pre-determined level of output, which leads to over- or under-recovery as actual output differs from the estimate. (SM 14.9)

Definition

Product (inventoriable) costs

Under marginal costing, variable manufacturing costs and under absorption costing, total manufacturing costs (variable and fixed) constitute inventoriable or product costs. (SM 1.35)

ANSWER FRAME

"Discuss the points of difference between absorption costing and marginal costing" (ICAI's five, in order)

No.
1

Marginal costing
Only variable costs are considered for product costing and inventory valuation.

Absorption costing
Both fixed and variable costs are considered for product costing and inventory valuation.

No.
2

Marginal costing
Fixed costs are regarded as period costs; the profitability of different products is judged by their P/V ratio.

Absorption costing
Fixed costs are charged to the cost of production; each product bears a reasonable share of fixed cost, so its profitability is influenced by the apportionment of fixed costs.

No.

3

Marginal costing

Cost data presented highlight the total contribution of each product.

Absorption costing

Cost data are presented in the conventional pattern; net profit of each product is determined after subtracting fixed cost along with variable costs.

No.

4

Marginal costing

The difference in the magnitude of opening and closing stock does not affect the unit cost of production.

Absorption costing

The difference in the magnitude of opening and closing stock affects the unit cost of production due to the impact of related fixed cost.

No.

5

Marginal costing

Cost per unit remains the same irrespective of production, as it is valued at variable cost.

Absorption costing

Cost per unit reduces as production increases, because fixed cost per unit falls while variable cost per unit stays the same.

(SM 14.43–14.44) Add the difference in profit (the four cases) when the question asks about profits.

"Explain the difference in profit": the frame

1. **The cause:** the difference in stock valuation; under absorption costing stock carries fixed production overhead.
2. **The case that applies:** closing stock more than opening (absorption higher), less (marginal higher), equal (equal profits if the fixed cost in both stocks is the same).
3. **The amount:** fixed overhead in closing stock − fixed overhead in opening stock, with figures.
4. **The reconciliation statement**, from absorption profit to marginal profit.

Three answers to the paratha-bonus question

Loses marks

Profits for both plans with no working notes, closing stock valued at ₹34 under both methods.

Wrong stock value under marginal costing, and no way to award method marks.

Passes

Both statements correct, but no reconciliation and no explanation.

Part (iii) carries marks of its own.

Full marks

Working notes (unit costs, stock in units, under/over absorption); both statements in ICAI's layout; the reconciliation; the bonus; a comment naming the case (closing stock above opening) and its ₹ effect.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the commission that went into the godown**
   Fix: stock is valued at production cost only, under both methods; selling costs are charged to the period.

2. **Under-absorption counted on the packs sold**
   Fix: overhead is absorbed into production, so use units produced (4,60,000).

3. **Under-absorption put into the marginal costing statement**
   Fix: marginal costing charges actual fixed overhead (₹24,00,000) in full; no absorption, so nothing is under- or over-absorbed.

4. **"Equal stocks, so equal profits"**
   Fix: equal only if the fixed cost in both stocks is the same amount; here ₹12 against ₹15 a box makes a ₹36,000 difference.

RECAP

- Absorption costing charges all costs, fixed and variable, to products; marginal costing values products and stock at variable cost and charges fixed costs to the period.
- Product (inventoriable) cost: total manufacturing cost under absorption, variable manufacturing cost under marginal. Selling and administration costs are period costs under both.
- Fixed production overhead rate = budgeted fixed production overhead ÷ normal output; under/(over) absorption = actual − units produced × rate, adjusted in the absorption statement only.
- No stock: equal profits. Equal stocks: equal profits if their fixed cost is the same amount. Closing more than opening: absorption higher. Opening more than closing: marginal higher.
- Absorption profit − marginal profit = fixed overhead in closing stock − fixed overhead in opening stock.
- Reconciliation: absorption profit + fixed overhead in opening stock − fixed overhead in closing stock = marginal profit.
- The five ICAI differences: inventory valuation, fixed cost as period cost (P/V ratio), contribution highlighted, stock levels and unit cost, unit cost and volume.

Interactive version: PassBook app, lesson Marginal versus absorption costing and profit reconciliation

14.5 Limiting factor and product mix

Key idea

When a resource is scarce, rank products by contribution per unit of that resource, not by contribution (or profit) per unit of product. Fixed costs that stay the same whatever the mix never change the ranking.

FORMATS AND FORMULAS

Reproduce these three statements from memory; ICAI's suggested answers are built from them.

Formula

Contribution per unit of the limiting factor

$$\text{Profitability} = \frac{\text{Contribution per unit}}{\text{Key factor per unit}}$$

ICAI writes it as Profitability = Contribution ÷ Key factor (SM 14.28). Key factor per unit: hours, kg, metres or any other unit of the scarce resource one unit of product uses.

∑ **Formula**

Recovering the quantity of the factor

$$\text{Hours per unit} = \frac{\text{VOH per unit}}{\text{VOH rate per hour}}$$

$$\text{Kg per unit} = \frac{\text{Material cost per unit}}{\text{Price per kg}}$$

VOH = variable overhead. Labour hours per unit = labour cost per unit ÷ wage rate per hour, in the same way.

Statement 1: ranking

Particulars	Product A	Product B	Product C
Selling price per unit	xxx	xxx	xxx
Less: Variable cost per unit	xxx	xxx	xxx
Contribution per unit	**xxx**	**xxx**	**xxx**
Key factor per unit (hours, kg)	xx	xx	xx
Contribution per unit of key factor	**xx**	**xx**	**xx**
Rank	II	I	III

Statement 2: optimum product mix

Product (in rank order)	Units	Key factor used	Key factor left
Minimum commitment, if any	xxx	xxx	xxx
Rank I (up to demand)	xxx	xxx	xxx
Rank II (up to demand)	xxx	xxx	xxx
Rank III (balance)	xxx	xxx	nil

Statement 3: profit

Product	Units	Contribution per unit (₹)	Total contribution (₹)
A	xxx	xx	xxx
B	xxx	xx	xxx
Total contribution			**xxx**
Less: Fixed costs			xxx
Profit			**xxx**

Definition

Limiting factor (key factor)

Limiting factor is anything which limits the activity of an entity. The factor is a key to determine the level of sale and production, thus it is also known as Key factor. (SM 14.36, 14.49)

ANSWER FRAME

"What is a limiting factor?"

1. Definition in ICAI's words: anything which limits the activity of an entity; it determines the level of sale and production, hence key factor.
2. Supply side: men, materials, machine, money; demand side: demand, nature of product, regulatory and environmental requirements.
3. Management's objective: optimise the key resource to the maximum possible extent.
4. Decision rule: rank products by contribution per unit of the key factor (profitability = contribution ÷ key factor).
5. A two-line example with figures.

The numerical: three answers to "Determine the optimum mix"

Loses marks

"Rank by contribution per pack: rolls, kachoris, samosas; profit ₹3,25,000."

Ranked per pack, not per fryer hour: the arithmetic is right, the mix is wrong, and only the contribution line earns marks.

Passes

Ranking per fryer hour, then "10,000 samosas, 5,000 kachoris, 4,000 rolls; profit ₹3,65,000."

Right answer, but the hours per pack appear from nowhere and there is no allocation table: a slip anywhere would leave no method marks.

Full marks

Working note on hours per pack; ranking statement per fryer hour; the check that the hours fall short; allocation in rank order within demand; profit statement with fixed costs deducted.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the paneer roll that hogged the fryer**
   Fix: divide each contribution by the scarce quantity it uses, and rank on that.

2. **Profit per unit dressed up as contribution**
   Fix: strip out absorbed fixed overheads; the total fixed cost is deducted once, at the end.

3. **The rank I product that ate the whole factory**
   Fix: allocate up to each product's maximum demand, then pass the balance down the ranking.

4. **Rupees read as hours**
   Fix: divide the rupees by the rate per hour (or per kg) before dividing the contribution.

RECAP

- Limiting (key) factor: anything which limits the activity of an entity; it decides the level of sale and production (SM 14.36).
- Supply side: men, materials, machine, money; demand side: demand, nature of product, regulatory and environmental requirements.

- Profitability = contribution ÷ key factor: rank by contribution per unit of the scarce resource.
- First check the resource really falls short of the full demand; if not, make everything with a positive contribution.
- Derive hidden quantities: hours = VOH ÷ rate per hour; kg = material cost ÷ price per kg.
- Allocate commitments first, then rank order up to each product's demand; the last product takes the balance.
- Fixed costs: ignored in ranking, deducted once in the profit statement; absorbed fixed overhead per unit is added back. A product's own fixed cost is saved only if it drops out altogether.
- Make or buy with a scarce resource: rank by saving per unit of the resource; buy outright anything cheaper outside. (Marks per minute, not marks.)

Interactive version: PassBook app, lesson Limiting factor and product mix

14.6 Make or buy, shut down and other short-term decisions

Key idea

Decide on relevant costs: future costs and revenues that differ between the options, including benefits given up. Costs already incurred, and fixed costs that continue whatever you choose, do not decide anything.

FORMATS AND FORMULAS

Reproduce these statements from memory: ICAI's suggested answers are built on this two-column comparison (the shut-down one is laid out as a "Statement showing operating loss").

Statement showing make or buy (per unit or in total)

Particulars	Make (₹)	Buy (₹)
Purchase price		xxx
Direct materials, labour, variable overheads	xxx	
Avoidable fixed costs	xxx	
Contribution forgone on the alternative use	xxx	
Relevant cost	**xxx**	**xxx**

Statement showing operating profit or loss: continue or shut down

Particulars	If continued (₹)	If shut down (₹)
Sales	xxx	
Less: Variable cost	xxx	
Contribution	**xxx**	
Less: Fixed cost	xxx	xxx (unavoidable)
Less: Additional shut-down costs		xxx
Profit / (loss)	**xxx**	**xxx**

∑ **Formula**

Shut-down point

$$\text{SDP} = \frac{\text{Fixed cost} - \text{Shut-down cost}}{\text{Contribution per unit}}$$

SDP = shut-down point in units; fixed cost = total fixed cost if the plant runs; shut-down cost = fixed costs that continue during the closure + additional costs of closing. Below this volume, closing loses less. In rupees, divide by the P/V ratio instead (SM practical problem 9: lock-up sales = avoidable fixed cost ÷ P/V ratio).

∑ **Formula**

Minimum price for a special order

$$\text{Minimum price} = \frac{\text{Relevant cost}}{\text{Units ordered}}$$

Relevant cost of the order = variable cost + costs specific to the order + contribution lost on any regular sales it displaces. With enough idle capacity the last part is nil.

Cost terms ICAI asks you to classify (Chapter 1, SM 1.34–1.35)

Term	ICAI's meaning
Differential cost	The change (increase or decrease) in total cost, variable and fixed, due to a change in activity level, technology, process or method of production
Imputed cost	Notional cost involving no cash outlay, such as interest on capital not actually paid
Opportunity cost	The value of sacrifice made or benefit of opportunity foregone in accepting an alternative course of action
Out-of-pocket cost	The portion of total cost that involves cash outflow; used in pricing in recession and make or buy; avoided if the proposal is not accepted

Term	ICAI's meaning
Shut down costs	Fixed costs which continue to be incurred, and cannot be avoided, during the temporary closure of a plant (rent, rates, depreciation)
Sunk costs	Historical costs incurred in the past; they play no role in decision making in the current period

Definition

Relevant cost

A cost is treated as relevant only if (a) it is a future cost and (b) it differs under two options under consideration. (SM 14.33)

ANSWER FRAME

"Identify the type of cost and state whether it is relevant"

Write a three-column table, as ICAI's suggested answer does: **Situation | Type of cost | Relevant / Non-relevant**. One line per situation; give the ICAI term (opportunity, imputed or notional, sunk, committed, shut-down, out-of-pocket, differential), and the relevance follows from the two tests: future, and different between the options.

"Explain the framework for decision making"

1. **Identification of problem**: the goal set to reach the objectives, such as low-cost production; a cost leader must achieve the 3Es: economy in inputs, efficiency in process and operations, effectiveness in output (SM 14.32).
2. **Identification of options**: all possible options are explored. ICAI's list for low-cost production (SM 14.32–14.33): (a) purchase of inputs from a specialised market: local vs import; (b) make the input in its own factory: make or buy; (c) bulk purchase to avail a discount: how much to purchase; (d) make in-house: make vs outsource; (e) bulk processing: how much to produce; (f) an efficient machine: old machine vs new machine; (g) optimisation of key resources: product mix decisions.
3. **Evaluation of the options** against the objective: financial measures (impact on profit, market share, return on investment) and non-financial factors (customer satisfaction, impact on existing market or customers, ethics); two tasks: identification of the costs and benefits of each option (controllability, relevance) and estimation of their amount (variability, traceability).
4. **Selection of the option**: the best option is selected and implemented.

"Should the special order be accepted?"

1. **Relevant cost of the order**: variable cost, costs specific to the order, and the contribution lost on any regular sales it displaces; fixed costs that continue (or are already recovered by budgeted sales) are excluded.
2. **Price against relevant cost**: the margin the order adds to profit.
3. **Capacity**: whether the order fits the idle capacity or displaces regular sales.
4. **Non-financial factors** (SM 14.33): impact on the existing market or customers (regular buyers may ask for the same price), customer satisfaction, ethics of the decision.
5. **Recommendation**: accept, reject, or accept with conditions, with the reason.

The numerical: shut down or continue

Loses marks

"Loss ₹1,08,000 if open daily, so close the café."

Compares a loss with nothing: closing has a cost too.

Passes

Both columns, the right decision, no shut-down point.

The second half of the question carries marks of its own.

Full marks

Contribution per unit; a column for each option (continued / shut down) with unavoidable fixed and closing costs; decision in words; shut-down point by formula, in units (and as a % of capacity where asked).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the apportioned rent that killed the bun maska**
   Fix: compare its contribution with its avoidable fixed costs only.

2. **Full cost against the buy price**
   Fix: count only the variable cost, the fixed costs that would really disappear, and any contribution the capacity could earn elsewhere.

3. **The special order with no room for it**
   Fix: displaced regular sales and price concessions are opportunity costs: add the contribution lost to the cost of the order.

4. **The feasibility report that keeps billing you**
   Fix: sunk and committed costs are the same whichever option is chosen: leave them out.

RECAP

- Relevant cost: a future cost that differs between the options (SM 14.33); sunk, historical and committed costs are irrelevant; opportunity cost is relevant; notional cost only if a benefit is really forgone.
- Framework: identify the problem, identify options, evaluate (controllability, relevance; variability, traceability; non-financial factors), select.
- Make or buy: buy price against variable cost + avoidable fixed cost + contribution forgone on the alternative use.
- Discontinue: drop only if the avoidable fixed cost (plus rent from freed space) exceeds the contribution lost; apportioned common costs stay.
- Shut down: compare the loss if continued with the shut-down cost (unavoidable fixed + closing costs).
- Shut-down point = (total fixed cost − shut-down cost) ÷ contribution per unit; in rupees ÷ P/V ratio.
- Special order: accept if price exceeds variable + specific costs; with no idle capacity add the contribution of displaced sales. Minimum price = relevant cost ÷ units.
- If it would sit in both columns at the same figure, it cannot decide anything. (The rent will be there tomorrow.)

Interactive version: PassBook app, lesson Make or buy, shut down and other short-term decisions

14.7 Multi-product break-even analysis

Key idea

With products sharing fixed costs, the break-even point exists only for a stated sales mix. Weight the contributions by the mix, divide the fixed cost by the weighted figure, then split the result in the same mix.

FORMATS AND FORMULAS

Reproduce ICAI's statement from memory: it is the backbone of every value-mix answer (SM 14.77–14.78).

Particulars	A	B	C	Total
Selling price (₹)	xxx	xxx	xxx	
Less: Variable cost (₹)	xxx	xxx	xxx	
Contribution per unit (₹)	xx	xx	xx	
P/V ratio	x%	x%	x%	
Sales mix	x%	x%	x%	100%
Contribution per rupee of sales (P/V ratio × sales mix)	**x%**	**x%**	**x%**	**x%**

Then: total contribution = total sales × composite P/V ratio; less fixed costs = profit; break-even sales = fixed costs ÷ composite P/V ratio.

∑ **Formula**

Composite contribution per unit (mix in units)

$$\text{Composite contribution} = \sum (C \times w)$$

C = a product's contribution per unit; w = its units weight, its units in the mix ÷ total units in the mix. The result is a contribution per (average) unit. ICAI's Example 4: 4 units of J (₹40) to 3 of K (₹20) gives ₹40 × 4/7 + ₹20 × 3/7 = ₹31.43 (SM 14.16).

∑ **Formula**

Composite break-even point

$$\text{BEP (units)} = \frac{\text{Common fixed cost}}{\text{Composite contribution}}$$

$$\text{Product BEP} = \text{BEP} \times w$$

∑ **Formula**

Composite P/V ratio (mix in values)

$$\text{Composite P/V} = \sum (\text{P/V} \times \text{mix \%})$$

$$= \frac{\text{Total contribution}}{\text{Total sales}}$$

Mix % = each product's share of total sales value.

∑ Formula

Composite break-even sales

$$\text{Composite BES} = \frac{\text{Fixed cost}}{\text{Composite P/V}}$$

$$\text{Product BES} = \text{BES} \times \text{mix } \%$$

ICAI's Illustration 11 also shows a second, equivalent method: share the fixed cost among the products in the ratio of their weighted contributions, then each product's BEP = its share of fixed cost ÷ its own contribution per unit (SM 14.39).

ANSWER FRAME

"Advise whether the company should change over to the new mix"

1. Statement of contribution to sales ratio at the existing mix (P/V ratio × sales mix for each product, total).
2. Existing total contribution, profit and break-even sales.
3. The same statement and figures for the proposed mix, with its own sales and fixed costs.
4. Comparison: composite P/V ratio, break-even sales, profit (and margin of safety).
5. Conclusion in words, with the reason (higher composite P/V ratio and profit), and the assumption that the mix will be achieved.

Three answers to "Find the break-even point of each product"

Loses marks

Namkeen: ₹10,20,000 ÷ ₹24 = 42,500 packs; mathri: ₹10,20,000 ÷ ₹28 = 36,429 packs.

Each product is made to carry all the fixed cost, and the mix is ignored.

Passes

Composite contribution ₹25.50, composite BEP 40,000 packs.

Correct, but stops before splitting it between the products, which the question asked for.

Full marks

Working note on the mix (contribution of a batch), composite BEP by formula, split by the ratio in units and rupees, total checked against fixed cost ÷ composite P/V.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the P/V ratios that were simply averaged**
   Fix: weight each P/V ratio by its share of sales value, or divide total contribution by total sales.

2. **The right weights on the wrong figures**
   Fix: units weights go with contribution per unit; value weights go with P/V ratios. Convert one into the other with the prices if needed.

3. **Splitting the answer in the wrong ratio**
   Fix: split composite units by the units ratio (25,000 and 15,000 packs), then multiply by the prices.

RECAP

- Several products sharing a fixed cost break even only for a stated sales mix (CVP assumption: the mix stays constant).
- Mix in units: composite contribution = Σ contribution per unit × units weight; composite BEP = common fixed cost ÷ composite contribution; split by the units ratio.
- Batch method: contribution of one batch of the mix; fixed cost ÷ that = batches.

- Mix in values: composite P/V = Σ P/V ratio × sales mix % (contribution per rupee of sales) = total contribution ÷ total sales; BES = fixed cost ÷ composite P/V; split by the sales mix %.
- A shift towards high-P/V products lowers break-even sales and raises profit; a replacement product means a new composite.
- Margin of safety = actual (or budgeted) sales − composite break-even sales; profit = MOS × composite P/V ratio.
- "Sales mix 5 : 3" with no unit stated: ICAI's answers treat it as units; state the assumption. (Break even on thalis, not on rotis.)

Interactive version: PassBook app, lesson Multi-product break-even analysis

CHAPTER 15

Budgets and Budgetary Control

15.1 Budgets, budgetary control, the budget manual and key factor

Key idea

A budget is a plan in numbers for a definite future period, approved in advance; budgetary control is the continuous comparison of actual results with it, with responsibility fixed for the differences. The key factor sets the ceiling for the whole budget, so it is identified before the budgets are built.

FORMATS AND FORMULAS

Reproduce these definitions from memory; ICAI asks for them in 1-mark slices of a 5-mark question.

Definition

Budget centre

A section of an organisation developed for the purpose of budgetary control, and intended to facilitate formulation of various budgets with the help of the head of the department. (SM 15.70)

Definition

Budget period

The period covered by a budget. There is no general rule governing its selection; the Budget Committee determines the length suitable for the business, normally a calendar year or a period co-terminus with the financial year, divided into months, quarters or periods coinciding with the trading activity of the business. (SM 15.18)

Definition

Key factor

A factor which sets a limit to the total activity: usually one or two (sometimes more than two), such as non-availability of power or lack of demand. It must be identified and its influence on production or sales estimated properly while preparing the budget. (SM 15.16)

Definition

Master budget

CIMA: "the summary budget, incorporating its component functional budgets, which is finally approved, adopted and employed." (SM 15.55)

∑ **Formula**

Budgeted activity under a key factor

$$\text{Limit} = \frac{\text{Quantity available}}{\text{Quantity per unit}}$$

$$\text{Budgeted output} = \text{the lowest of the limits}$$

Work out the limit of demand and of each resource (in units of output); the factor giving the lowest limit is the key factor. Conditions: one product, or a mix that stays fixed. With several products competing for one scarce resource, rank them by contribution per unit of that resource instead (limiting factor and product mix).

ANSWER FRAME

"Discuss the objectives of a budgetary control system"

The nine points ICAI's answer gives, in order (SM 15.7–15.8); a 4-mark question takes any four with their explanation:

1. Portraying with precision the overall aims of the business and determining targets of performance for each section or department.
2. Laying down the responsibilities of each executive, so that everyone knows what is expected and how he will be judged.
3. Providing a basis for comparing actual performance with the predetermined targets and investigating

deviations, which helps in adopting corrective measures.

4. Ensuring optimum use of available resources to maximise profit or production, subject to the limiting factors.
5. Co-ordinating the activities of the business, centralising control and yet enabling management to decentralise responsibility and delegate authority.
6. Engendering a spirit of careful forethought, assessment of what is possible and an attempt at it: dynamism without being reckless.
7. Providing a basis for revision of current and future policies.
8. Drawing up long range plans with a fair measure of accuracy.
9. Providing a yardstick against which actual results can be compared.

"Advantages" and "limitations"

Advantages (SM 15.12)	One-line explanation
1. Efficiency	Business activities are conducted in an efficient manner
2. Control on expenditure	A yardstick for measuring the performance of individuals and departments

Advantages (SM 15.12)	One-line explanation
3. Finding deviations	Actual is compared with budget and deviations are communicated to management
4. Effective utilisation of resources	Men, material, machinery and money are planned for
5. Revision of plans	Helps review current trends and frame future policies
6. Implementation of standard costing	Creates suitable conditions for a standard costing system
7. Cost consciousness	Encourages cost consciousness and maximum utilisation of resources
8. Credit rating	Well-ordered budget plans receive greater favour from credit agencies

Limitations (SM 15.13–15.14)	One-line explanation
1. Based on estimates	Made on conditions at the time; revision is needed if they change
2. Time factor	Not executed automatically; management must not expect too much in the initial period
3. Co-operation required	Staff co-operation is usually not available initially; success depends on willing co-operation and teamwork
4. Expensive	Needs a proper organisation structure and consumes qualified manpower and equipment
5. Not a substitute for management	Only a managerial tool, to be intelligently applied
6. Rigid document	Should be flexible enough to take in changes in internal and external factors

"Distinguish between a forecast and a budget"

Point	Forecast	Budget
Nature	An assessment of probable future events	A financial or quantitative plan to be pursued over a period
Commitment	Denotes some degree of flexibility	A commitment, a definite target management seeks to attain
Order	Made first, on sales, production cost and finance	Prepared on the basis of the forecasts
Sales	Projects the available customer demand: reflects the environment	Shows how management intends to react to it (SM 15.31)
Common ground	Both relate to a defined period of time	Both relate to a defined period of time

Three answers to "What is a key factor?"

Loses marks

"The most important factor in the budget."

Vague: no limit, no example, no consequence.

Passes

"A factor which sets a limit to the total activity, such as lack of demand or shortage of power."

The definition with an example, but not what the budget does with it.

Full marks

The definition; "usually one or two, sometimes more"; examples (power, demand); it must be identified and its influence on production or sales estimated while preparing the budget; where sales are the limit they are the principal budget factor.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the 9,60,000-pack sales budget**
   Fix: find the limit set by each factor first; the lowest is the key factor and caps every budget.

2. **Calling the forecast a budget**
   Fix: a forecast is an assessment of probable events and stays flexible; a budget is the target management commits to and acts to achieve, built on the forecast.

3. **The budget manual that is only a list of budgets**
   Fix: a budget manual sets out responsibilities, routines, forms and records for budgetary control (CIMA): approvals, timetable, organisation chart, account codes, revision procedure, key assumptions. The budgets themselves are not the manual.

4. **Mixing up the three lists of steps**
   Fix: read the question's verb and noun, then write the matching list; for "budgetary control technique", the five steps plus key factor, manual and period.

RECAP

- Budget (CIMA): a financial and/or quantitative statement, prepared and approved before a defined period, of the policy to be pursued to attain a given objective.
- Forecast = assessment of probable events (flexible); budget = commitment, a definite target.
- Budgetary control: establish budgets, compare actual with budget continuously, revise when circumstances change, fix responsibility.
- Budget committee (functional heads) acts through the budget officer; the Chief Executive is ultimately responsible; the board approves the master budget.
- Preparation: objectives, key factor, controller, budget manual, budget period, standard of activity.

- Key factor: limits the total activity (power, demand); identify it first; the lowest limit binds and can move once removed.
- Budget manual: responsibilities, routine, forms and records for budgetary control: 14 typical contents.
- Components: physical, cost, profit, financial budgets; period-wise: long-term (3 to 10 years), short-term (one or two years), current (months, weeks). (The hall seats 400.)

15.2 Functional budgets and the master budget

Key idea

Every functional budget is built from the one before it: sales, then production (sales + closing stock − opening stock), then materials (usage + closing stock − opening stock = purchases), labour and overheads. The master budget is their summary, approved and adopted as the standard to work to.

FORMATS AND FORMULAS

Reproduce every format from memory: ICAI's suggested answers lay the statements out exactly like this, with a working note for every figure brought in.

Formula

Production budget (units)

$$P = S + C - O$$

P = units to be produced; S = budgeted sales; C = desired closing stock; O = opening stock, all in units of finished goods. If good output is less than output (post-production rejects), divide the good units needed by (1 − rejection rate).

∑ **Formula**

Material usage and purchase budgets

$$U = P \times q \qquad B = U + C - O$$

U = usage; P = units to be produced; q = material per unit; B = quantity to be bought; C and O = closing and opening stocks of the material. Purchase cost = units to be purchased × unit price. A closing stock given as "N days' consumption" is usage × N ÷ days in the period.

∑ **Formula**

Hours to be paid in a labour budget

$$\text{Productive} = \frac{\text{Standard hours}}{\text{Efficiency ratio}}$$

$$\text{Paid} = \text{Productive} + \text{Down-time}$$

All in hours. Standard hours = units to be produced × standard hours per unit. Hours paid beyond the normal hours (workers × normal hours each) are overtime.

Production budget	Product A (units)	Product B (units)
Budgeted sales	xxx	xxx
Add: Desired closing stock	xxx	xxx
Total quantity required	**xxx**	**xxx**
Less: Opening stock	(xxx)	(xxx)
Units to be produced	**xxx**	**xxx**

Direct material purchase budget	Material X	Material Y
Units required for production	xxx	xxx
Add: Desired closing stock	xxx	xxx
Total requirement	**xxx**	**xxx**
Less: Opening stock	(xxx)	(xxx)
Units to be purchased	**xxx**	**xxx**
Unit price (₹)	xxx	xxx
Purchase cost (₹)	**xxx**	**xxx**

Cost of goods sold budget	Amount (₹)
Direct materials used	xxx
Direct labour	xxx
Factory overhead	xxx
Total manufacturing costs	**xxx**
Add: Finished goods (opening)	xxx
Less: Finished goods (closing)	(xxx)
Cost of goods sold	**xxx**

Budgeted income statement	Amount (₹)
Sales	xxx
Less: Cost of goods sold	xxx
Gross margin	**xxx**
Less: Selling and distribution expenses	xxx
Less: Administrative expenses	xxx
Profit before interest and taxes	**xxx**
Less: Interest	xxx
Profit before tax	**xxx**
Less: Income tax	xxx
Net profit	**xxx**

That is ICAI's budgeted income statement (SM 15.56). The budgeted balance sheet beside it shows share capital and retained income, represented by fixed assets less depreciation and the current assets (raw materials, finished goods, debtors, cash) less creditors. When a question gives only sales and costs, ICAI's "Master Budget" is the shorter statement of Illustration 6: sales, less direct materials and wages (prime cost), fixed and variable factory overhead (works cost), gross profit, less administration, selling and distribution expenses, net profit (SM 15.57–15.58).

Theory parts ICAI asks

"List the functional budgets" (the chapter's own question asks for eight): sales, production, plant utilisation, direct material usage, direct material purchase, direct labour, factory overhead, production cost, ending inventory, cost of goods sold, selling and distribution cost, administration expenses, research and development cost, capital expenditure, cash. Write at least the number asked, in ICAI's order.

"Explain the production budget": definition (forecast of production, in two parts: volume and cost); based on the four (sales budget, capacity, planned change in finished stocks, outside purchase policy); the stock-level factors; stabilised production versus stabilised stocks; a format.

"State the considerations for the capital expenditure budget": the five considerations and, if asked, the six merits (Under the hood).

The numerical parts: three answers to the same wages budget

Loses marks

"Wages = 18,900 hours × ₹120 = ₹22,68,000."

Standard hours paid at the normal rate: no efficiency ratio, no down-time, no overtime.

Passes

Productive hours 21,000, down-time 5,250, hours paid 26,250 × ₹120 = ₹31,50,000.

The hours are right, but the 6,250 overtime hours are paid without their premium.

Full marks

Standard hours by product as a working note; ÷ 90%; + 25% down-time; normal hours (workers × normal hours each) at ₹120 and the excess at ₹180; total ₹35,25,000, with each step labelled.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the opening stock that was added**
   Fix: need = use + closing − opening, for finished goods and for materials alike.

2. **Material bought for the sales instead of for production**
   Fix: material usage comes from the **production** budget, which already includes the finished-goods stock change; then adjust for the material stocks.

3. **The closing stock taken on the wrong base**
   Fix: read the base the question gives (next month's sales for products, next month's usage or N days' consumption for materials); for "N days", divide by the working days in the period.

4. **Overtime without the premium**
   Fix: hours beyond the normal week are paid at the basic rate plus the premium (₹120 × 150% = ₹180 in the packing example). Down-time is added to productive hours before the overtime is found.

RECAP

- Functional budgets: one for each function (15 in ICAI's list), subsidiary to the master budget; the sales budget comes first.
- Production = budgeted sales + desired closing stock − opening stock (units); based on sales, capacity, stock change, outside purchase policy.
- Stabilised production: plant fully used, carrying cost varies; stabilised stocks: lowest carrying cost, capacity may be idle.
- Material usage = production × standard quantity; purchases = usage + closing stock − opening stock; governed by EOQ and the re-order point with safety stock.
- Labour: standard hours ÷ efficiency ratio = productive hours; + down-time = hours paid; beyond normal

hours is overtime at the premium rate.

- Cost of goods sold = materials + labour + factory overhead + opening finished goods − closing finished goods (without the stock adjustments: production cost budget).
- Master budget (CIMA): the summary budget, incorporating its component functional budgets, which is finally approved, adopted and employed: budgeted P&L and balance sheet.
- Need = use + closing − opening, every time. (Barfi first, khoya second.)

Interactive version: PassBook app, lesson Functional budgets and the master budget

15.3 Flexible budgets

Key idea

A fixed budget is for one level of activity and does not change; a flexible budget recognises fixed, semi-variable and variable costs and changes with the level of activity attained. Judge actual costs only against the flexible budget for the actual level.

FORMATS AND FORMULAS

Reproduce the format from memory; ICAI's suggested answers show every level side by side, the fixed portions worked out in notes.

Formula

Flexible budget allowance

$$\text{Allowance} = F + v \times A$$

F = fixed cost (including the fixed part of semi-variable costs); v = variable cost per unit of activity; A = activity attained, in units, hours, or sales value (for costs given as a percentage of sales).

Formula

High and low points method

$$v = \frac{\text{Change in expense}}{\text{Change in output}}$$

$$F = \text{Total expense} - v \times \text{Output}$$

v = variable cost per unit; F = fixed cost. Use two levels at which the fixed cost is the same; the fixed part comes out equal at both.

Definition

Fixed budget

CIMA: "a budget designed to remain unchanged irrespective of the level of activity actually attained." (SM 15.19)

Flexible budget for ...	60%	80%	100%
Output (units)	xxx	xxx	xxx
Sales (A)	**xxx**	**xxx**	**xxx**
Variable costs, item by item	xxx	xxx	xxx
Variable part of semi-variable costs	xxx	xxx	xxx
Total variable cost (B)	**xxx**	**xxx**	**xxx**
Contribution (A – B)	**xxx**	**xxx**	**xxx**
Fixed costs, item by item	xxx	xxx	xxx
Fixed part of semi-variable costs	xxx	xxx	xxx
Step costs at the level	xxx	xxx	xxx
Total fixed cost (C)	**xxx**	**xxx**	**xxx**
Profit (A – B – C)	**xxx**	**xxx**	**xxx**

When only costs are asked ("flexible budget showing the total cost"), stop at total cost (B + C); ICAI often adds the cost per unit or per hour at each level.

ANSWER FRAME

"Distinguish between fixed and flexible budgets"

ICAI's five points, in order (SM 15.22–15.23):

Point

1. Change

Fixed budget

Does not change with the actual volume achieved: rigid or inflexible

Flexible budget

Can be re-cast for the activity level achieved: not rigid

Point

2. Levels

Fixed budget

One level of activity, one set of conditions; assumes no change, which is unrealistic

Flexible budget

Various budgets for different levels of activity

Point

3. Variance analysis

Fixed budget

All costs related to one level, so it does not give useful information

Flexible budget

Each cost analysed by behaviour, so it gives useful information

Point	Fixed budget	Flexible budget
4. Cost and price	Cost ascertainment and price fixation mislead when budgeted and actual levels differ significantly	Facilitates cost ascertainment, price fixation and quotations
5. Comparison	Comparing actual with budget is meaningless when activity levels differ	A meaningful basis of comparison of actual with budget

"Explain briefly the concept of flexible budget"

1. The definition (the behaviour of fixed and variable costs; CIMA's wording with semi-variable costs).
2. Unlike a static budget: results for different activity levels; a series of static budgets.
3. Costs classified as fixed, variable and semi-variable, the last split into fixed and variable.
4. Where it is used: new ventures, businesses dependent on nature, labour-intensive industries.
5. A three-line numerical illustration (the café).

"Which budgeting system would you recommend?" (actuals far above a fixed budget)

ICAI's suggested answer (MTP September 2025 Series II Q6(b)): recommend a **flexible budget**; define it (both

definitions); state that unlike static budgets it shows expected results for different activity levels; then list its advantages: sales, costs and profit calculated easily at various levels; simple adjustment to changes in business conditions; helps select the production level that earns the predetermined profit; shows the quantity to be produced to earn a determined profit.

"Suitability for flexible budget" (SM 15.21–15.22)

1. Seasonal fluctuations in sales and/or production, for example the soft drinks industry.
2. A company which keeps introducing new products or changes the design of its products frequently.
3. Industries engaged in make-to-order business, like ship building.
4. An industry influenced by changes in fashion.
5. General changes in sales.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the league table that punished the busiest café**
   Fix: flex the budget to the 36,000 cups actually sold (₹3,62,000): the café stayed ₹8,000 within it. The same error works the other way: Vijay Nagar's "saving" of ₹18,000 against its fixed budget is only lower volume.

2. **The budget rate per unit carried to another level**
   Fix: fixed costs do not shrink: allowance = fixed ₹70,000 + ₹5 × 15,000 = ₹1,45,000.

3. **Semi-variable costs treated as wholly variable or wholly fixed**
   Fix: split it first, by the percentage the question gives or by the high and low points method (here ₹8,40,000 fixed and ₹12 a bottle); flex only the variable part. And pick two levels with the same fixed cost: the rusk unit's 2025-26 figure includes a second oven line.

4. **8.33 vans**
   Fix: vehicles and batches are hired whole: round up (9 vans), and cost the step.

RECAP

- Fixed (static) budget: one level of activity, unchanged whatever is attained; for stable, non-seasonal businesses and fixed expenses.
- Flexible budget (CIMA): recognises fixed, semi-variable and variable costs and changes with the level of activity attained; a series of static budgets.
- Budgeted cost = fixed cost + variable cost per unit × activity; split semi-variables first (high and low points: change in expense ÷ change in output).
- Add the question's steps: price changes at a level, extra fixed costs beyond a capacity, vehicles or batches

rounded up.

- Judge performance against the flexible budget for the actual level: a fixed budget's "saving" may be only lower volume.
- Never carry a cost per unit from one level to another: fixed costs do not shrink.
- Fixed vs flexible: change, levels, variance analysis, cost and price, comparison.
- Suitable where sales are seasonal, products or designs keep changing, work is made to order, fashion rules, or sales change generally. (Fuel for the distance you actually drove.)

Interactive version: PassBook app, lesson Flexible budgets

15.4 Cash budget

Key idea

A cash budget is a month-by-month plan of cash receipts and cash payments: only cash, and only in the month it moves. It shows in advance when cash will be short (so borrowing is arranged in time) and when it will be in excess (so it can be invested).

FORMATS AND FORMULAS

ICAI's cash budget format

The Paper 6A Study Material prints this format, headed "... Co. Ltd., Cash Budget, Period ..." (FM Study Material 9.43–9.44); the suggested answer to RTP May 2026 Case Scenario 1 uses the same layout in Paper 4. *Reproduce it from memory.*

Particulars	Month 1 (₹)	Month 2 (₹)	Month 3 (₹)
Receipts:			
1. Opening balance	xxx	xxx	xxx
2. Collection from debtors	xxx	xxx	xxx
3. Cash sales	xxx	xxx	xxx
4. Loans from banks	xxx	xxx	xxx

Particulars	Month 1 (₹)	Month 2 (₹)	Month 3 (₹)
5. Proceeds from issue of shares, debentures etc.	xxx	xxx	xxx
6. Miscellaneous receipts	xxx	xxx	xxx
7. Other items	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**
Payments:			
1. Payments to creditors	xxx	xxx	xxx
2. Wages	xxx	xxx	xxx
3. Overheads (factory rent, administration, selling and distribution)	xxx	xxx	xxx
4. Interest	xxx	xxx	xxx
5. Dividend	xxx	xxx	xxx
6. Corporate tax	xxx	xxx	xxx
7. Capital expenditure	xxx	xxx	xxx
8. Other items	xxx	xxx	xxx
Total	**xxx**	**xxx**	**xxx**
Closing balance [Surplus (+) / Shortfall (–)]	**xxx**	**xxx**	**xxx**

The second layout: net cash flow first

Several Paper 4 suggested answers (MTP Series I for January 2025 Q4(b), MTP Series II for May 2025 Q5(b)) keep the opening balance out of the receipts:

Particulars	Month 1 (₹)	Month 2 (₹)
Total receipts (A)	xxx	xxx
Total payments (B)	xxx	xxx
Net cash flow (A – B)	**xxx**	**xxx**
Add: Opening balance	xxx	xxx
Closing balance	**xxx**	**xxx**

Both earn full marks. Pick one and never mix them: an opening balance counted inside the receipts *and* added again below is counted twice.

The working notes

Every cash budget answer stands on two schedules, shown as working notes:

Credit sales of	Amount (₹)	Collected in month 1	Collected in month 2
Month −1	xxx	xxx (2nd month %)	—
Month 0	xxx	xxx (1st month %)	xxx (2nd month %)
Month 1	xxx	—	xxx (1st month %)
Collection from debtors		**xxx**	**xxx**

The payments to creditors follow the same pattern, with purchases in place of sales.

∑ **Formula**

Closing balance

$$\text{Closing balance} = \text{Opening balance} + \text{Total receipts} - \text{Total payments}$$

Each month's closing balance is the next month's opening balance.

∑ **Formula**

Collection from debtors in a month

$$\text{Collection}_t = \sum_{k \geq 0} \text{Credit sales}_{t-k} \times p_k$$

p_k = the share of a month's credit sales collected k months after the sale ($k = 0$ is the month of sale). The shares add up to 100% less the bad debts.

∑ **Formula**

Collections and payments from opening and closing balances

$$\text{Receipts from debtors} = \text{Opening debtors}$$
$$+\text{Credit sales} - \text{Closing debtors}$$

$$\text{Purchases} = \text{Cost of goods sold}$$
$$+\text{Closing stock} - \text{Opening stock}$$

$$\text{Payment to creditors} = \text{Opening creditors}$$
$$+\text{Credit purchases} - \text{Closing creditors}$$

Used when a question gives budgeted balances instead of collection patterns (FM Study Material 9.48). Bad debts written off are also deducted from the receipts.

Formula

Surplus or deficit against a minimum balance

$$\text{Surplus or (deficit)} = \text{Opening balance} + \text{Receipts} - \text{Payments} - \text{Minimum cash balance}$$

A deficit is borrowed (or investments liquidated); a surplus is invested or repays the borrowing. The closing balance is then the minimum balance, or above it (FM Study Material 9.46).

Definition

Cash budget

A detailed budget of cash receipts and cash payments incorporating both revenue and capital items for the budget period. It represents the cash requirements of the business during the budget period: the plan of receipts and payments of cash, analysed to show the monthly flow of cash, drawn up in such a way that the balance can be forecasted at regular intervals. (SM 15.54)

The theory parts

Objectives of preparing a cash budget (SM 15.55), in ICAI's order:

1. The probable cash position resulting from planned operations is assessed, so the excess or shortage of cash becomes clear; this helps in arranging short-term borrowings in advance for a shortage, or investments when cash is in excess.
2. Cash can be coordinated in relation to total working capital, sales investment and debt.
3. A sound basis for credit and for current control of the cash position is established.
4. The effect of sudden and seasonal requirements, large stocks, delay in collection of receipts, etc. on the cash position of the organisation is revealed to the management.

Advantages of a cash budget (SM 15.55):

1. It aids in securing the optimum working capital needed for smooth running of the operations, and in planning payments to shareholders (the Study Material prints "option working capital").
2. It eases the strains of a cash shortage.
3. It facilitates temporary investment of cash wherever, and to whatever extent, it is found in excess.
4. It provides for normal growth.

"Define cash budget": the definition above, then its two parts (receipts and disbursements, with three examples of each) and that it is a financial budget prepared month by month.

The numerical: three answers to the same question

Loses marks

A single table of figures: "Collection from debtors, June: 44,00,000."

If one lag is wrong, nothing shows the method, and nothing earns method marks.

Passes

The cash budget with every line, but no working notes; depreciation correctly left out without a word.

The examiner cannot see whether 44,00,000 came from the right months.

Full marks

Working notes first (collections, creditors, wages, overheads), each showing which month's figure is paid when; then the budget in one consistent layout; a closing line on any shortfall against the minimum balance, and one sentence on items deliberately left out (depreciation, loss on sale).

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: depreciation paid out of the bank**
   Fix: deduct depreciation (and any amortisation or provision) from the expense before it enters the payments.

2. **The lag that slipped a month**
   Fix: write the time line first, label each credit sale with the month it is collected, and check one figure by hand: April collects February's second instalment and March's first.

3. **Collecting the bad debts anyway**
   Fix: the collection percentages add up to less than 100%; the bad debts never become receipts.

4. **The opening balance counted twice**
   Fix: choose one layout (opening balance inside the receipts, or net cash flow plus opening balance) and keep it for every month.

5. **The asset's profit in place of its price**
   Fix: a cash budget records the ₹2,50,000 actually received; profit or loss on sale is not cash.

RECAP

- Cash budget (SM 15.54): a detailed budget of cash receipts and cash payments, revenue and capital, for the budget period; two parts, receipts and disbursements; a financial budget, usually short-term and monthly.
- Only cash, only when it moves: credit sales in the months collected, purchases in the months paid; depreciation, bad debts and profit or loss on sale stay out.
- Working notes first: collections from debtors, payments to creditors, wages and overheads, each re-timed to the month of payment.
- Closing balance = opening balance + receipts − payments; it becomes next month's opening balance.
- Two layouts: opening balance inside the receipts (FM Study Material 9.43), or net cash flow plus opening balance; never both.
- Against a minimum balance: borrow the shortfall so the minimum is restored, repay or invest out of later surpluses.
- Objectives (SM 15.55): probable cash position (borrow or invest in time), coordination with working capital, basis for credit and cash control, effect of seasonal needs and delayed collections revealed.
- Profit is a promise, cash is a fact: read the fuel gauge before the trip.

Interactive version: PassBook app, lesson Cash budget

15.5 Zero based budgeting and performance budgeting

Key idea

ZBB: every activity, old or new, must justify its cost from the first rupee; without justification the budget allowance is zero, and justified decision packages are ranked and funded in order of priority. Performance budgeting: the budget is presented in terms of functions, programmes and activities, linking the money (inputs) to the results expected (outputs), with responsibility fixed for those results.

ANSWER FRAME

"Explain zero-based budgeting and its stages" (5 marks)

1. The definition, word for word, including "though the activities ... are not being undertaken for the first time" and "without justification, the budget allowance is zero".
2. How it differs from traditional budgeting: from scratch, not from trends or historical levels; last year's figures are not accepted, cut or increased.
3. Activity-based; cost-benefit justification; evaluated and prioritised on objectives, funds and regulatory requirement.
4. Where it suits: corporate and non-corporate entities; social programmes; discretionary costs.

5. The four stages in ICAI's order: names only for "mention", a line on each for "explain".

"Distinguish between traditional budgeting and ZBB"

The six points of the table above (SM 15.61–15.62), written as a table, each row a pair of statements: traditional on one side, ZBB on the other.

"What should an enterprise consider to adopt performance budgeting?"

The five imperatives of the third worked example (objectives in concrete terms; translated into functions, programmes, activities and tasks; norms and indicators in physical units; decentralised responsibility; an accounting and reporting system for review).

Three answers to "Explain ZBB"

Loses marks

"In ZBB the budget starts from zero every year."

One line; no definition, no justification, no stages.

Passes

"ZBB requires each cost element to be justified for activities undertaken for the first time. Stages: identify, evaluate, rank, allocate."

The stages are there, but the definition has lost its "not", which reverses its meaning, and nothing is explained.

Full marks

Definition word for word, with "not" and "without justification, the budget allowance is zero"; the contrast with incremental budgeting; activity-based with cost-benefit justification; where it suits; the four stages in ICAI's order, each with its line when the question says "explain"; "priority-based budgeting" named at the ranking stage.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: "zero-based means a zero budget"**
   Fix: "zero" is the starting point of the argument, not the result.

2. **ZBB "for new activities only"**
   Fix: the definition says the opposite: each cost element is justified **though** the activities are **not** being undertaken for the first time. Continuing activities are the whole point.

3. **Calling ZBB quick and cheap**
   Fix: creating decision packages and ranking them on new data is very tedious for management: that is ICAI's first limitation, and the false option in ICAI's own MCQ.

4. **Performance budget, performance report, flexible budget**
   Fix: a performance budget presents funds by functions, programmes and activities with the physical results each must achieve, and fixes responsibility for those results.

RECAP

- ZBB (SM 15.59): each cost element specifically justified, though the activities are not being undertaken for the first time; without justification the budget allowance is zero.
- Starts from scratch, not from trends; activity-based; cost-benefit justification; prioritised on objectives, funds and regulatory requirement.
- Suits corporate and non-corporate entities (mid-day meal, street lights, drinking water); in companies,

discretionary costs (R&D, training, advertisement).

- Stages: identify and describe decision packages, evaluate, rank (hence priority-based budgeting), allocate resources.
- Six advantages: systematic evaluation, critical functions, cost-benefit allocation, waste eliminated, departmental budgets linked to corporate objectives, management by objectives. Two limitations: tedious, may not be implemented properly.
- Performance budgeting: funds presented by functions, programmes and activities with quantitative data on accomplishments; links inputs to outputs; synonymous with responsibility accounting.
- ARC: function, programme, activity, project; PB is still annual but output-oriented. To adopt it: concrete objectives, translated into functions to tasks, physical norms and indicators, decentralised responsibility, a reporting system.
- The jingle gets money when it shows what the money buys.

15.6 Control ratios, budget variances, motivation, feedback and feed-forward

Key idea

Activity ratio = efficiency ratio × capacity ratio: output against budget is speed times time. A budget variance is read against a budget flexed to the actual activity, with its cause and its controllability. Control works best when actuals are compared during the period (feed-forward) against targets the managers accept as achievable.

FORMATS AND FORMULAS

Reproduce the six formulas and the report layout from memory. (SM 15.68)

Formula

Efficiency ratio

$$\text{Efficiency ratio} = \frac{\text{Standard hours}}{\text{Actual hours}} \times 100$$

Standard hours = standard hours for the actual output (standard hours earned).

∑ **Formula**

Activity ratio

$$\text{Activity ratio} = \frac{\text{Standard hours}}{\text{Budgeted hours}} \times 100$$

∑ **Formula**

Calendar ratio

$$\text{Calendar ratio} = \frac{\text{Available working days}}{\text{Budgeted working days}} \times 100$$

∑ **Formula**

Standard capacity usage ratio

$$\frac{\text{Budgeted hours}}{\text{Maximum possible hours}} \times 100$$

Maximum possible hours in the budgeted period.

∑ **Formula**

Actual capacity usage ratio

$$\frac{\text{Actual hours worked}}{\text{Maximum possible hours}} \times 100$$

Maximum possible working hours in the period.

∑ **Formula**

Actual usage of budgeted capacity ratio (the capacity ratio)

$$\frac{\text{Actual working hours}}{\text{Budgeted hours}} \times 100$$

ICAI's MTPs call it the capacity ratio or capacity utilisation ratio.

∑ **Formula**

How three of them fit together

$$\text{Activity ratio} = \text{Efficiency ratio} \times \text{Capacity ratio}$$

$$\text{Efficiency ratio} = \frac{\text{Activity ratio}}{\text{Capacity ratio}}$$

True whenever the capacity ratio is actual hours ÷ budgeted hours, because the actual hours cancel.

Which ratio from which data

You are given	Find first	Then
Employees (or machines), hours a day, days	Maximum and budgeted hours	Both capacity usage ratios
Output and standard time (or rate) a unit	Standard hours	Efficiency and activity ratios
A capacity usage ratio	Hours = ratio × maximum hours (budgeted from the standard ratio, actual from the actual)	The other ratios
Two of efficiency, capacity and activity	The third: activity = efficiency × capacity	The missing hours from the ratio's definition

Definition

Budget ratios

A ratio is a mathematical relationship between two or more related figures. Budget ratios provide information about the performance level, i.e., the extent of deviation of actual performance from the budgeted performance and whether the actual performance is favourable or unfavourable. (SM 15.67, 15.71)

Definition

Controllable and uncontrollable variances

Controllable variances are those which can be controlled under normal operating conditions if a responsibility centre takes preventive measures and acts prudently. Uncontrollable variances are those which occur due to conditions beyond the control of a responsibility centre and cannot be controlled even though all preventive measures are in place. (SM 13.11–13.12)

The budget report

The layout of ICAI's own answer comparing a master budget, a flexible budget and actual results (SM 15.92):

Particulars	Master budget (₹)	Flexible budget (₹)	Actual (₹)	Variance (₹)
Units	xxx	actual units	actual units	
Sales	xxx	xxx	xxx	xxx (F/A)
Variable costs, item by item	xxx	xxx	xxx	xxx (F/A)
Contribution	xxx	xxx	xxx	
Fixed overhead	xxx	xxx	xxx	xxx (F/A)
Net profit	**xxx**	**xxx**	**xxx**	**xxx (F/A)**

The variance column compares actual with the flexible budget, not with the master budget.

ANSWER FRAME

"Discuss feedback and feed-forward control" (4

marks)

Point	Feedback control	Feed-forward control
Timing	Variances identified after the completion of the budget period	Actual results continuously monitored and compared during the period
Nature	Ex-post corrective	Ex-ante preventive
Action	Corrective action, responsibility fixed, future targets modified	Targets kept realistic, reviewed and reset if necessary
Information needed	Data available only after finalisation of books; MIS not robust	Robust MIS supported by an integrated ERP system
Cost and fit	Less expensive, but leaves the organisation on the looser side	Very expensive; suits a dynamic business environment where information gives a competitive edge

"How can budgets motivate executives?"

The four considerations in ICAI's order, each with a line: performance measurement (communicated targets, achievement used in evaluation, rewards); achievable targets (neither very easy nor too tough; unrealistic targets demotivate); optimum utilisation of resources

(beyond the comfort zone, up to the effort zone); involvement in the budgeting process (participative budgeting gives ownership; a balance, not involvement at every stage).

The ratio numericals: three answers

Loses marks

"Efficiency 95%, activity 104.5%, capacity 110%."

Bare figures; one wrong hours figure and nothing earns method marks.

Passes

Each ratio with its formula and figures, but the budgeted hours appear without the lines × hours × days working.

Correct, with the base of every ratio unexplained.

Full marks

Working notes for maximum, budgeted, actual and standard hours and for the days; each ratio with formula, figures and result (ICAI prints 109.375% in full, SM 15.69); a one-line check (efficiency × capacity = activity) and a comment marking each ratio favourable or unfavourable.

COMMON MISTAKES

⚠ **Common mistakes and their fixes**

1. **Hall of Shame: the activity ratio on actual hours**
   Fix: activity compares output with the **budget**: standard hours ÷ budgeted hours. Check with E × C = A.

2. **Maximum capacity used as the budget**
   Fix: the maximum capacity (installed lines, maximum employees) gives the maximum possible hours; the lines or employees budgeted to work ("actual working" in ICAI's data) give the budgeted hours.

3. **The efficiency ratio upside down**
   Fix: standard hours are always the numerator of the efficiency and activity ratios.

4. **"Favourable, so well done"**
   Fix: compare with the flexed budget, then ask the cause: an unspent campaign, skipped maintenance or a lower activity level is not efficiency.

RECAP

- Efficiency = standard hours ÷ actual hours; activity = standard ÷ budgeted; calendar = available days ÷ budgeted days; all × 100.

- Standard capacity usage = budgeted ÷ maximum hours; actual capacity usage = actual ÷ maximum; actual usage of budgeted capacity (the capacity ratio) = actual ÷ budgeted.
- 100% or more is favourable; below 100%, unfavourable. Activity = efficiency × capacity.
- Budgeted hours come from the employees budgeted to work; maximum hours from maximum capacity; deduct normal downtime if given.
- Budget variances: compare with the flexed budget; favourable is not necessarily efficient; exception reports show both directions; controllable variances are the manager's to answer.
- Feedback: after the period, ex-post corrective, cheaper. Feed-forward: continuous, ex-ante preventive, needs robust MIS and ERP, suits dynamic businesses.
- Motivation: performance measurement, achievable targets, optimum utilisation of resources, involvement. (Rain cut the overs; the batters were fine.)